Tracking Artificial Intelligence Sentiment in the U.S. Labor Market

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Abstract

This study aims to track U.S. labor market sentiments toward artificial intelligence (AI) from multiple aspects by using large-scale datasets of employee reviews from Glassdoor, executive disclosures from earnings conference calls, individual-level career profiles from LinkedIn, and firm-level job postings, spanning 2021 to 2025. Through a textual analysis of employee job reviews, we find that employee sentiment towards AI is significantly more negative than their overall sentiment towards firms. Importantly, we document a strong positive association between firm productivity and employee AI sentiment. In contrast, management AI sentiment (measured based on management discussions of AI in conference calls) is highly optimistic but does not explain productivity, suggesting that employee AI sentiment plays an important role in unlocking the benefits of AI. By analyzing the contents of employee reviews, we discovered several factors contributing to the negative impact of AI on employee sentiment, among which concerns about job security have the most negative impact. AI-related layoffs negatively affect employee AI sentiment and thus could offset the benefits of adopting AI. Consistent with employee concerns, we find that firms with more AI-skilled employees have a higher overall turnover rate, lower hire rate, and slower employment growth. These results are also true for firms without excessive hiring after the COVID pandemic, suggesting that the effects are not mainly due to AI washing. Finally, we document a significant and increasing wage premium and greater upward mobility for skilled AI employees, suggesting a talent war for skilled AI employees among firms and significant rewards for AI skills in the labor market. Overall, the study provides a comprehensive analysis of AI's impact on the U.S. labor market by using the most recent data.

Keywords: Artificial Intelligence, Employee Sentiment, Firm Productivity, Job Displacement, Wage Premium, Labor Market Mobility

1. Introduction

The rapid advancement of artificial intelligence (AI) is fundamentally transforming the labor market. Since the launch of ChatGPT in November 2022 and the subsequent rise of agentic AI in 2025, AI is evolving from a technical tool into a pervasive force reshaping how work is structured, performed, and valued. Recent announcements of AI-related layoffs have further led to a heated debate of AI's impact on the labor market. This study aims to track the labor market impact of AI from three main perspectives, including employee and management sentiment towards AI, job displacement effects of AI, and labor market premium for AI-skilled employees.

Our analyses begin with an analysis of over 21 million employee job satisfaction reviews from Glassdoor from 2021 to 2025. The share of Glassdoor reviews mentioning AI grows exponentially over the period from 0.21% in 2021 to 1.20% in 2025, with sharp accelerations following both the ChatGPT launch and the onset of the agentic AI era. Through a textual analysis, we find that employee AI sentiment, measured by the sentiment of AI-related employee review comments, is significantly more negative than the sentiment of employees' overall comments on their employers. Also, we find that employees who mention AI in their reviews have significantly lower job satisfaction ratings, after controlling for firm and quarterly fixed effects, seniority, and employment status. This negative effect intensifies following the ChatGPT launch. These findings highlight a negative employee sentiment toward AI. Importantly, we document a strong positive association between employee AI sentiment and firm productivity, suggesting that employee AI sentiment plays an important role in unlocking the benefits of AI and that the negative employee AI sentiment likely hurts productivity.

To reveal factors that contribute to the negative employee AI sentiment, we provide two sets of analyses. First, by analyzing the content of AI-related comments, we classify these reviews into four themed categories: 1) the impact of AI on job security, 2) the adoption of AI and its impact on operational productivity, 3) corporate AI leadership, and 4) AI upskilling. While the sentiment of comments in all four categories is more negative than employees' overall sentiment towards firms, job security related issues have the most negative impact on employee sentiment. Second, a regression analysis on determinants of employee AI sentiment further reveals that employee AI sentiment is significantly negatively associated with AI-driven layoff announcements. Thus, AI-related layoffs actually offset the benefits of adopting AI by negatively affecting employee AI sentiment. Management AI expertise, strong corporate leadership, and the availability of career opportunities in a firm are positively correlated with employee AI sentiment.

Further, through analyzing approximately 10,000 earnings conference call transcripts for U.S. public firms, we find that executives consistently express strong optimism about AI in earnings conference calls, and this strong optimism remains stable across years. This executive optimism, however, bears no significant relationship to firm productivity or employee AI sentiment. These findings suggest that management optimism does not effectively translate into higher employee optimism or better performance. In contrast, as discussed above, ground-level employees' AI perspectives carry real informational content that executive narratives do not.

As job security concerns play a central role in shaping employee AI sentiment, we dive deeper into the impact of AI on job displacement. We obtain U.S. job postings from Revelio Labs, encompassing tens of millions of listings with detailed salary and skill requirements, and individual-level employment histories from LinkedIn, which we use to identify job contents, career transitions, and worker characteristics, including gender, seniority, and ethnicity. Consistent with prior research (e.g., Kalyani et al. 2025), we find a sharp increase in job postings requiring AI skills. But, as firms employ more AI-skilled workers, they experience significantly lower hire rates, higher employee turnover rates, and slower employment growth, consistent with AI directly displacing human labor demand. These findings hold across industries, seniority levels, genders, and ethnic groups. Importantly, this pattern is unlikely to be driven by AI washing behavior (i.e., employees blame AI for layoffs that are due to post-covid excessive hiring). Specifically, both firms that aggressively expanded headcount after COVID and those that did not exhibit similarly higher turnover and lower hiring as their AI talent pools grow. These findings are consistent with employees' concerns about the job displacement effect of AI.

Finally, we analyze the impact of AI skills on wages and job market mobility. Consistent with prior literature (Alekseeva et al. 2021), we find a salary premium for AI-skilled employees. Also, this wage premium increases over time. Further, AI-skilled employees have a higher turnover rate and a greater upward labor market mobility than non-AI-skilled employees, suggesting a talent war for AI-skilled employees among firms. Specifically, AI-skilled workers who transition between firms are significantly more likely to receive promotions and move to larger organizations, reflecting strong and persistent demand for AI skills and a growing divide in career mobility between AI and non-AI workers.

Our paper contributes to several strands of literature. First, we provide the first analysis of how employees, the direct users of AI inside firms, perceive AI adoptions and contribute to the growing body of research on the productivity impact of AI (e.g., Brynjolfsson et al., 2023; Babina, 2024) by providing evidence on a significant link between employee AI sentiment and firm productivity. By using survey data, Acemoglu et al. (2023) find that firms with more AI utilization are more productive. Babina et al. (2024) use job postings and employee resumes to measure firm-level AI investment based on the number of AI skilled employees and document positive effects of AI investment on sales growth. Brynjolfsson et al. (2023) find that access to AI tools increases customer support agents' productivity by 14% on average. This improvement concentrates among novice and low-skilled workers. AI tools have minimal impact on the productivity experienced by highly skilled workers. Our findings contribute to the literature by suggesting that employee sentiment towards AI plays a critical role in unlocking the benefits of AI for firm productivity. We also document factors that shape employee AI sentiment, which have direct practical implications for firms and managers. In contrast, we find that management AI sentiment does not significantly explain firm predictivity. Therefore, employee AI sentiment carries predictive content for productivity that executive-level disclosures do not.

Second, we extend the literature on the job displacement effect of AI. Felten et al. (2021) construct occupation-level AI exposure indices and find that AI exposure varies significantly across occupations. Eloundou et al. (2024) estimate that approximately 80% of the U.S. workforce could have at least 10% of their tasks affected by LLMs, and that 18.5% of workers face exposure exceeding 50% of their tasks — pointing to a potentially far more disruptive transition. Acemoglu et al. (2022) examine job postings from 2010 to 2018 and find that as establishments adopt AI, they post fewer positions for non-AI jobs. But the aggregate impacts of AI-labor substitution on employment are limited. Tracking task-level AI exposure from 2010 to 2023, Hampole et al. (2025) find significant evidence of the substitution effect at the task level. But overall employment effects are modest because AI-driven labor replacement, which is concentrated in a narrow set of highly exposed tasks, is partially offset by workers reallocating effort across other tasks. We extend the literature by looking at the most recent data and providing firm-level evidence on the job displacement effect of AI through analyzing firm hire rate, turnover rate, and total employment growth.

Third, we also contribute to the literature on the wage impact of AI skills. Alekseeva et al. (2021) use job posting data to document that AI-related positions pay about 11% more than comparable non-AI positions, though their analysis ends before the generative AI era. Consistent with prior literature, we find that AI skills are associated with significant wage premiums and that the wage premium is larger and increasing. We also find that AI-skilled workers have higher upward labor market mobility (e.g., more promotion opportunities and more opportunities to move to larger firms). These findings provide a more complete understanding of the labor market incentives for employees to obtain AI skills.

2. Data and Methodology

2.1 Data Sources

Our analysis draws on four primary data sources spanning the period 2021-2025. We first obtain all English-language employee reviews posted on Glassdoor between January 2021 and December 2025, comprising over 21 million reviews. For each review we extract an overall satisfaction rating (1-5), a business outlook indicator (positive, neutral, negative), employment status (regular, part-time, temporary, apprentice, contract, freelance, intern, per-diem, PhD, reserve, self-employment, seasonal, trainee, and unknown), and textual comments for "Summary," "Pros," "Cons," and "Advice to Management." We match reviewed companies to public firm identifiers where possible, yielding a subsample of approximately 5.7 million reviews linked to publicly traded firms. Our second data source is earnings conference call transcripts for U.S. public firms from 2021 to 2025. These transcripts capture executives' prepared remarks and responses to analyst questions, providing a window into how firm leadership publicly discusses AI strategy, adoption, and investment plans.

We further obtain job postings from Revelio Labs for all public firms posted between 2021 and 2025.[1] Each posting includes the job title, description, advertised salary (when available), firm identifier, and posting date. The full dataset encompasses tens of millions of postings per year. Because our AI-keyword list preparation and AI-related content identification procedures require textual analysis, we retain only postings with non-missing job descriptions. We also obtain individual-level worker records from Revelio Labs for all individuals with a LinkedIn profile. We aggregate these records into a firm-role-quarter panel covering 1,500 job positions classified by Revelio Labs, capturing the number of employees who joined, the number who departed, and the total headcount in each position. We obtain information about the worker's seniority level, gender, ethnicity, and remote work status.

2.2 LLM-Augmented Classification Pipeline

A key methodological innovation of this paper is the development of a multi-stage, LLM-augmented classification pipeline for identifying and categorizing AI-related content across multiple text sources.

2.2.1 Stage 1: Keyword Generation

We draw a stratified random sample of 200,000 job descriptions from the Revelio Labs data and use GPT-based models to identify AI-related terms and phrases. This process yields an initial dictionary of over 1,600 unique keywords. We then manually review the list, removing ambiguous terms whose primary usage falls outside the AI domain, for example, "alignment,"

[1] In Revelio Lab database, Alphabet Inc. only covers employees of the parent holding company or Google but does not include employees of Google. In additional tests, we include Google's employees into Alphabet Inc. and results are similar.

"diffusion," "gate," and "whisper." This leaves us a comprehensive yet clean list of 862 terms for identifying AI-related content in employee LinkedIn profiles. The final dictionary covers a broad range of AI-related topics, including core AI terminology (e.g., "artificial intelligence," "ai"), generative AI (e.g., "genai"), machine learning (e.g., "ml"), neural network architectures (e.g., "rnn"), large language models (e.g., "llm"), genetic algorithms, prompt engineering, retrieval-augmented generation (RAG), natural language processing, conversational AI and chatbots (e.g., "conversational interface"), computer vision and image processing, cloud-based AI services, robotics and autonomous systems, etc. The complete keyword list is provided in Appendix C. We use this comprehensive dictionary to identify AI-related content in LinkedIn job descriptions and Glassdoor employee reviews.[2]

To identify AI related content in conference call transcripts, we adopt a separate keyword generation process because the vocabulary of executive earnings discussions differs substantially from that of job descriptions. Conference call content is more formal and carefully prepared for public understanding, rather than casually expressing opinions related to work or targeting specialized talent for a job. We use GPT-5 to read 10,000 conference call transcripts and generate a distinct keyword list, then repeat the manual review process to produce a clean set of 357 terms after removing ambiguous and redundant entries.

2.2.2 Stage 2: NLP-Based Automatic Text Analytics

Using the above keyword dictionaries, we use natural language processing (NLP) techniques to automatically extract AI-related content from both Glassdoor reviews and conference call transcripts. The document is first split into sentences using sentence segmentation.

[2] To validate the applicability of the dictionary to Glassdoor reviews, we repeat the process by using a sub-sample of Glassdoor reviews and found that the AI keywords identified based on Glassdoor review were already captured in the job description keyword list.

For each keyword occurrence, the sentence containing the keyword is extracted as a context segment, optionally including adjacent sentences to provide additional narrative information. By design, our keyword lists are intentionally broad (862 terms for reviews and 357 for conference calls) to maximize recall. This over-inclusive approach inevitably produces some false positives, which we address in the subsequent validation stage.

2.2.3 Stage 3: LLM Validation and Classification

Each candidate passage flagged in Stage 2 is submitted to GPT-5 for validation through structured multi-task prompting. For employee reviews and earnings conference calls, the model determines whether the passage genuinely discusses AI, filtering out false positives and identifying the AI-specific content of the review. For earnings conference calls, the model additionally assesses whether the passage contains information pertaining to AI-related investments. Throughout the pipeline, we draw stratified random samples from each classification stage, submit them to GPT, and manually verify the returned results. We iteratively adjust prompts based on error analysis until achieving above 95% confident accuracy levels. This human-in-the-loop approach ensures that our automated classifications are reliable while maintaining the scalability needed to process millions of text passages. For example, when manually checking GPT's identification of AI-related content in Glassdoor employee reviews, we find in the first round that the model cannot correctly identify metaphorical language. In one review, the employee wrote: "No matter how much effort you devote to this job, they will lay you off as if a heartless algorithm." GPT mistakenly labels this as an AI-related review, prompting us to add instructions that address metaphorical usage. For job postings and AI-related job identification, we skip this validation step because the dataset is prohibitively large and the keywords were originally extracted from the same source content.

To construct reliable sentiment measures, we apply the Harvard General Inquirer IV-4 (HIV4) dictionary to the AI-related content identified through the keyword screening stage and subsequently validated by GPT. Specifically, for each review or conference call passage confirmed as AI-related, we compute employee AI sentiment by applying the HIV4 lexicon to the extracted AI-specific text to identify positive and negative words. Specifically, we calculate a polarity score for each employee review, ranging from -1 to 1, as (positive word count - negative word count) / (positive word count + negative word count). Scores approaching 1 indicate more positive sentiment, and scores approaching -1 indicate more negative sentiment. This approach produces our primary sentiment measures for both employee AI sentiment and executive AI sentiment. We also use GPT-5 to independently measure the sentiment of the AI-related content by classifying the content as positive, neutral, or negative. However, because GPT's internal sentiment classification mechanism is not fully transparent, we treat the sentiment labels returned by GPT as an alternative measure for robustness checks rather than as our main measure.

2.2.4 Stage 4: Thematic Categorization

For employee reviews, we develop a hierarchical categorization scheme through an iterative process. We first manually collected an initial set of categories and feed them along with the comprehensive keyword dictionary into GPT as guidance. GPT then repeatedly reads and classifies AI-related reviews using this input as a reference framework. We regroup and consolidate the emerging categories, and arrive at a final taxonomy comprising four categories:

(i) *AI Job Insecurity*, encompassing concerns about job displacement, layoffs, and workforce reduction due to AI;

(ii) *AI Leadership*, covering discussions at the intersection of leadership and artificial intelligence;

(iii) *AI Productivity & Operation*, capturing discussions of AI-driven efficiency gains, process automation, and operational changes;

and (iv) *AI Upskill*, capturing discussions of employees' experiences with and opportunities to use AI tools, as well as broader efforts toward AI-related skill development and training.

We provide representative examples for each subcategory in Table A1. More detailed guidance on tuning the LLM to identify AI-related content is provided in Appendix D.

3. Empirical Analyses

3.1 Employee AI Sentiment

We begin by documenting the prevalence of AI-related content in employee reviews. Table 1 reports the quarterly share of reviews containing AI-related comments in our Glassdoor employee review sample. The share rises steadily from 0.21% in 2021 Q1 to 1.20% in 2025 Q4, reflecting the rapid adoption and use of AI in the workplace. There are two transformative turning points in the trajectory: the ChatGPT launch and the onset of the Agentic AI era. Following the ChatGPT launch in November 2022, the AI-related review percentage accelerated from approximately 0.23% to 0.33% over the subsequent year. A second, sharper acceleration occurs in early 2025 as AI systems shifted from conversational assistants to autonomous agents capable of multi-step reasoning, tool use, and workflow execution, with the percentage rising from 0.49% in 2024Q4 to 1.20% in 2025Q4. Figure 1a visualizes this time trend, clearly marking the two AI milestones. The exponential growth pattern in the Agentic AI period suggests that the emergence of autonomous AI agents has substantially broadened employee engagement with AI technologies.

Figure 1b tracks public firm AI-related corporate investment events over the same period, revealing a parallel acceleration in firm-level AI commitment, particularly after the ChatGPT launch and into the Agentic AI era. Figure 1c further shows that the frequency of AI-attributed layoff events from 2022 Q2 rose steadily alongside investment activity, with a renewed upward trend emerging in 2025 as autonomous AI systems began displacing and restructuring human roles at scale. To construct the cumulative AI investment and layoff measures, we collect all RavenPack news article headlines containing "AI" or "Artificial Intelligence" in combination with investment-related keywords (e.g., invest, investment, spend, million, billion) or layoff-related keywords (e.g., lay off, layoff, laid off), which we categorize as AI investment news and AI layoff news, respectively. We then apply GPT to standardize the firm names mentioned in each headline and to verify whether each article genuinely pertains to an AI investment or layoff event. For investment and layoff news, GPT additionally extracts the reported investment amount or number of affected employees. To remove duplicate coverage of the same underlying event, we first discard articles reporting identical investment amounts or layoff figures within a five-day window of the earliest report, and then manually review the remaining articles to retain only the first news item per event.

In Figure 1c, we also classify AI related layoffs into real AI displacement or potential AI washing based on a firm's AI talent pool and hiring behavior since Covid. Real AI displacement refers to the situation where a firm cuts its headcount due to either past or planned AI investment and adoption. For example, if a firm lays off employees because it has already invested intensively in AI or plans to increase its investment in AI, the layoff is considered real AI displacement. Otherwise, the layoff is potentially AI washing, which refers to the situation where a firm cuts its headcount due to over-hiring after the COVID Pandemic or other non-AI related reasons. Empirically, we classify its layoffs as real job displacement, only if a firm has more AI-skilled employees (see section 3.3 for definitions of AI skilled employees) and lower employment growth after COVID compared with their industry peers. Such firms have invested heavily in AI, as shown by their AI skilled employees, and did not over hire compared to peers. Otherwise, it is classified as potential AI washing (manager blaming AI for layoffs due to overhiring or other issues). 56 cases of 86 AI-related layoffs that have data for this analysis are identified as real displacement rather than AI washing based on the criteria. The detailed list of these layoffs and their classifications are in the appendix Table B2.

Figure 1d plots the distribution of Cumulative Abnormal Returns (CARs) for AI investment announcements and AI layoff announcements. The top panels show CARs for a three-day window surrounding the layoff announcement and the bottom panels show CARs for a seven-day window surrounding the layoff announcement. We find that while some firms experience significant jumps in stock prices after the announcement, the stock market reaction is actually negative for many firms. On average, abnormal returns following AI investment announcements is approximately 1 percentage point. However, abnormal returns following these AI layoff announcements are statistically insignificant and close to 0 in economic magnitude. Therefore, the market does not always react positively to AI investment and AI layoff announcements, suggesting that these events are likely associated with significant costs.

Figure 2a decomposes the AI-related reviews by industry. Unsurprisingly, the Information sector leads by a wide margin, with AI-related review shares climbing from 0.55% in 2021 to 2.29% in 2025, more than four times the rate of any other industry. Manufacturing and Finance & Real Estate follow. Perhaps more striking is the depth that AI is reaching that even traditionally low-tech industries, such as Construction, Agriculture, and Mining, show notable increases by 2025, suggesting that AI's workplace impact extends well beyond the technology sector.

We then examine the content of AI-related employee reviews. Again, we classify AI related reviews into four categories: *AI Job Insecurity* captures employee concerns about job displacement, layoffs, and workforce reduction driven by AI adoption; *AI Leadership* documents discussions covering the role of corporate leadership in AI adoption; *AI Productivity & Operations* covers discussions about how AI affects day-to-day productivity and business processes, including efficiency gains and process automation; and, *AI UpSkill* describes discussions of training, skill development, and opportunities to work with or learn AI-related technologies. Figure 2b tracks the quarterly number of AI-related reviews by each category. All four categories grow over time, with *AI Productivity & Operations* and *AI UpSkill* being the most frequently mentioned issues. Notably, *AI Job Insecurity* sees a sharp rise over recent years, reflecting a growing unease by employees about how AI is going to affect the labor market. Together, these trends highlight three dominant themes in the employee AI conversation: concerns about job displacement, opportunities and pressure to develop AI-related skills, and the evolving impact of AI on operational productivity.

Figure 3 presents the sentiment distribution of AI-related employee reviews across each of the four thematic categories. The sample includes 82,160 reviews that mentioned AI related issues. First, we measure employee *Overall Sentiment* towards firms based on the sentiment of employees' overall comments about their employers. At the aggregate level, overall employee sentiment across all 82,160 reviews is predominantly positive, with 89.7% positive and only 5.1% negative. This high positivity rate reflects the general tone of employee reviews rather than their specific views on AI. *AI Sentiment* is measured specifically on the AI-related content within each review, the picture shifts considerably: only 53.7% of AI-specific comments are positive, and 12.4% are negative, indicating that employees hold more mixed and critical views when discussing AI directly.

The category-level breakdown reveals further heterogeneity. *AI UpSkill* is the most positively skewed category, with 61.8% positive and only 8.0% negative sentiment across 27,863 reviews, suggesting that employees largely welcome opportunities to develop AI-related skills. *AI Productivity & Operations* similarly leans positive, though with moderately higher negative shares of 14% across 48,323 reviews. In contrast, *AI Job Insecurity* and *AI Leadership* are notably more negative. *AI Job Insecurity* registers 21.2% negative sentiment across 4,333 reviews, while *AI Leadership* records 17.2% negative sentiment across 1,641 reviews. Together, these patterns suggest that while employees are generally optimistic about AI's operational and developmental potential, concerns about job security and the role of corporate leadership represent widespread sources of employee dissatisfaction.

Table 2 presents further regression results of how mentioning AI related issues in employee comments relates to employee job satisfaction. The dependent variable *Job Satisfaction* is the employee's overall job satisfaction rating scaled from 1 to 5. The independent variable of interest, *Employee AI Mention*, is a binary indicator equal to 1 if an employee review contains AI-related content and 0 otherwise. Column (1) reports the results using the full sample of over 21 million U.S. employee reviews from 2021 to 2025. The coefficient on *Employee AI Mention* is -0.134 (t-statistic = -15.33) and is significant at the 1% level, indicating that employees whose reviews mention AI report significantly lower overall satisfaction with their employers. The regression controls for *Seniority*, which is the reviewer's self-reported seniority level, scaled from 1 (lowest) to 7 (highest). The coefficient on *Seniority* is positive and significant, suggesting that more senior employees tend to report higher satisfaction. We also control for fixed effects of employment status, quarter, and firm.

Column (2) introduces an interaction term to examine how the public launch of ChatGPT moderates the results. To capture this dynamic effect, we define *PostGPT* as an indicator equal to 1 for reviews posted in 2023 or later, following the public launch of ChatGPT. *Employee AI*

*Mention*PostGPT* captures how the relation between AI mentioning and job satisfaction changes after the public launch of ChatGPT (2023 and later). The coefficient on this interaction is negative and highly significant at -0.119 (t = -7.68). This pattern suggests that ChatGPT accelerates the widespread adoption of AI and deepens employee dissatisfaction.

Columns (3) to (6) replicate the analysis using subgroups. To ensure that the reason why employees do not mention AI in their comments is not that their employers do not use AI, Column (3) limits the sample to firms with at least one AI-related review, and Column (4) further restricts the sample to firm-quarters with at least one AI-related employee review. The coefficients remain stable and highly significant across both subsamples, ruling out concerns that our results are driven by selection into AI adoption. Columns (5) and (6) further restrict the sample to public firm employees, with and without firm-level financial controls. These control variables can not be calculated for private firms due to data limitations. The estimates remain large and significant at -0.271 (t = -15.13) and -0.302 (t = -13.57) after controlling for the set of financial controls, respectively, indicating that the sentiment penalty is not driven by differences in firm size, leverage, or growth opportunities. Overall, these results suggest that AI-related issues are associated with lower job satisfaction, confirming a general negative employee sentiment towards AI.

Table 3 examines whether employee AI sentiment explains firm-level total factor productivity, restricting the sample to firm-years with at least one AI-related employee review. We estimate total factor productivity (TFP) using three standard approaches. *TFP_LP* is estimated using the Levinsohn and Petrin (2003) method, which uses intermediate inputs (cost of goods sold) as a proxy for unobserved productivity shocks to correct for simultaneity bias. *TFP_OP* is estimated using the Olley and Pakes (1996) method, which uses capital expenditures as a proxy and additionally corrects for selection bias from firm exit. *TFP_OLS* is the residual from an OLS regression of log sales on log capital and log labor. All TFP measures are estimated separately within each two-digit SIC industry and demeaned by industry-year means. We then construct a firm-level employee AI sentiment measure, *Firm Employee AI Sentiment,* by taking the annual firm-level average of employee AI sentiment scores of all employees who mentioned AI related issues in their reviews. As an alternative measure, we also construct a measure, *Firm Employee AI Sentiment GPT,* based on the annual firm-level average of the GPT calculated employee AI sentiment. Firm-level controls include *Size* (the natural logarithm of total assets), *Tangibility* (net property, plant, and equipment scaled by total assets), *Sales_Growth*, *Leverage*, *Capex* (capital expenditures scaled by total sales), and *Management AI sentiment* (the sentiment of management discussions of AI during conference calls, which are explained with more details in the next section). All continuous control variables are winsorized at the 1st and 99th percentiles.

Column (1) uses *TFP_LP* as the dependent variable without controls and finds a positive and significant coefficient on *Firm Employee AI Sentiment* of 0.014 (t = 2.62), indicating that firms where employees express more positive sentiment toward AI tend to exhibit meaningfully higher total factor productivity. Column (2) adds firm-level controls, and the coefficient remains stable at 0.015 (t = 2.87), confirming that the result is not driven by differences in firm size, tangibility, leverage, or growth opportunities. Among the control variables, *Sales_Growth* is consistently positive and significant, while *Tangibility* is negative and significant across all specifications, consistent with asset-heavy firms facing greater structural constraints on productivity. Interestingly, *Management AI sentiment* does not bear a significant relationship with productivity. Columns (3) and (4) replicate the analysis using *TFP_OP* and *TFP_OLS* as alternative productivity measures and find similarly positive and significant coefficients of 0.017 (t = 2.31) and 0.014 (t = 2.02), respectively, demonstrating robustness to the choice of TFP estimation method. Column (5) replaces the Harvard IV-based sentiment measure with the GPT-based sentiment measure, *Firm Employee AI Sentiment GPT*, and the result remains positive and significant at 0.010 (t = 2.34). The consistency of the findings across three distinct TFP measures and two independently constructed sentiment measures significantly strengthens the inference that ground-level employee sentiment toward AI captures genuine variation in how effectively firms translate AI adoption into productivity.

Table 4 examines the determinants of firm-level employee AI sentiment, *Employee AI Sentiment*, by using all public firms' employee reviews that mention AI-related issues.[3] We adopt different sets of fixed effects across the four columns. First, we examine how corporate leadership's AI expertise affects employee AI sentiment. The coefficient on the percentage of corporate executives with AI skills identified based on their LinkedIn profiles, *Mgmt AI Expert*, is positive and significant across Columns (1) through (3) — 0.468 (t = 4.69), 0.548 (t = 5.23), and 0.477 (t = 3.91) — indicating that firms with more AI-skilled leadership foster meaningfully more positive employee attitudes toward AI. This effect turns insignificant once firm fixed effects are introduced in Column (4), likely because this variable lacks significant within-firm variations since most exercises do not experience turnovers over our sample period.

Next, we examine the roles of executive AI sentiment and corporate AI investment and layoffs. Management AI sentiment (*Managament AI Sentiment*) is measured based on earnings conference calls. Also, we manually match the firms mentioned in each headline of AI investment or layoff news to public firm records in Compustat. This step requires careful judgment, as many AI startups have names closely resembling those of public firms. For instance, "Lily's AI" could be confused with the public firm Eli Lilly, and we verify each match through additional news

[3] Our sample is limited to public firms, as the measures of AI investment, AI layoffs, management AI expertise and management AI sentiment are based on conference calls or other data which are only available among public firms.

searches to ensure accurate firm identification. The resulting variables, *AI Investment* and *AI Layoff*, capture the natural logarithm of the cumulative number of AI investment and AI-attributed layoff events, respectively, recorded for each firm from 2021 through the end of each time period used in our analysis. In contrast to leadership AI expertise, management AI sentiment, and publicly announced AI investment (*AI Investment*) bear no significant relationship with employee AI sentiment across all specifications. The most consistently negative determinant of employee AI sentiment is *AI Layoff*, publicly announced AI related layoffs. Its coefficients range from -0.055 to -0.093 and remain significant across all four columns, including the within-firm estimator. Thus, AI-attributed layoff announcements significantly erode employee sentiment toward AI.

Further, we include the seniority of the employee who gives the review and the employee's Glassdoor ratings on corporate leadership and career opportunities in the regression. *Seniority*, *Leadership*, and *Career Opportunity* are all positively associated with employee AI sentiment, suggesting that good corporate leadership and perceived career prospects positively shape employee AI sentiment in the workplace.

3.2 Management AI Sentiment

The tests above show no significant relation between management AI sentiment and productivity. But the sample in Table 4 requires each firm to have at least one employee AI related comments. To mitigate concerns that the insignificant effect of management AI sentiment is due to the sample restriction, we provide further analyses of management's AI sentiment using larger samples without requiring employee AI related comments. Specifically, we construct three measures from earnings conference call transcripts. Our first measure captures the mentioning of AI. For each conference call, we first identify whether the executives mention AI-related content in at least one conference call during the fiscal year, *Management AI Mention*. Our second measure, *Management AI Investment Mention*, is an indicator equal to 1 if executives explicitly mention investing in or increasing AI spending on at least one conference call during the fiscal year. Our third measure, *Management AI Sentiment*, is constructed by first calculating the polarity score of AI-related content for each conference call based on Harvard IV dictionary and then averaging across all conference calls held during the fiscal year.

Figure 4 documents the divergence between executive AI sentiment in earnings conference calls and employee AI sentiment in Glassdoor reviews over the sample period. Executives consistently discuss AI in a strongly positive tone, with average sentiment scores remaining stable between 0.4798 and 0.4651 on a [-1, 1] scale throughout 2021 to 2025. Employee AI sentiment, by contrast, remains steadily low over the same period. The resulting gap between the two series is approximately 0.008.

Table 5 examines whether executive AI discourse in earnings conference calls explains firm productivity in a sample of all firms with earnings conference calls. Columns (1) report results for *Management AI Mention* and *Management AI Investment Mention* for *TFP_LP,* the most common and robust methods for estimating firm-level productivity*,* and the results are insignificant: the coefficients are 0.000 (t = 0.04) and -0.009 (t = 1.46) for *TFP_LP*, respectively, indicating that whether executives mention AI or AI investments in earnings calls bears no meaningful relationship with firm productivity. Columns (2) to (4) further examine whether the sentiment of executive AI discourse carries predictive power, restricting the sample to firm-years in which executives discuss AI-related content. Across columns using different total factor productivity measures, *Management AI Sentiment* remains insignificant: the coefficients are 0.012 (t = 1.01), 0.019 (t = 1.40), and 0.018 (t = 1.32), respectively. Taken together, the results in Table 5 indicate that neither the occurrence nor the tone of executive AI discourse in investor communications carries informative content about firm-level productive performance, consistent with the view that executive AI optimism reflects deliberate investor-facing framing rather than genuine operational progress.

Taken together, the results in Tables 3 and 5 reveal a fundamental asymmetry in the information content of employee vs. management AI discussions. Employee sentiment toward AI, measured consistently across HIV4-based and GPT-based approaches, provides a meaningful and robust signal of firm productivity. Executive AI discourse, by contrast, bears no positive relationship with productivity despite its overwhelmingly optimistic tone. This finding carries important implications for investors, analysts, and policymakers who rely on earnings calls for signals about AI's productive impact.

3.3 Job Displacement

Since concerns about job displacement have the most negative impact on employee AI sentiment, we further test whether such concerns are valid and investigate the impact of AI on the job market. We identify whether an employee is skilled or not based on the inclusion of AI related keywords in the employee’s job title or job description on LinkedIn. This approach is consistent with the method used in Babina et al. (2024), which identifies AI skilled employees based on their resumes. Because job descriptions are self-reported, we also try to use only job titles as an alternative way to identify AI skilled employees in additional tests. Further, while we can not fully address the potential self-reporting bias, we do not see strong reasons why the issue would explain our results. In addition, workers can list AI-related skills on their LinkedIn Profiles. But we decide not to use such self-reported skills to identify AI skilled employees, because such data in the Revelio Lab database are only a snapshot of the most recent information and do not cover historical records.

Figure 5a first shows the demographic composition of AI skilled workers and non-AI skilled workers. Panel A shows a clear gender gap: males account for 69.8% of AI skilled workers compared to 56.1% of non-AI talents. Panel B reveals that AI talent occupies significantly more senior roles: 8.9% of AI workers hold top-executive positions (seniority 6-7) versus 5.6% for non-AI workers, and 40.2% are mid-level managers compared to 28.9% for non-AI. This seniority skew likely reflects both the rapid career advancement that valuable AI skills command and the market's recognition of their scarcity. Panel C shows that AI positions disproportionately employ Asian/Pacific Islander workers (25.7% versus 11.1%). Further, Panel D shows that AI positions are considerably more likely to be compatible with remote work. The greater workplace flexibility afforded to AI workers likely stems from the same source: firms that place a high value on AI skills are also more willing to accommodate the work arrangements that attract and retain them. These differences in the workforce composition imply potential challenges for workers from non-AI positions to find new AI jobs. Figure 5b displays the share of AI-skilled employees as a proportion of total talent across ten industries in 2025. The Information sector leads with the highest concentration of AI talent (11.23%). Construction (1.41%) and Transportation and Warehousing (2.27%) have the lowest AI talent penetration.

To measure AI-driven labor replacement, we calculate the key independent variable, *Ln_#AI*, as the natural logarithm of the total number of AI-skilled employees in a given firm-role-quarter plus one. The log transformation is preferred over the raw headcount for two reasons. First, the distribution of AI employee counts is highly right-skewed, with a median of only 2 AI employees but a mean of 74, a 75th percentile reaching 18, and a top 1% exceeding 1,200, so the log transformation compresses extreme values and reduces the influence of a small number of firms with very large AI workforces. Second, the proportional interpretation of the log is economically appropriate: the relevant margin of AI adoption is not whether a firm adds one more AI employee in absolute terms, but whether it scales its AI workforce relative to its existing stock. The one is added before taking the log to retain firm-quarter observations with zero AI employees, preserving the extensive margin of non-adoption. Again, AI-skilled employees are identified based on the mention of AI-related keywords on LinkedIn profiles. We then construct three firm-role-quarter dependent variables using LinkedIn data. The first dependent variable, *Hire Rate*, is the average new-hire rate over the following two quarters, where the new-hire rate is the total number of new hires in a quarter divided by the average monthly headcount at the firm-role level. *Turnover* is the average turnover rate over the following two quarters, where the turnover rate is the total number of employees who left a firm's role in a quarter divided by the average monthly headcount. *Employee Growth* is defined as the average quarterly net change in employee headcount over the following two quarters, scaled by the current-quarter employee headcount.

Table 6 estimates the displacement effect of AI. Panel A reports the results of the firm-role-quarter level regression. For hire rates, the coefficient on *Ln_#AI* is -0.044 (t = -83.11) in column (1), indicating that a firm with more AI skilled employees has a significant reduction in subsequent hiring. Column (2) further tests whether the effect on hire rate changes after the launch of ChatGPT by interacting the number of AI-skilled employees with an indicator for the period after the ChatGPT launch. The results show that the displacement effect intensified following the ChatGPT launch, with the coefficient on the interaction term, *Ln_#AI*PostGPT*, being significantly negative -0.013, t = -34.03). For turnover rates, the coefficient on *Ln_#AI* is 0.039 (t = 69.19) in column (3), indicating that AI talent accumulation is associated with significantly higher overall turnover, consistent with a more competitive and unstable internal labor market. Column (4) shows that the turnover effect also intensifies after the ChatGPT launch (coefficient on the interaction term

*Ln_#AI*PostGPT* = -0.010, t = -39.47). For employment growth, the coefficient is -0.199 (t = -114.11) in column (5) and deepens further in the post-ChatGPT period (coefficient on the interaction term *Ln_#AI*PostGPT* = -0.018, t = -18.02), suggesting that AI adoption generates a sustained net contraction in employment growth.

Panel B tests whether the results are conditional on firms' hiring behavior during the COVID period. Specifically, we split the sample into two subsamples based on the employment growth rate of firms from December 2019 to December 2021. We find that the effects are present and similar in magnitude for both high- and low-COVID hiring firms, mitigating the concern that our results are driven by AI washing behavior (i.e., employees blame AI for layoffs that are actually due to post-COVID excessive hiring).

As discussed above, because job descriptions are self-reported, we use only job titles as an alternative way to identify AI skilled employees. Results are reported in Panel C and are generally consistent with those in Panel A. Panel D further aggregates the firm-year-role level data to the firm-year level and finds consistent results. The key independent variable *Ln_#AI* has a mean value of 1.68. For the *Hire Rate* test, the coefficient on *Ln_#AI* is −0.0159. Thus, on average, AI adoption is associated with a 2.67 (= −0.0159× 1.68) percentage-point decrease in the hire rate of firms in our sample. When *Ln_#AI* increases from the bottom quartile (0) to the top quartile (2.944), the *Hire Rate* decreases by 4.68 (=0.0159×2.944−0.0159×0) percentage points. These effects are economically significant given that the sample average hire rate in our sample is 4.54%. Similar, for the *Turnover* test, the coefficient on *Ln_#AI* is 0.036. Thus, on average, AI adoption is associated with a 6.05 (= 0.036× 1.68) percentage-point increase in the turnover rate of firms in our sample. When *Ln_#AI* increases from the bottom quartile (0) to the top quartile (2.944), the *Turnover Rate* increases by 10.60 (=0.036×2.944−0.036×0) percentage points. For the employment growth tests, the coefficient on *Ln_#AI* is -0.092. Thus, on average, AI adoption is associated with a 15.46 (= -0.092× 1.68) percentage-point increase in the turnover rate of firms in our sample. When *Ln_#AI* increases from the bottom quartile (0) to the top quartile (2.944), the *Turnover Rate* decreases by 27.08 (=-0.092×2.944−-0.092×0) percentage points

Table 7 shows the displacement effect across industries, seniority, gender, and ethnicity. Panel A shows that the negative hiring effect is pervasive and significant across all industries, with coefficients on *Ln_#AI* ranging from -0.039 to -0.052. Panel B shows that the effect of AI adoption on the hire rate is the strongest for Rank&File employees, followed by middle level management. The negative effect of AI adoption on the hire rate for Rank&File employees is three times as high as that for executives. In addition, the hiring effect of AI is significant across nearly all demographic groups. Males face a stronger negative effect than females. White employees experience the most negative hire rate effect, followed by Asian/Pacific Islander workers. These findings are likely due to White and Asian males dominate in industries that have greater exposure to AI. Panel C shows that the turnover effect is similarly consistent across industries, while Panel D again shows that the increase in turnover rate is the strongest for rank-and-file workers. These effects are similar to those documented in Panels A and B. Figures 6a and 6b present cross-sectional estimates of how AI talent accumulation affects new hire rates and turnover rates, respectively, across industries, seniority levels, gender, and race.

As remote work may correlate with AI adoption, Table 8 examines whether remote work flexibility has a similar effect on workforce dynamics. Specifically, we test whether firms offering more remote work arrangements experience lower employee hiring and growth due to the difficulty of managers providing training and mentorship to their employees (Emanuel et al., 2026). We use the firm-level cumulative percentage of full-time job postings that allow fully or partially remote work since the beginning of 2021 (*Flex_Work%*) as the independent variable. In Panel A, *Flex_Work%* is positively and significantly associated with *Hire Rate* (0.026, t = 2.85) and *Employee Growth* (0.059, t = 2.62) in columns (1) and (3), though it shows no significant relationship with *Turnover* (column 2). We further visualize these effects in Figure 7. The results suggest that firms with more remote job positions actually have higher hire rate and higher employment growth. Thus, remote work is unlikely to explain the negative effects of AI on hire rate and employment growth.

We further decomposing remote work flexibility into two components in columns (4) through (6): the firm-level cumulative percentage of full-time job postings that allow fully remote work since the beginning of 2021 (*Fully_Remote%*) and the firm-level cumulative percentage of full-time job postings that allow partially remote work since the beginning of 2021 (*Partial_Remote%*). We find that *Fully_Remote%* explain these patterns while *Partial_Remote%* shows no significant association with any of three outcomes.

Panel B further examines heterogeneity in the hiring effect across seniority groups. According to the first three columns, the positive hiring effect of *Flex_Work%* is concentrated among rank-and-file employees (0.034, t = 1.82) and mid-managers (0.044, t = 2.55), and insignificant for top executives (0.003, t = 0.15). After decomposing flexible work into fully remote and partial remote work in column (4) to (6), the effect of *Fully_Remote%* remains significant only for mid-managers (0.046, t = 2.45) and near-significant for rank-and-file employees (0.030, t = 1.55), while *Partial_Remote%* shows no significant association with the hire rate across any seniority group. These results together suggest that remote work arrangements do not appear to hurt the hire rate for junior level positions. Instead, firms with more remote work positions have higher hire rate for junior level positions. Thus, the workforce disruptions attributed to AI displacement are unlikely to be driven by firms' remote work policy.

3.4 Wage Premium and Job Market Mobility

We now turn to job market rewards for AI skills. Figures 8a and 8b show AI job posting shares over time and by industry. AI job postings as a share of total postings fluctuate between 2-5% over the sample period, with a marked increase in 2025 (reaching 5.18% in August 2025). The Information sector leads by a wide margin, with AI-related postings approaching 15% of all job postings by 2025, followed at a distance by Services and Manufacturing. Appendix Table B3 provides detailed monthly statistics on AI-related job postings. Appendix Table B4 provides the top 100 most expanding and contracting jobs from 2021 to 2025.

Given the rapid increase in AI skills, we expect a salary premium for AI-skilled workers. Figure 9a plots the monthly average salary for AI-related versus non-AI job postings from 2021 to 2025, illustrating a persistent and widening premium for AI roles. The annual premium rises from approximately 40% in 2021 to more than 45% by 2025, suggesting that the labor market continues to reward AI skills at an increasing rate. More statistics are reported in Table B3. AI job postings offer average salaries of approximately $100,000-$132,000, compared to $60,000-$74,000 for non-AI postings, representing a premium of 50-90%. Figure 9b breaks down the salary premium by industry. Wholesale & Retail Trade (72.9%) and Transportation (54.5%) exhibit the largest premiums, likely reflecting the relative scarcity of AI talent in these traditionally non-technical sectors, where such skills command an outsized wage rent precisely because they remain uncommon. These statistics do not control the differences in roles of different employees. Thus, the salary premium reflects the difference in the average salary of all AI skilled workers and the average salary of all non-AI workers in the same industry. This difference can be due to differences in roles of AI vs. non-AI employees.

Figures 10a and 10b further show the geographic distribution of AI job postings and salary premiums across U.S. states in 2025. Figure 10a shows that AI job posting shares are highest in Washington D.C. (7.4%), Washington (5.8%), and California (5.8%), reflecting the concentration of technology and defense-adjacent industries along the coasts, while states in the rural Midwest and Deep South generally fall below 2%. Figure 10b reveals that the AI salary premium is paradoxically largest outside established tech hubs, with West Virginia (114%), Alabama (104%), and Indiana (101%) recording the highest premiums, suggesting that AI skills command an outsized wage rent precisely where they remain scarce, while coastal states with already-elevated baseline salaries exhibit comparatively modest premiums.

Table 9 Panel A presents the salary premium regression analysis. The dependent variable is *Salary*, the average monthly salary per job role as disclosed in firm job postings. The key independent variable is *AI Job*, an indicator equal to 1 if the job posting contains AI-related keywords in its description, and 0 otherwise. We estimate specifications with role and monthly fixed effects, and progressively tighter firm-role fixed effects that absorb all time-invariant differences within firm-role level. Column (1), which includes role and monthly fixed effects, estimates an AI premium of $11,575 (t = 8.05), which is approximately 14% of non-AI workers' average salary ($82,104). Thus, AI skilled employees get paid 14 percent more than non-AI skilled workers with similar roles. Column (2) replaces role fixed effects with firm-role fixed effects, absorbing all time-invariant differences within each firm-role pair, and the premium remains substantial at $5,291 (t = 9.13), which is approximately 6.44% of non-AI workers' average salary ($82,104). This result demonstrates that AI skills generate a genuine wage premium: holding the firm, the role, and the time period constant, AI skilled employees get paid 6 percent more than non-AI skilled workers with the same role in the same firm, and this premium cannot be explained away by compositional differences in the types of firms or roles that demand them.

Finally, we examine the labor market mobility of AI workers who transition between firms. Figure 11 presents the turnover rate comparison between AI and non-AI talent. AI talent has a substantially higher average quarterly turnover rate (7.3%) than non-AI talent (4.6%). This elevated mobility likely reflects the intense cross-firm competition for scarce AI talent, which pulls AI workers toward better opportunities.

Figure 12 shows the distribution of origin firm sizes for transitioning workers. AI talent disproportionately originates from very large firms: 71.5% of AI workers who change jobs leave firms with 10,000 or more employees, which are calculated based on LinkedIn firm profile,[4] compared to 33.5% for non-AI workers. These are consistent with the expectations that AI talents concentrate in large firms. Figure 13 presents Sankey diagrams of job transition flows for AI vs. non-AI talent across firm size categories. We still find that AI talent who change jobs is heavily concentrated in large firms. Further, 66.6% of AI workers leaving large firms transition to other large firms, compared to 60.2% for non-AI workers, suggesting that AI talent circulates primarily within the large-firm ecosystem and is less likely to move to smaller firms compared with non-AI workers. Thus, once AI workers are embedded in large firms, they tend to stay within that tier of firms, reinforcing the advantage of large firms in accumulating and retaining AI capabilities.

Table 9 Panel B provides regression results to complete the figures. Specifically, we compare career outcomes of AI vs. non-AI-skilled workers who leave the same firm-role in the

[4] LinkedIn reports firm employee counts based on several categories without the exact employee count. The highest bracket for employee count on LinkedIn is greater than 10,000 employees. Thus, we use 10,000 employees as our cutoff to identify large firms.

same quarter. To examine career outcomes at job transition, we track individual worker records for U.S.-to-U.S. job moves from 2021 onward. The dependent variables are *Promotion* (a binary indicator equal to 1 if the seniority level of the new job increases at the new firm), *Demotion* (a binary indicator equal to 1 if the seniority level of the new job decreases at the new firm), *Parallel* (a binary indicator equal to 1 if the seniority level of the new job does not change at the new firm), and *Move to Large Firm* (a binary indicator equal to 1 if the worker moves to a large firm with 10,000 or more employees). The key independent variable is *AI Talent,* an indicator for AI skilled workers. The coefficient on this variable compares the career outcome of AI workers with that of non-AI workers who left the same firm-role in the same quarter. All specifications include firm-role and left-quarter fixed effects, with standard errors clustered at the firm level. We find that AI workers are 1.6 percentage points more likely to receive a promotion at their new firm (t = 7.76), 1.8 percentage points less likely to make a lateral move (t = -6.70), and 5.7 percentage points more likely to transition to a large firm with 10,000 or more employees (t = 19.97). The coefficient on *Demotion* is positive but statistically insignificant. Taken together, these results indicate that AI skills confer substantial and persistent career advantages, with AI workers enjoying both upward mobility and preferential access to large, well-resourced employers upon changing jobs.

4. Conclusion and Discussion

This study provides a comprehensive analysis of artificial intelligence's impact on the U.S. labor market from 2021 to 2025, spanning the ChatGPT revolution and the rise of the Agentic AI era. By integrating over 21 million employee reviews, 10,000 conference call transcripts, and tens of millions of job postings, we construct a detailed portrait of AI's workforce effects that covers employee sentiment, executive communication, wage premiums, and workforce dynamics.

Our findings yield several important takeaways. First, employee sentiment toward AI is substantially more negative than executive sentiment toward AI, and this gap carries real informational content. Employee AI sentiment explains firm productivity, while management AI sentiment does not. This finding has implications for investors and analysts, who might be over-weighting executive AI optimism due to the high observability of management sentiment while neglecting the more informative ground-level employee perspective which are not as salient.

Second, we analyze the primary sources of dissatisfaction, and concerns about job insecurity are associated with the largest negative factor. AI layoffs negatively affect employee AI sentiment and thus actually offset the benefits of adopting AI. Concerns about the displacement effect are also supported by our results. Firms with larger AI talent pools experience significantly lower overall hire rates, higher overall employee turnover, and slower employment growth, consistent with AI displacing human labor demand.

Third, AI skills generate significant benefits for workers. AI skilled workers enjoy strong career upward mobility, with higher promotion rates and access to large firms. AI skills are associated with an increasing wage premium. Thus, the labor market is now dividing into two branches: AI-skilled workers benefit from strong demand and upward mobility, while non-AI workers face increasing competitive pressure.

These findings carry several policy implications. First, firms should be more cautious with AI related layoffs, which offsets the productivity benefits of AI adoption. Also, organizations should not blindly follow the AI trend by investing heavily on the basis of executive enthusiasm alone, particularly when leadership lacks a clear understanding of which tasks AI can meaningfully improve and how to integrate it into existing workflows. Firms with AI-savvy leadership demonstrate significantly better outcomes, underscoring the value of technical fluency at the executive level. Second, the informativeness of employee AI discussion offers a practical lesson for investors and other capital market participants: employees' opinions often represent firsthand assessments of how a firm is deploying AI, and this informational advantage is especially valuable amid the current period of rapid and uncertain AI adoption. Third, this study confirms that AI exerts real displacement pressure on the broader labor market and reveals employee anxiety about being replaced. Policymakers should be attentive to the scale of this disruption and take proactive steps to help the workforce adapt, whether by facilitating reskilling to meet the evolving task demands of the AI era, adopting shorter workweeks to rebalance the labor market, and developing broader safety nets to cushion the societal costs of large-scale displacement.

Our study is subject to several limitations. First, although our empirical analysis draws on large-scale data, the time span remains relatively short, and the full consequences of AI adoption may take longer to materialize. Second, while we examine the overall impact of AI on employee sentiment, firm productivity, and labor market displacement, AI also affects many other dimensions of the labor market beyond our scope. We encourage future research to explore related topics such as the effects of AI on skill formation, occupational mobility, and wage distribution. Third, our LLM-based classification pipeline, while carefully validated, can be further refined as language models continue to improve. Finally, we do not draw definitive causal conclusions in this study. Future work could exploit quasi-experimental variation in AI adoption to strengthen causal identification.

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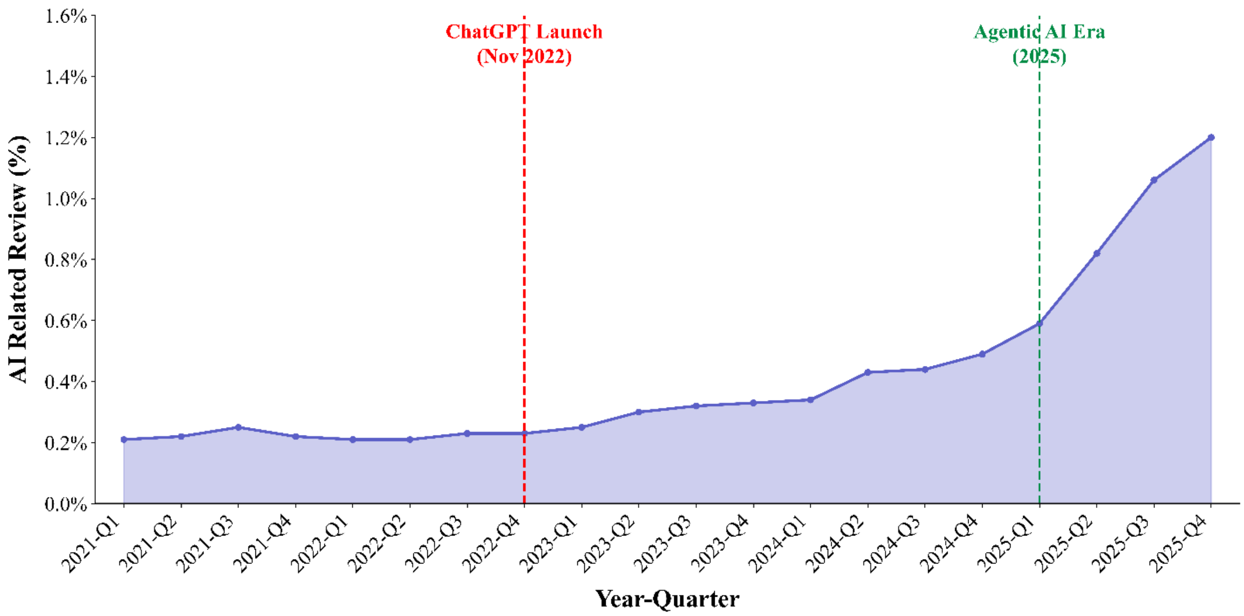


Figure 1a. AI Review Over Time: This chart illustrates the percentage of English-language reviews containing AI-specific comments. The timeline identifies two pivotal moments: the ChatGPT launch (Q4 2022), which introduced generative AI to the masses, and the onset of the Agentic AI era (Q1 2025), when AI systems shifted from conversational assistants to autonomous agents capable of multi-step reasoning, tool use, and workflow execution.

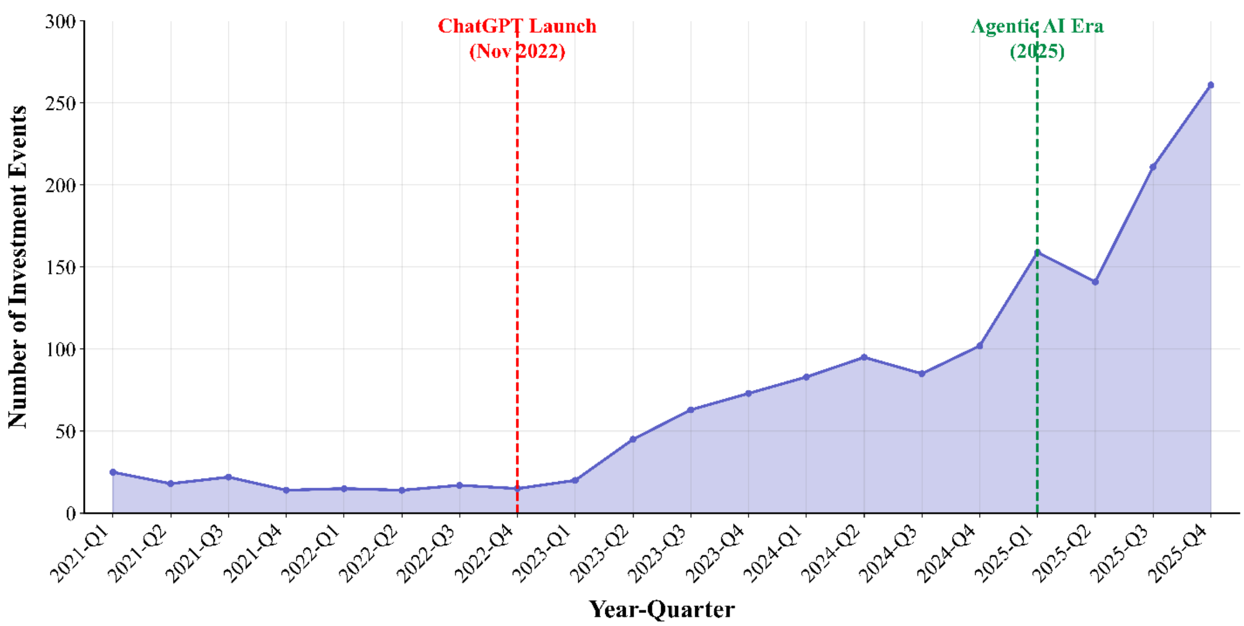


Figure 1b. Public Firm AI Investment Events Over Time: This chart illustrates the number of AI-related corporate investment events by quarter, including funding rounds, acquisitions, and strategic AI initiatives announced by public firms.

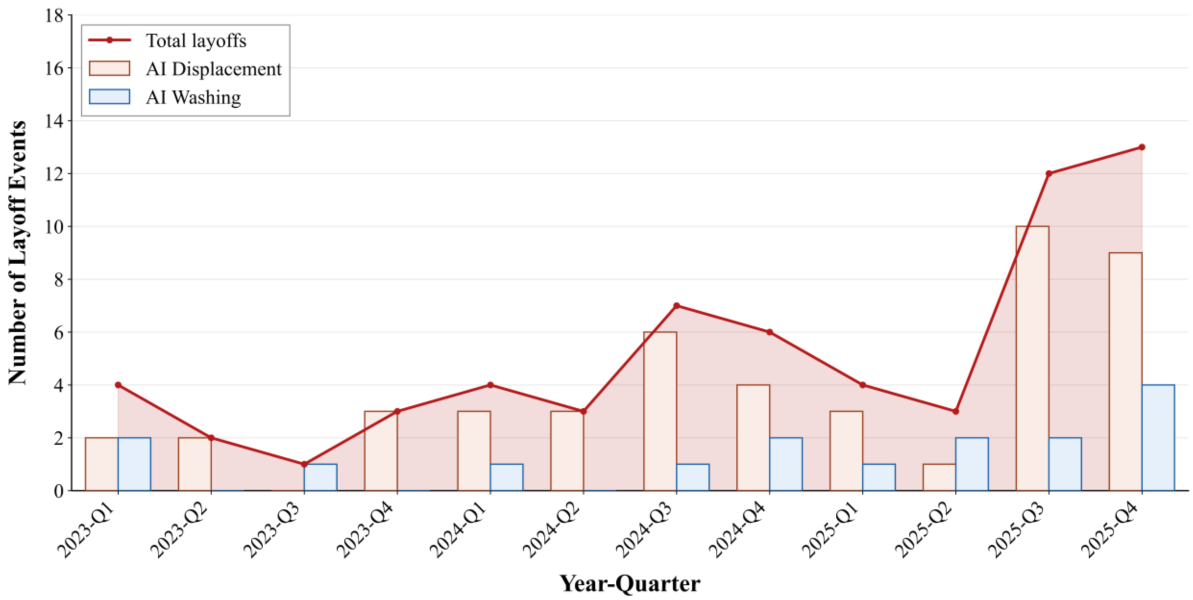


Figure 1c. Public Firm AI-Related Layoff Events Over Time: This line illustrates the number of AI-related layoff events by quarter among firms. We further classify layoffs into two categories based on their AI talent percentage and the employee growth rate after 2021. Layoffs are classified as AI Displacement (shown as pink bars in the figure) when the announcing firm exhibits above-industry AI talent concentration alongside below-industry employment growth rate after 2021; remaining layoffs are classified as potential AI Washing (shown as blue bars).

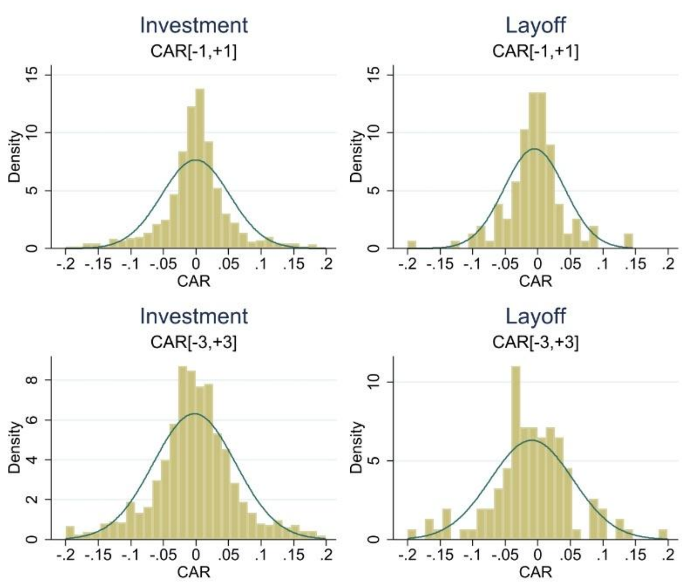


Figure 1d. Distribution of Cumulative Abnormal Returns. This figure plots the distribution of Cumulative Abnormal Returns (CARs) for AI investment and AI layoff announcements. We collect a sample of publicly announced AI investments and AI layoffs by publicly traded firms. The top panels show CARs for a three-day window surrounding AI investment announcements and AI layoff announcements and the bottom panels show CARs for a seven-day window surrounding AI investment announcements and AI layoff announcements. The solid curve is the fitted normal distribution.

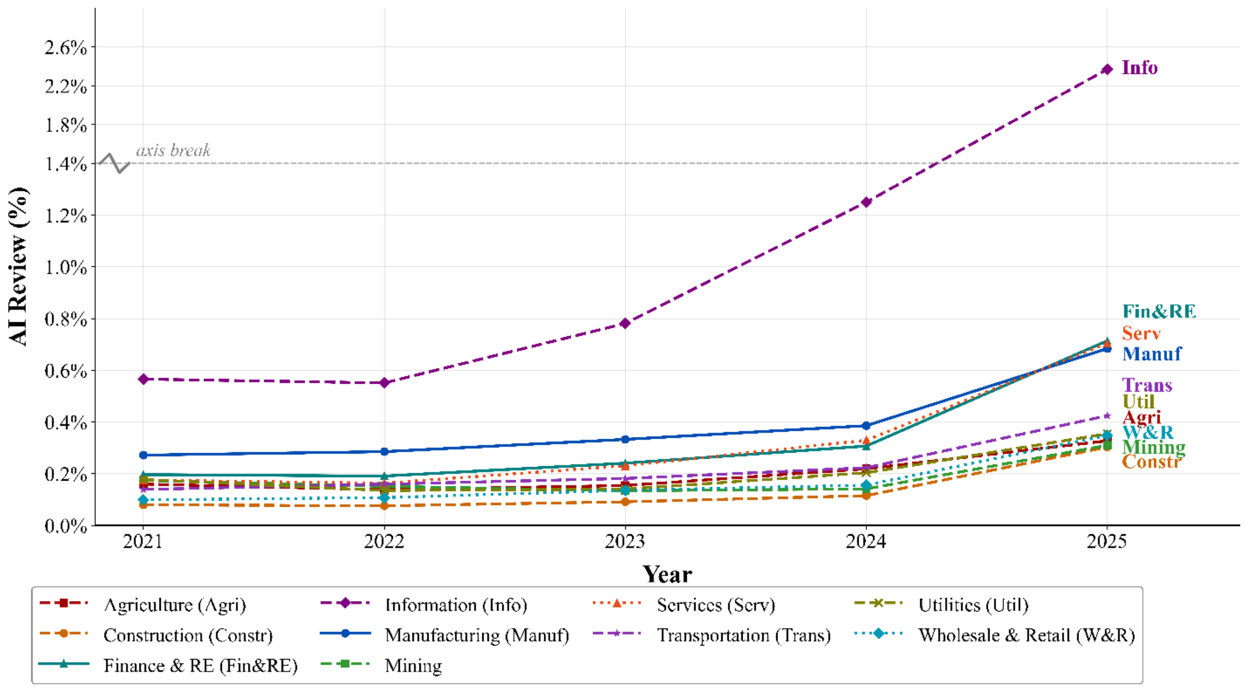


Figure 2a AI-Mentioned Review% by Industry: This figure shows the percentage of AI-related employee reviews relative to total employee reviews in each industry across our sample period. Industries are classified by NAICS codes: Agriculture (NAICS 11), Construction (NAICS 23), Manufacturing (NAICS 31-33), Wholesale & Retail Trade (NAICS 42, 44-45), Transportation (NAICS 48-49), Services (NAICS 54-56, 61-62, 71-72, 81), Information (NAICS 51), Finance & Real Estate (NAICS 52-53), Utilities (NAICS 22), and Mining (NAICS 21).

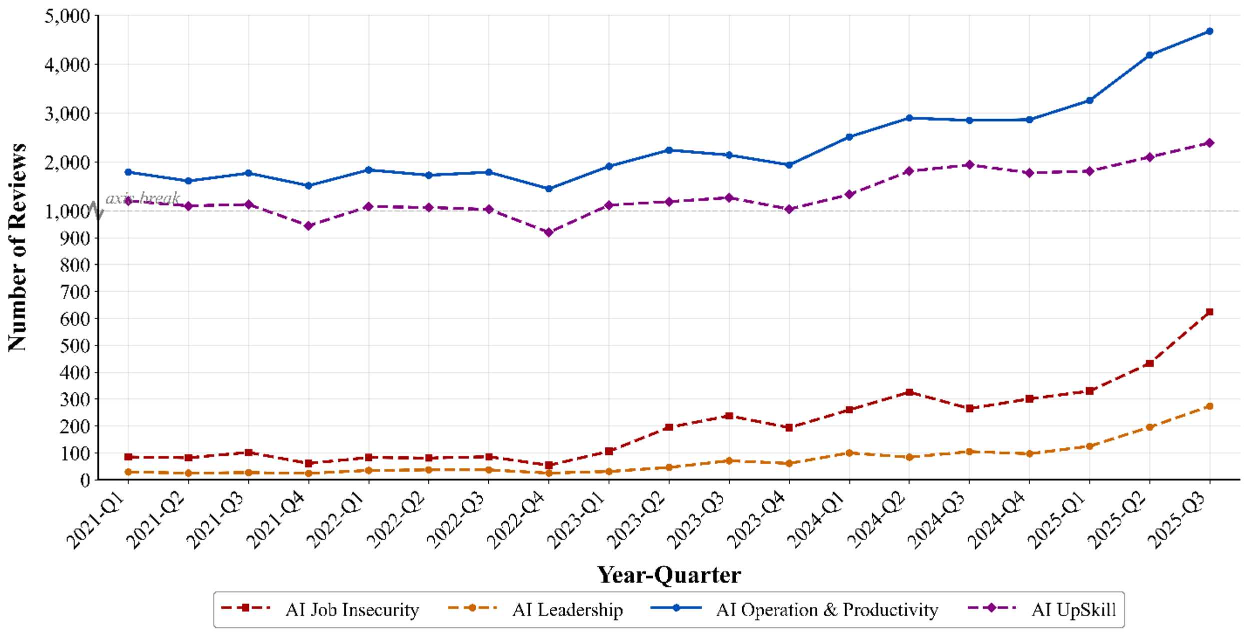


Figure 2b Frequency of AI-Mentioned Reviews by Content Over Time: This figure shows the number of AI-mentioned employee reviews by content category across each quarter from 2021 to 2025. "AI Job Insecurity" captures reviews mentioning job displacement or workforce reduction due to AI. "AI Leadership" captures discussions discussing both the leadership and AI adoption. "AI Operation & Productivity" captures references to AI-driven efficiency gains and operational conditions. "AI UpSkill" captures references to learning, training, and skill development related to AI.

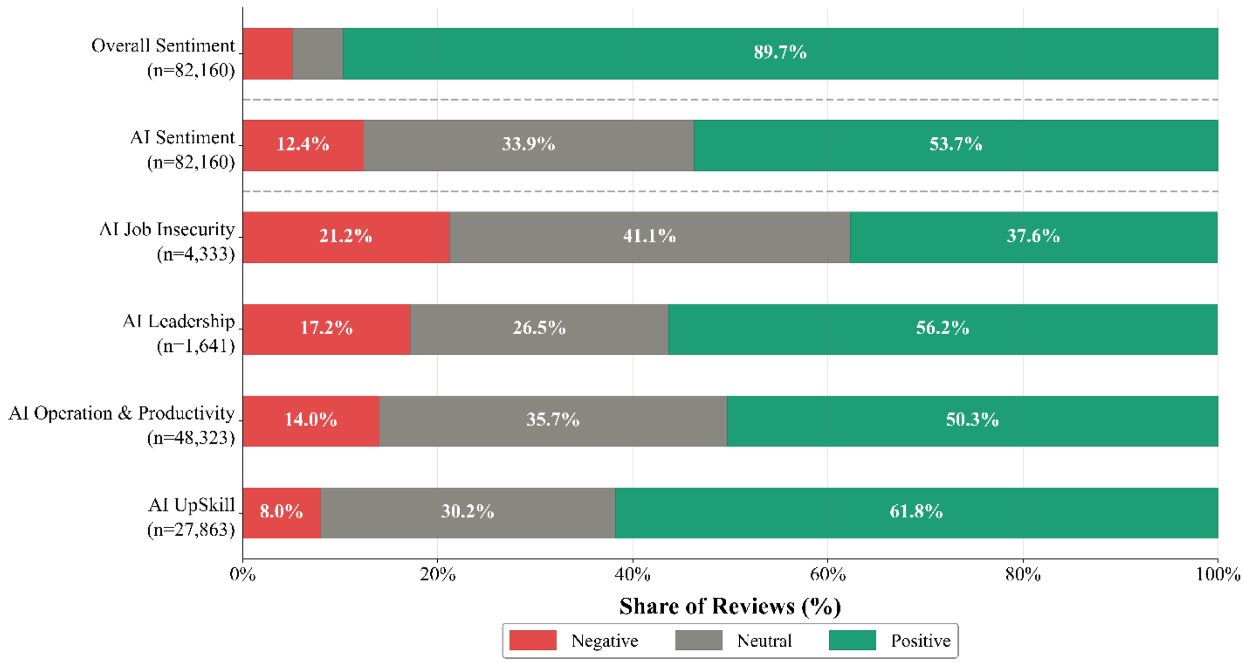


Figure 3 Sentiment Distribution: This figure displays the sentiment distribution within each AI-related review category. The sentiment is generated using the Harvard General Inquirer IV-4 (HIV4) dictionary, a widely used lexicon-based sentiment analysis tool that classifies words into positive (Positive, n=1,634 words) and negative (Negative, n=2,004 words) categories. The sample includes Glassdoor reviews of employees who mention AI in their comments. The top two bars show the sentiment of employees' overall comments on firms (Overall, n=82,160) and the sentiment of employees' AI-related comments (AI Sentiment, n=82,160). Within each bar, the share of negative (red), neutral (gray), and positive (green) sentiment is shown as a percentage of total reviews in that category. "AI Job Insecurity" captures reviews mentioning job displacement or workforce reduction due to AI. "AI Leadership" captures discussions discussing both the leadership and AI adoption. "AI Operation & Productivity" captures references to AI-driven efficiency gains and operational conditions. "AI UpSkill" captures references to learning, training, and skill development related to AI.

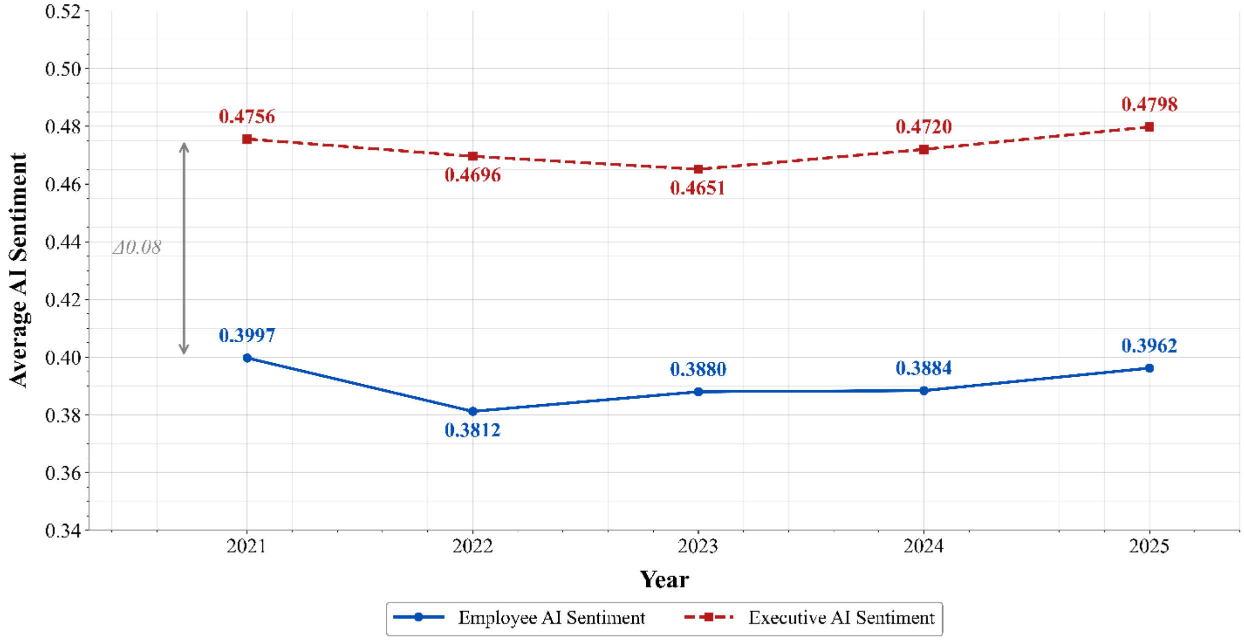
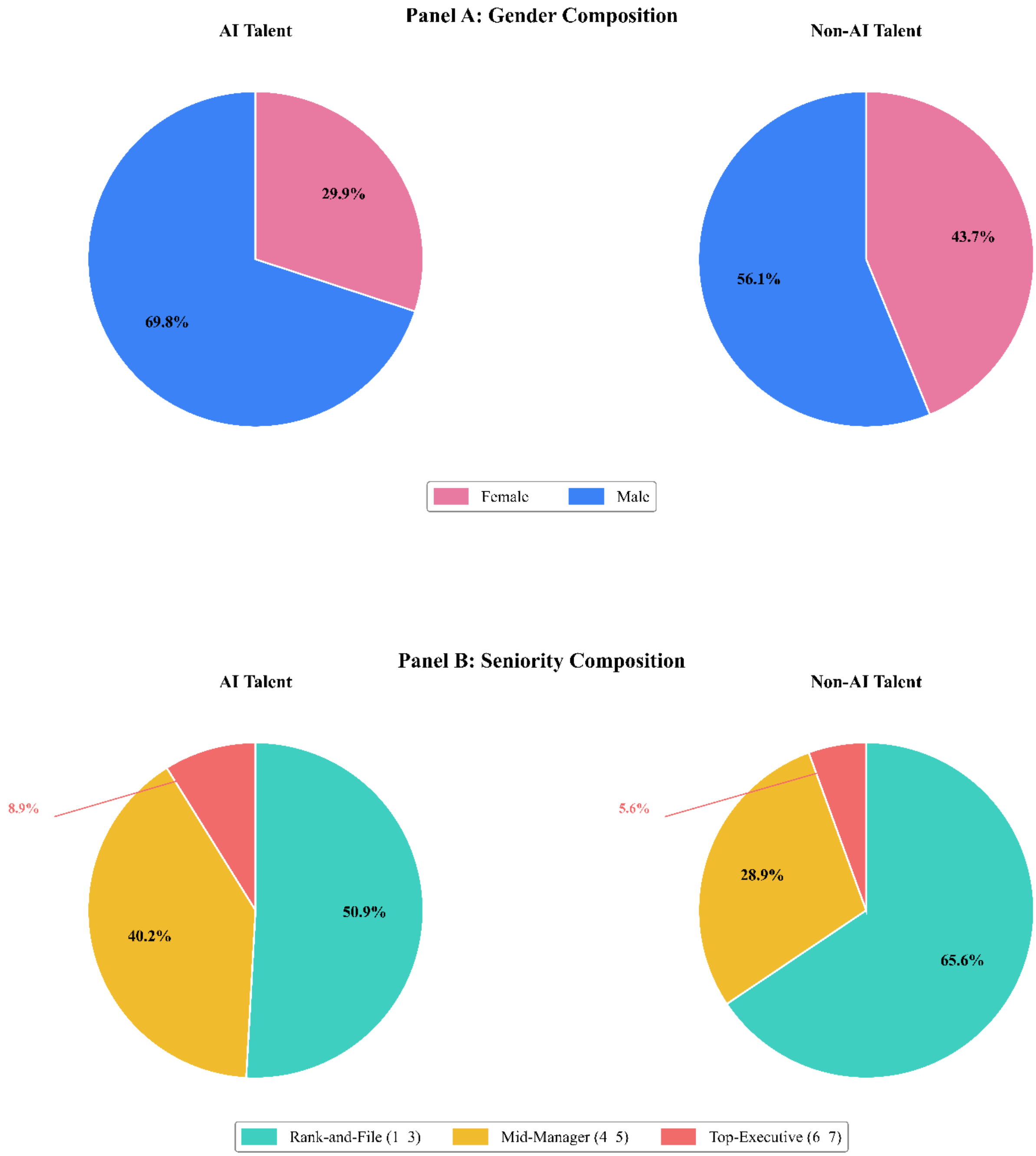


Figure 4 Employee vs. Executive AI Sentiment: This figure displays the average employee and executive sentiment towards AI. Both series are constructed using the Harvard General Inquirer IV-4 (HIV4) dictionary. Each AI-related mention is assigned a sentiment polarity score ranging from -1 to 1, calculated as (positive word count - negative word count) / (positive word count + negative word count), where a score closer to 1 indicates more positive sentiment and a score closer to -1 indicates more negative sentiment. For employees, mentions are drawn from AI-related Glassdoor reviews; for executives, mentions are drawn from earnings conference call transcripts. Data span 2021–2025.


Panel A: Gender Composition
AI Talent
Non-AI Talent
29.9%
69.8%
43.7%
56.1%
Female
Male
Panel B: Seniority Composition
AI Talent
Non-AI Talent
8.9%
40.2%
50.9%
5.6%
28.9%
65.6%
Rank-and-File (1 3)
Mid-Manager (4 5)
Top-Executive (6 7)

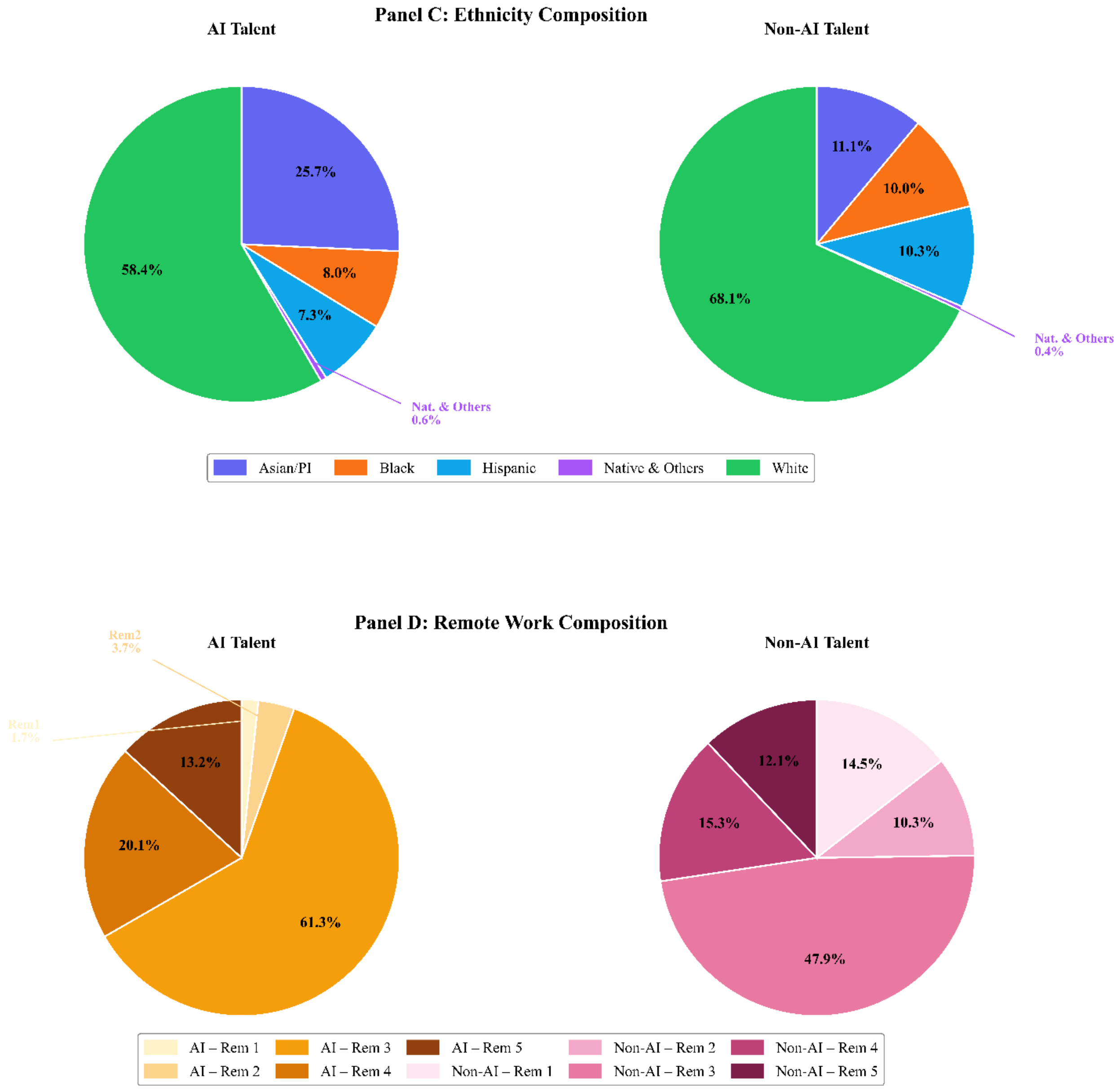


Figure 5a Workforce Composition: This figure shows the average workforce composition of AI versus non-AI talent over 2021Q1 to 2025Q2. Panel A shows gender shares: female and male. Panel B shows seniority levels, where levels 1-3 correspond to rank-and-file, levels 4-5 to mid-manager, and levels 6-7 to top-executive. Panel C shows ethnicity group shares across Asian/Pacific Islander, Black, Hispanic, Multiple, Native, and White. Panel D shows remote work sustainability levels, ranging from 1 (least remote) to 5 (most remote); AI talents are shown in amber shades and non-AI talents in rose shades. All values represent average headcount shares across the sample period.

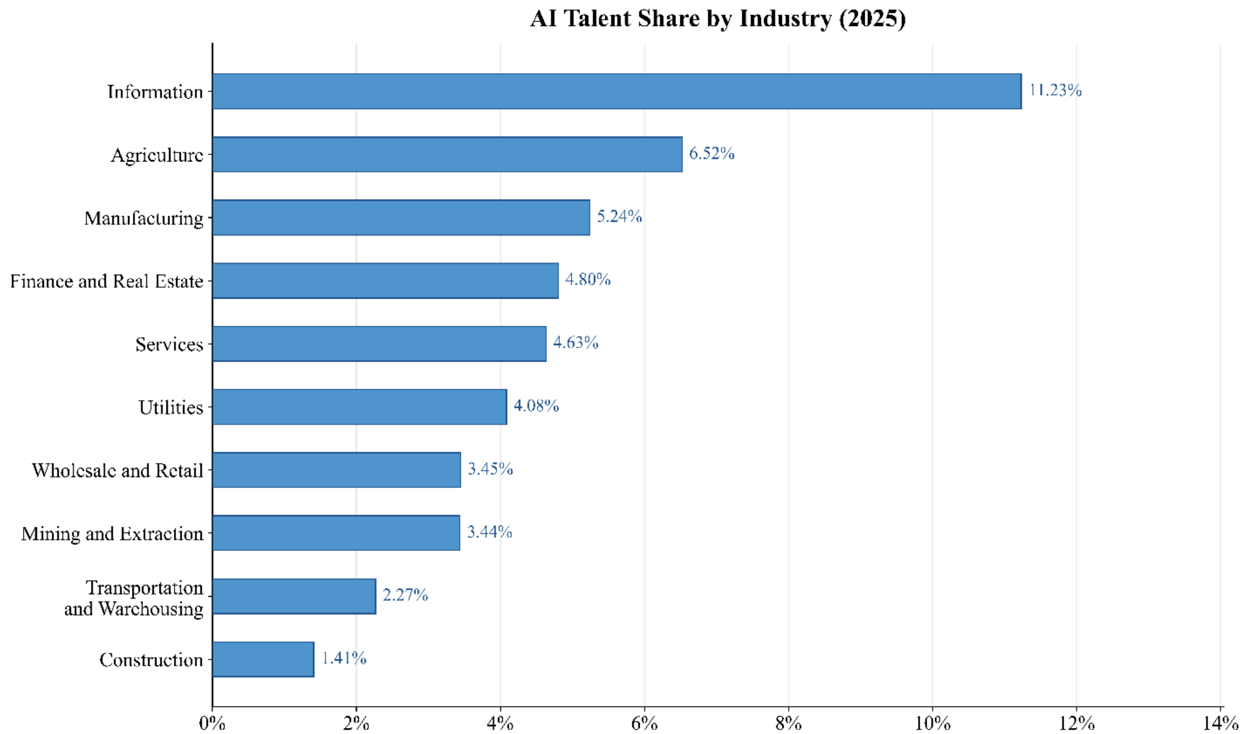


Figure 5b Industrial AI Talent Share: This figure presents the share of AI-skilled employees as a proportion of total talent across ten industries in 2025. Industries are ranked in ascending order by AI talent share. The Information sector leads with the highest concentration of AI talent (11.23%), followed by Agriculture (6.52%), Manufacturing (5.24%), and Finance and Real Estate (4.80%). Construction (1.41%) and Transportation and Warehousing (2.27%) have the lowest AI talent penetration.

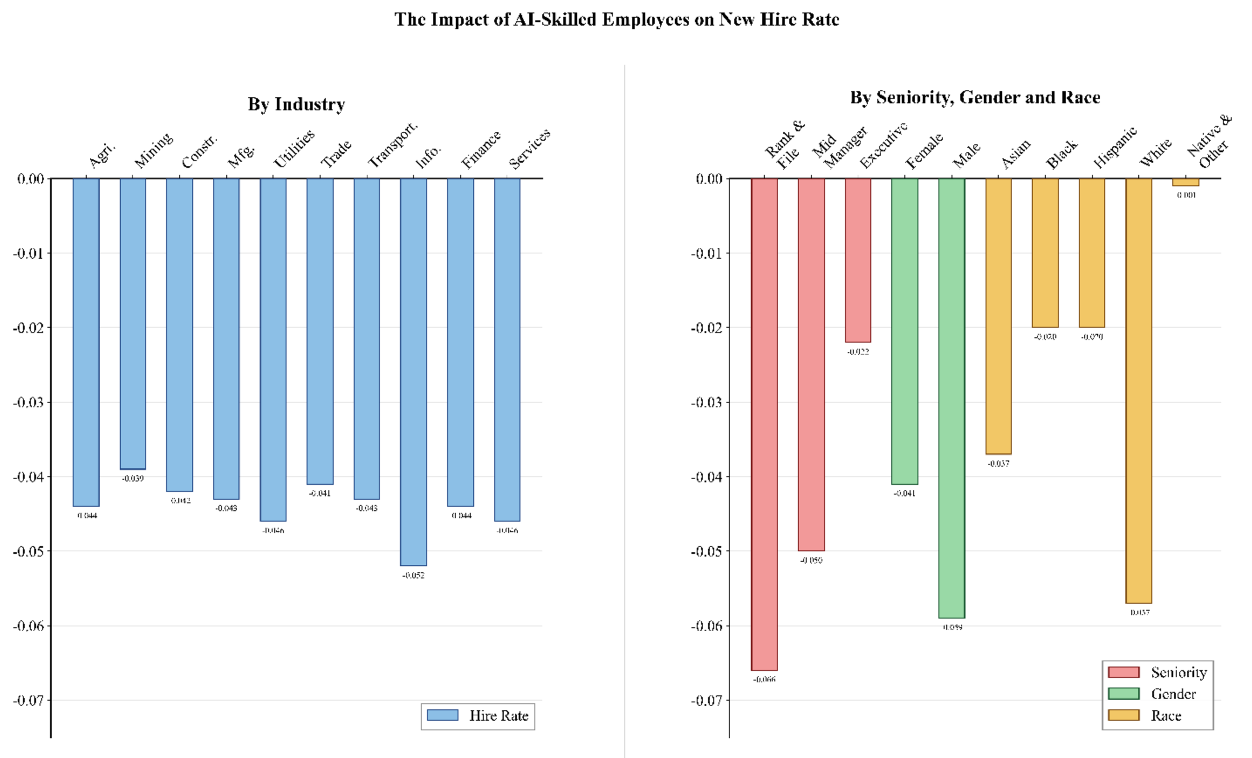


Figure 6a The Impact of AI-Skilled Employees on New Hire Rate: This figure presents cross-sectional estimates of how AI talent accumulation affects new hire rates across industries, seniority levels, gender, and race. The left panel reports estimates by industry (two-digit NAICS codes) and the right panel by seniority, gender, and race. The coefficients on *Ln_#AI* (the natural logarithm of the number of AI-skilled employees in a firm) from Table 7 under each specification are reported.

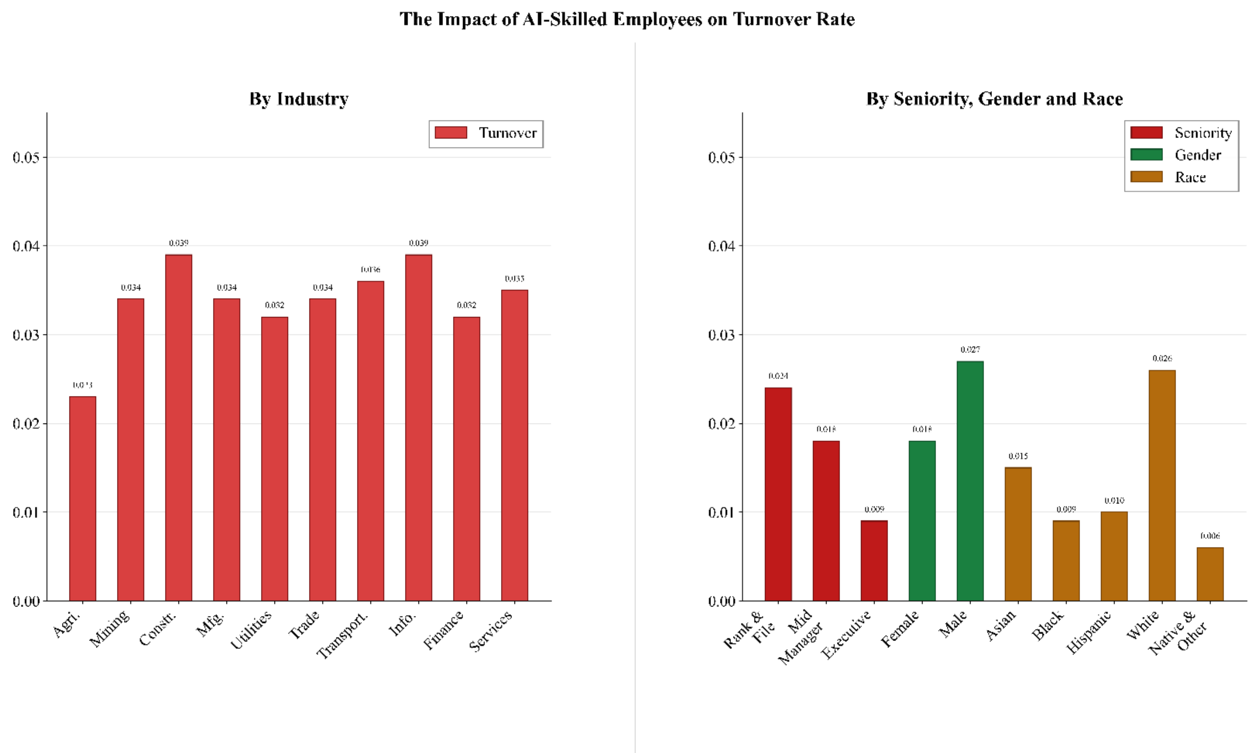


Figure 6b The Impact of AI-Skilled Employees on Turnover Rate: This figure presents cross-sectional estimates of how AI talent accumulation affects employee turnover rates across industries, seniority levels, gender, and race. The left panel reports estimates by industry (two-digit NAICS codes) and the right panel by seniority, gender, and race. The coefficients on *Ln_#AI* (the natural logarithm of the number of AI-skilled employees in a firm) from Table 7 under each specification are reported.

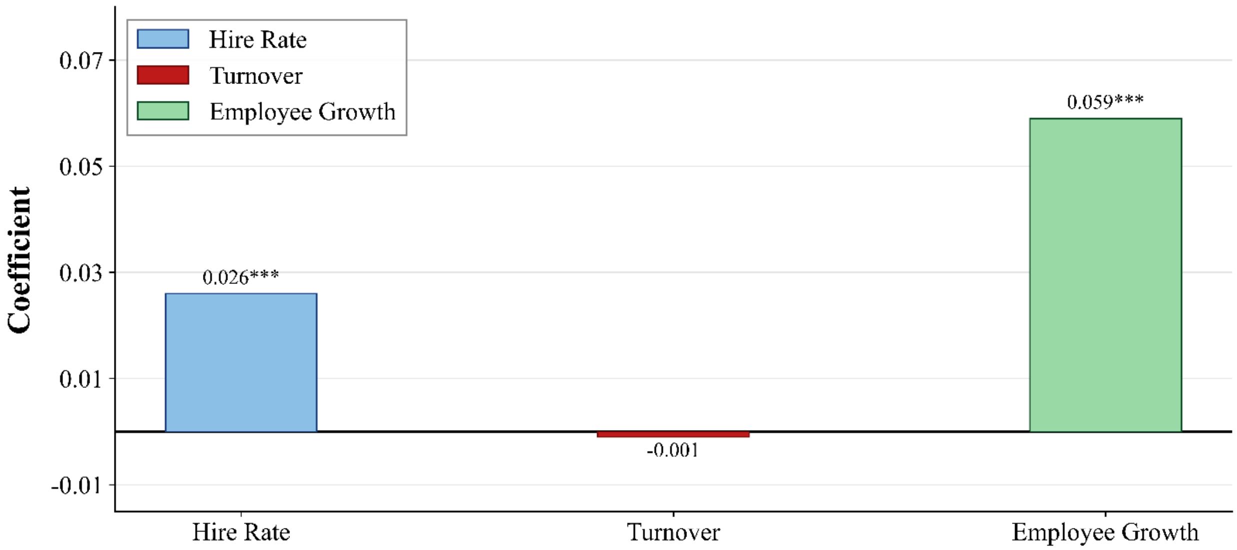


Figure 7a Impact of Remote Work on Overall Workforce Outcomes: This figure displays regression coefficients from Panel A of Table 8, showing the estimated effect of *Flex_Work%* on hire rate, turnover, and employment growth. *Flex_Work%* is the cumulative percentage of full-time job postings allowing fully or partially remote work from 2021 Q1 through the end of each firm's current quarter. *Hire Rate* is the average ratio of new joiners to beginning-of-quarter headcount over the next two quarters. *Turnover* is the average employee turnover rate over the next two quarters. *Employee growth* is the percentage change in employee headcount over the next two quarters. All regressions include firm and quarter fixed effects, with standard errors clustered at the firm level. *** denotes statistical significance at the 1% level.

Figure 7b Impact of Remote Work Flexibility on Hire Rate by Employee Seniority: This figure displays regression coefficients from Panel B of Table 8, showing the estimated effect of *Flex_Work%* on hire rates separately for rank-and-file employees (seniority levels 1–3), mid-managers (levels 4–5), and top executives (levels 6–7). *Flex_Work%* is the cumulative percentage of full-time job postings allowing fully or partially remote work from 2021 Q1 through the end of each firm's current quarter. Hire rate is defined as the average ratio of new joiners to beginning-of-quarter headcount over the next two quarters. All regressions include firm and quarter fixed effects, with standard errors clustered at the firm level. *** denotes statistical significance at the 1% level.

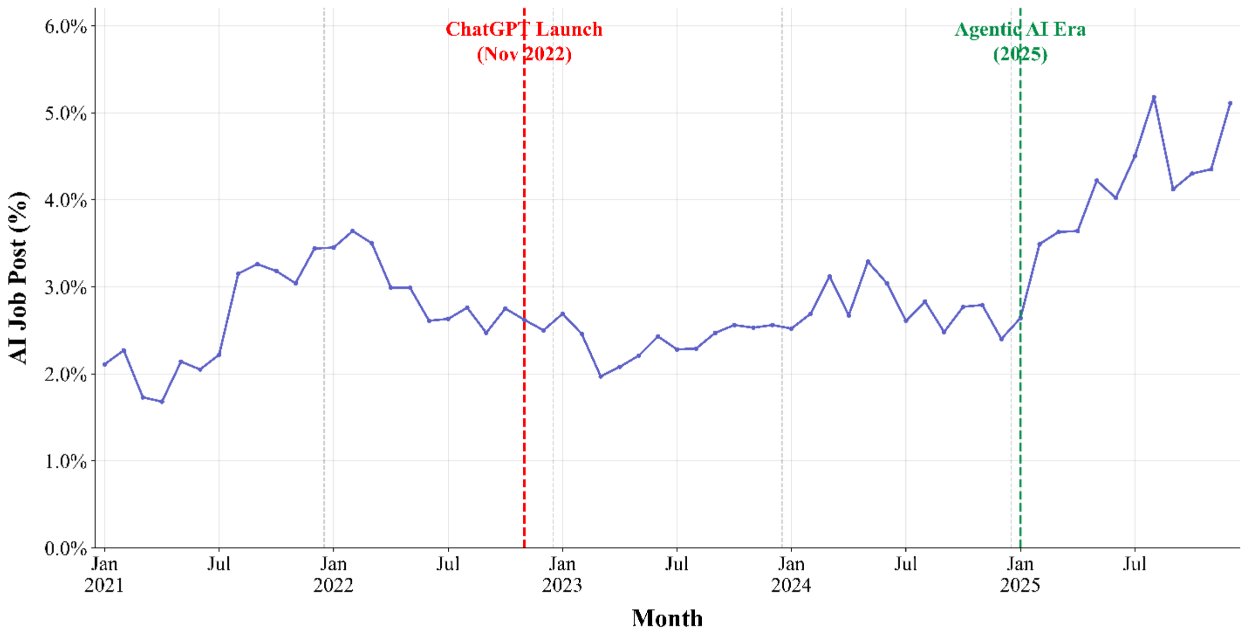


Figure 8a AI Job Posting% Over Time: This figure displays the monthly share of AI-related job postings as a percentage of total job postings from January 2021 to December 2025.

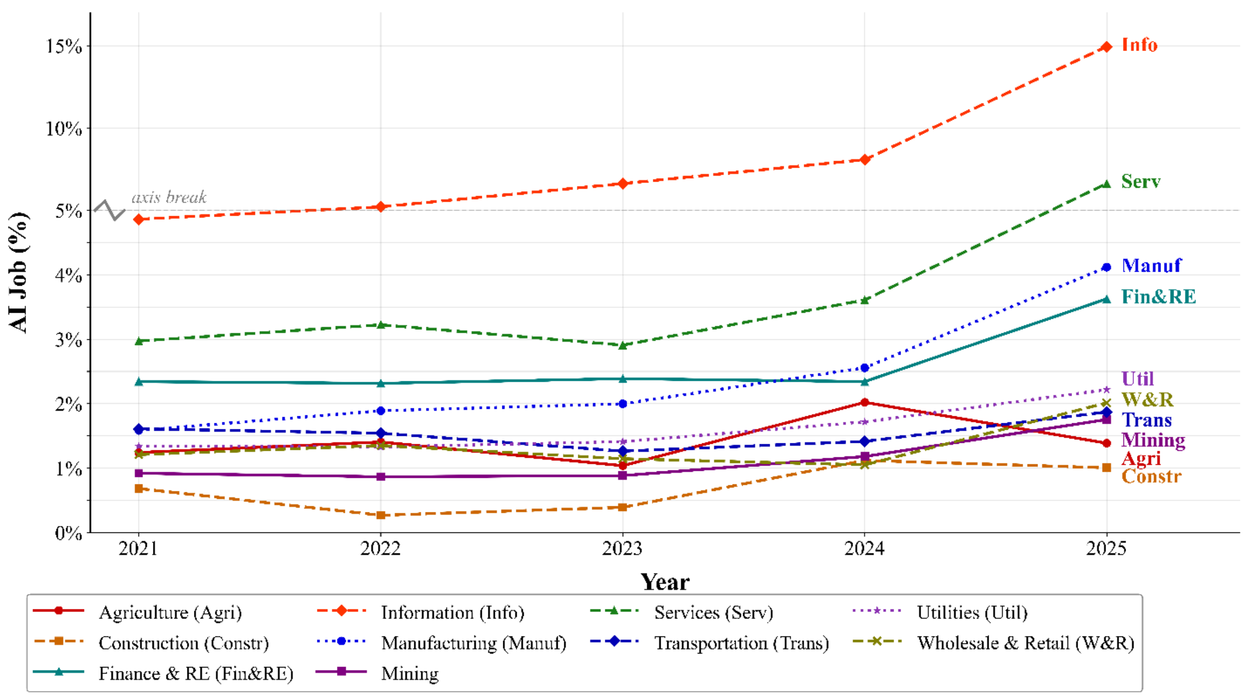


Figure 8b AI Job Posting% by Industry: This figure shows the monthly average percentage of AI-related job postings relative to total job postings per firm in each industry, from 2021 to 2025. Industries are classified by NAICS codes: Agriculture (NAICS 11), Construction (NAICS 23), Manufacturing (NAICS 31-33), Wholesale & Retail Trade (NAICS 42, 44-45), Transportation (NAICS 48-49), Services (NAICS 54-56, 61-62, 71-72, 81), Information (NAICS 51), Finance & Real Estate (NAICS 52-53), Utilities (NAICS 22), and Mining (NAICS 21).

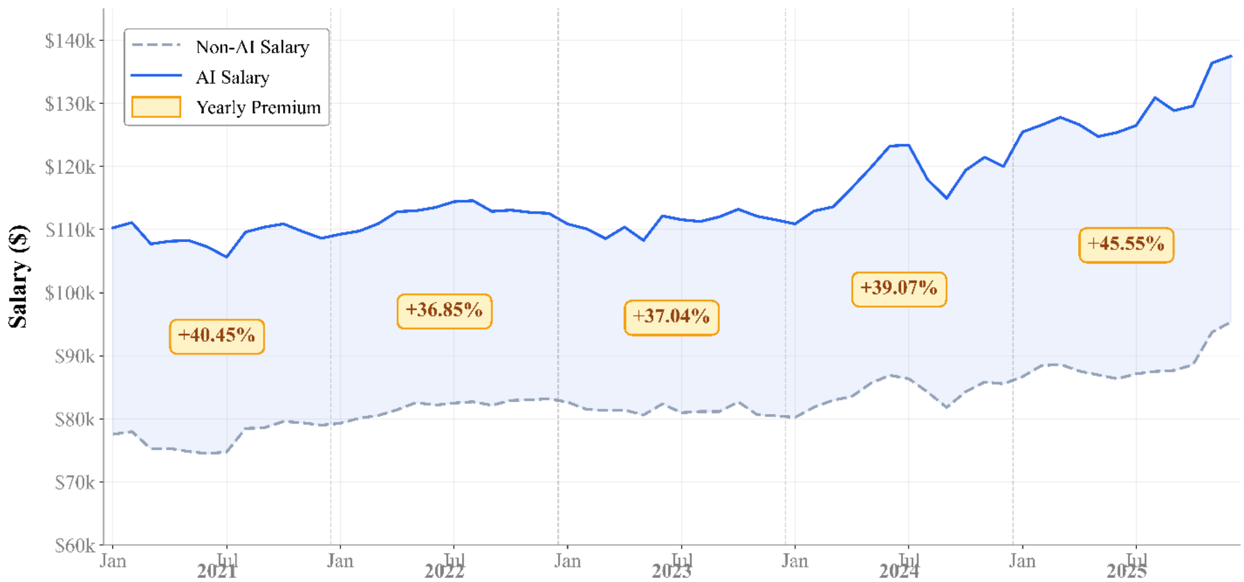


Figure 9a AI Salary Premium: This figure compares the monthly average salary for AI-related job postings versus non-AI job postings from January 2021 to December 2025. The shaded area represents the salary gap between the two categories. Yearly premium badges show the percentage by which AI salaries exceed non-AI salaries in each calendar year. The AI salary premium has grown from approximately 40% in 2021 to over 45% in 2025.

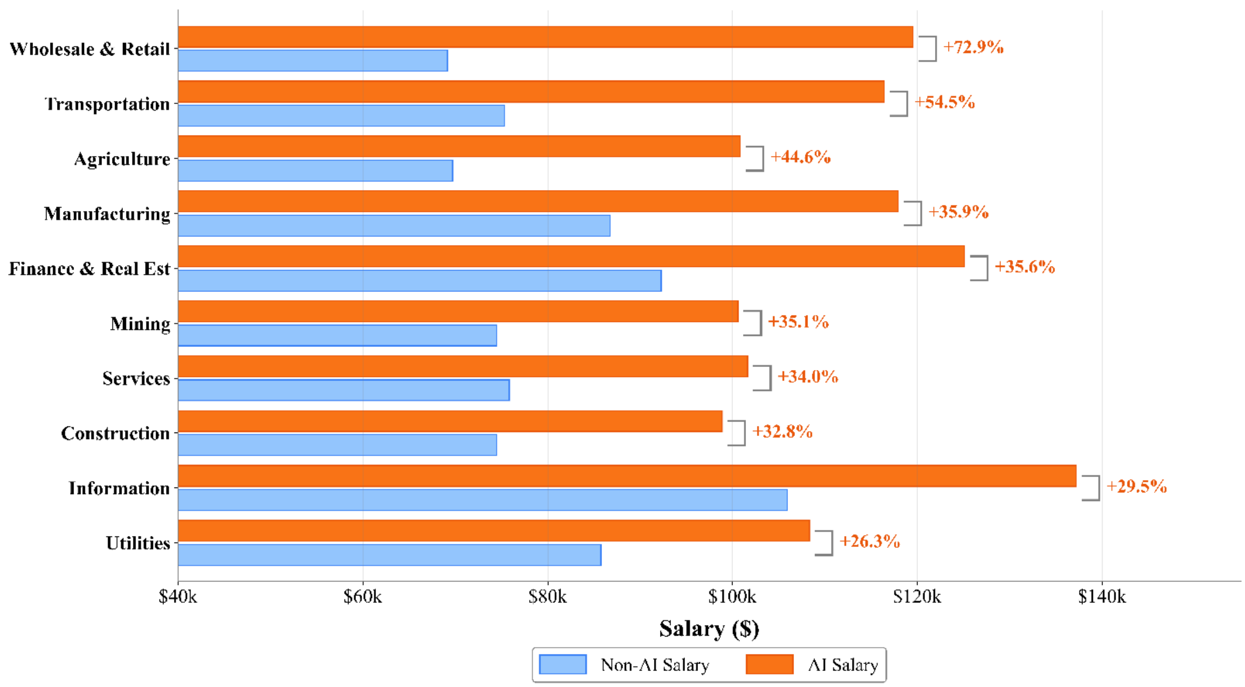


Figure 9b AI Salary Premium by Industry: This figure compares the average posted salary for AI-related versus non-AI job postings by industry, aggregated over 2021-2025. Industries are sorted by the AI salary premium percentage (descending). The premium label to the right of each pair shows the percentage by which AI job salaries exceed non-AI salaries within that industry.

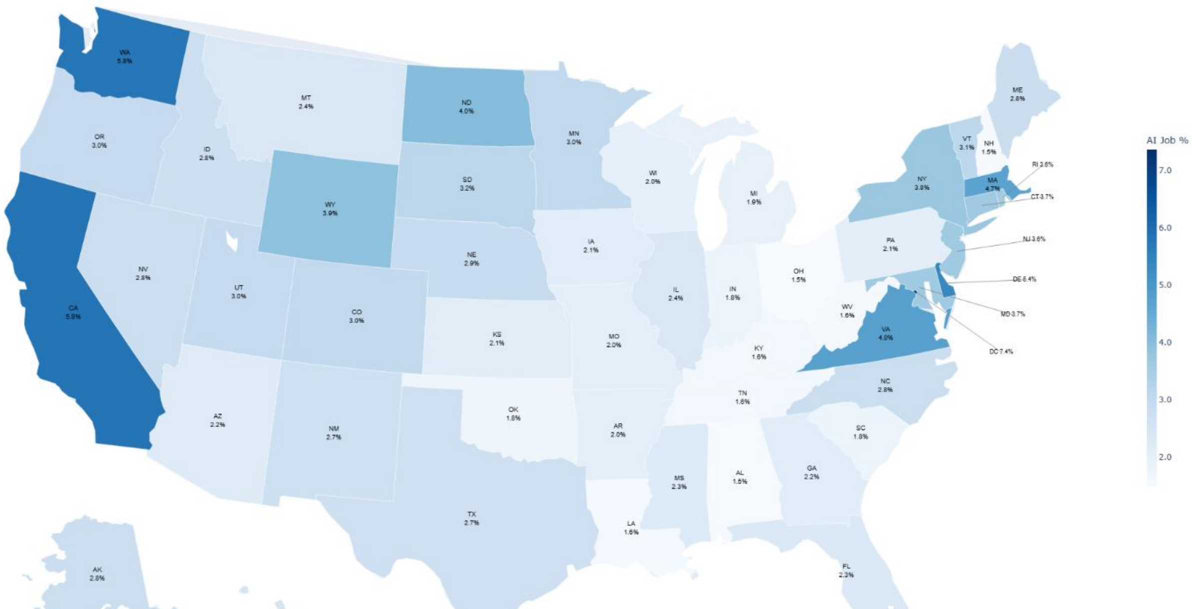


Figure 10a Share of AI-Related Job Postings by State, 2025: This map displays the percentage of total job postings classified as AI-related for each U.S. state in 2025, weighted by the number of job postings within each state.

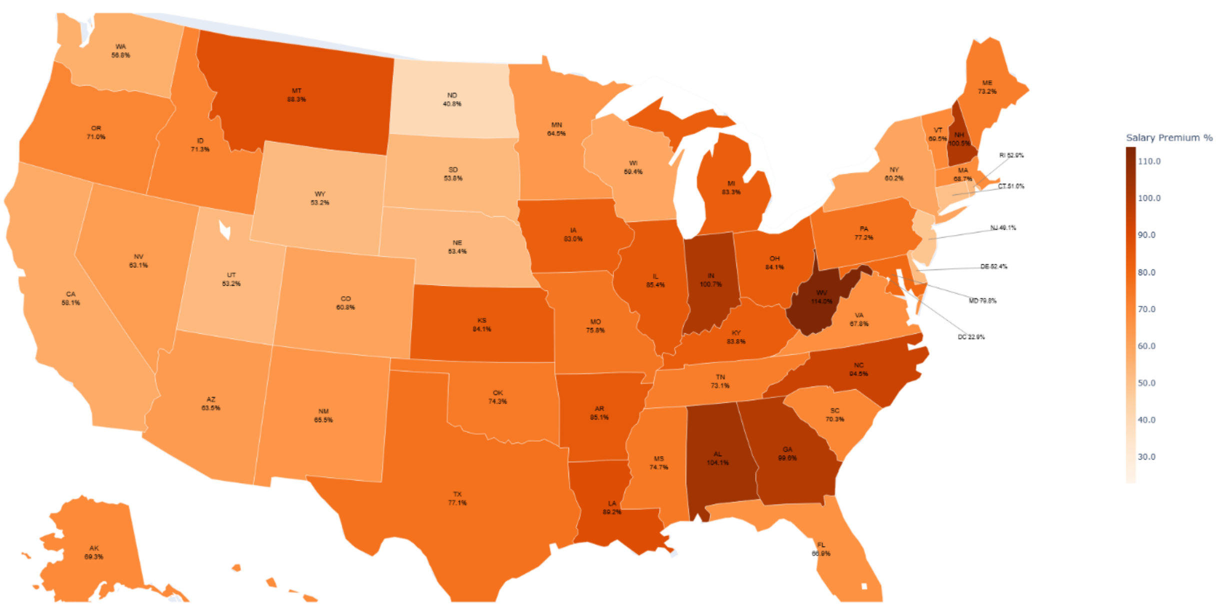


Figure 10b AI Job Salary Premium by State, 2025: This map presents the salary premium of AI-related jobs over non-AI jobs for each U.S. state in 2025, calculated as the percentage difference between the weighted average salaries of AI and non-AI job postings.

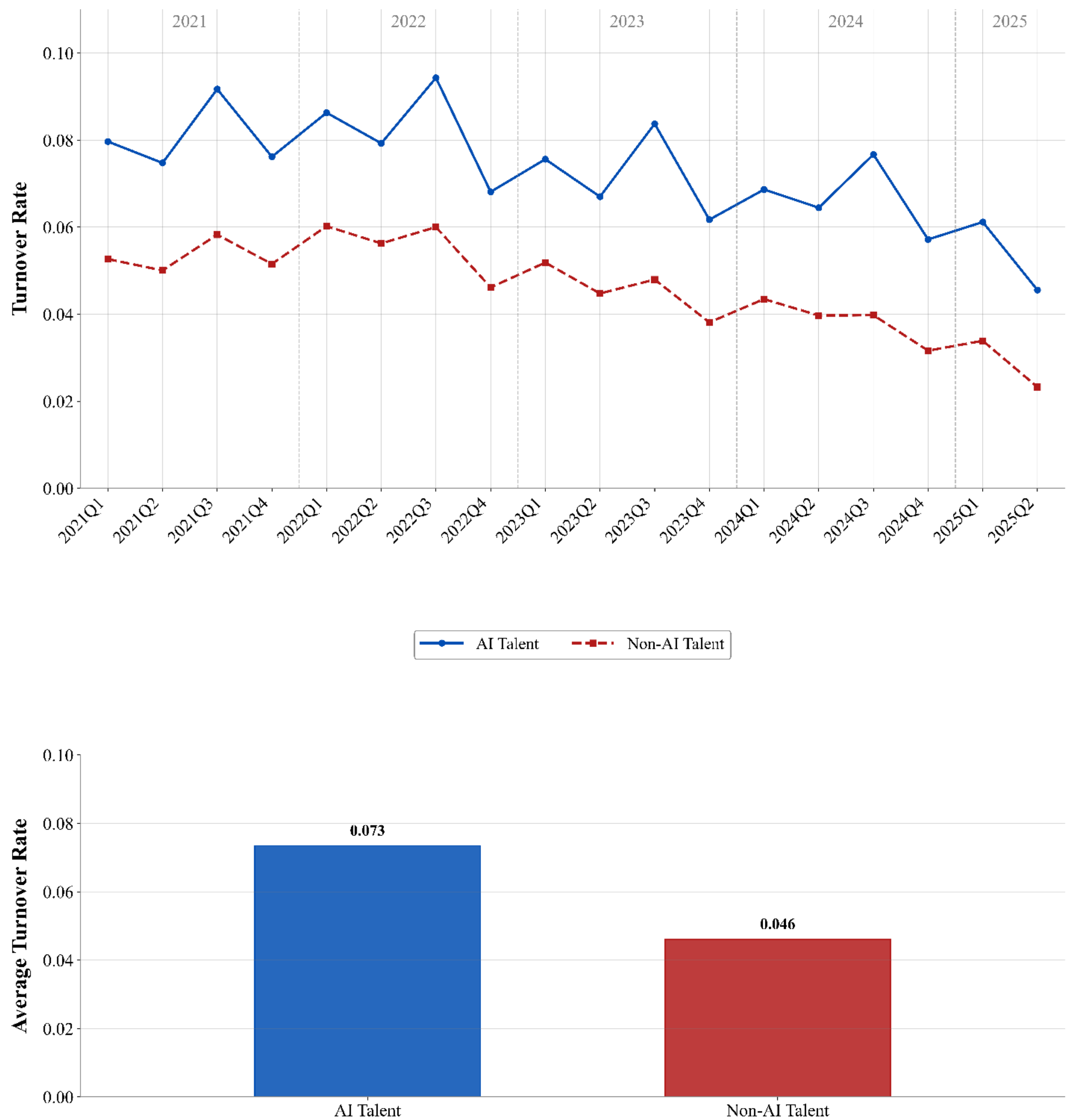


Figure 11 AI vs No-AI Turnover: Turnover rate is defined as the quarterly share of employees who left their employer relative to the total number of talents in the firm-role, separately computed for AI and non-AI talents. The bar chart shows the sample average across 2021 Q1 to 2025 Q2.

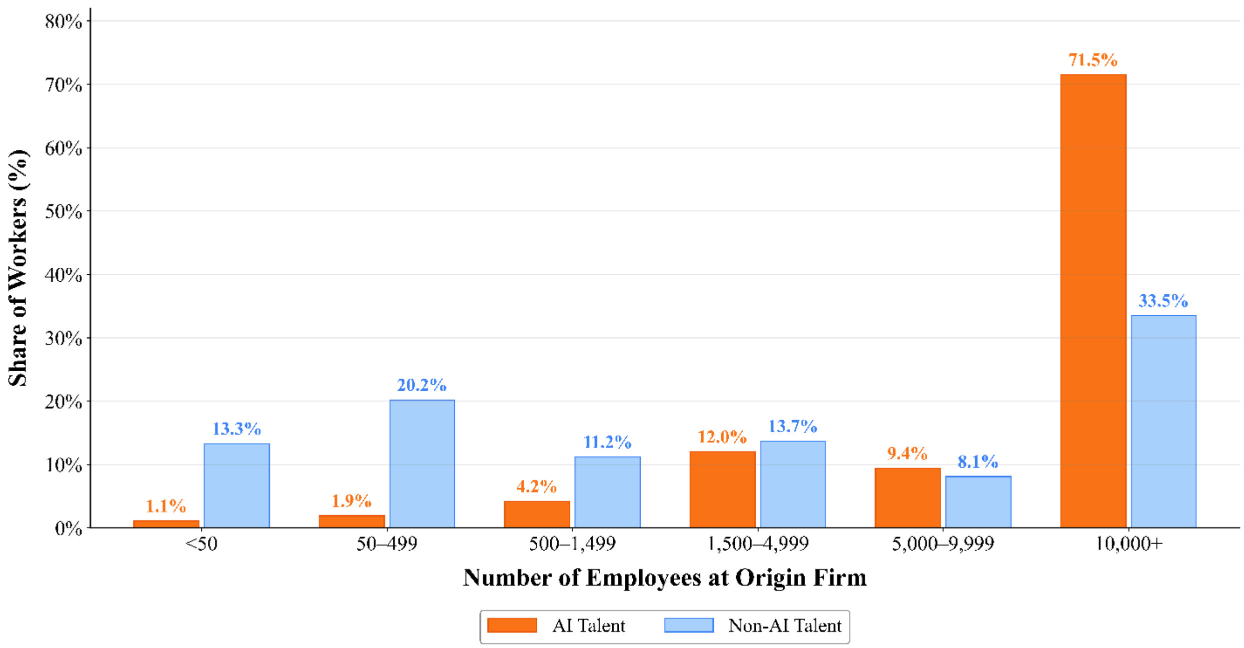


Figure 12 Origin Firm Size Distribution at Job Transition: This figure shows the distribution of origin firm size (by number of employees) for workers transitioning between jobs, comparing AI talent versus non-AI talent. AI workers disproportionately originate from large firms (10,000+ employees: 71.5% vs. 33.5%), while non-AI workers are more evenly distributed across firm sizes. Data covers US workers with LinkedIn profiles, 2021-2025.

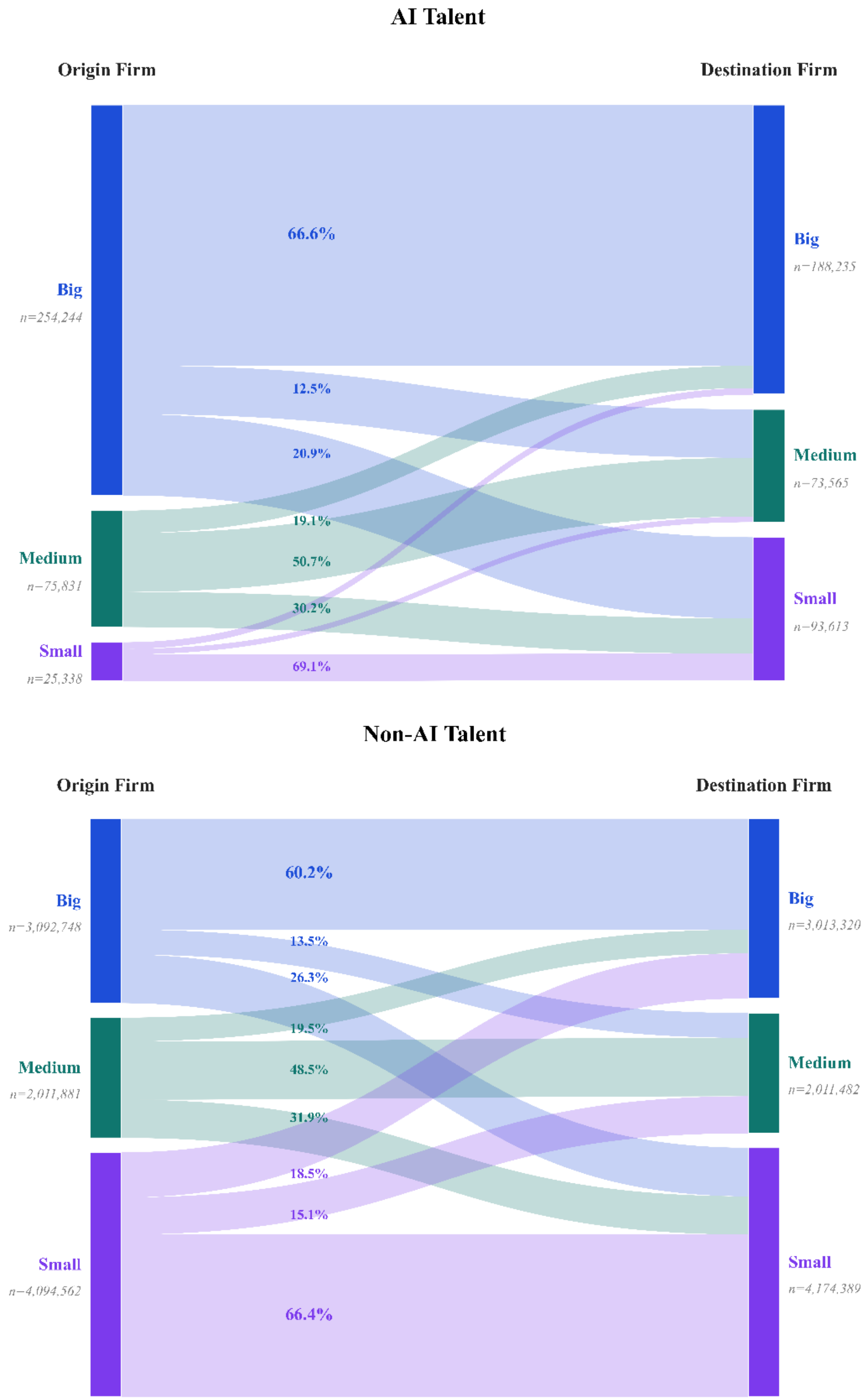


Figure 13 Job Transition Flow: The Sankey diagrams show job transition flows for AI talent and non-AI talent between firms of different sizes: Small (<1,500), Medium (1,500-9,999), and Big (≥10,000). Flow widths are proportional to worker counts, and percentages indicate within-origin shares (e.g., the share of workers leaving Big firms that go to other Big firms). Total transitions: 355,413 for AI talent and 9,199,191 for non-AI talent. Data cover US workers with LinkedIn profiles, 2021-2025.

Table 1: AI-Related Review Percentage Change

Year-Quarter	*Total Reviews*	*AI-Related Reviews*	*Percentage (%)*
2021-Q1	1,502,668	3,109	0.21%
2021-Q2	1,308,918	2,821	0.22%
2021-Q3	1,223,222	3,031	0.25%
2021-Q4	1,141,206	2,546	0.22%
2022-Q1	1,453,914	3,042	0.21%
2022-Q2	1,366,973	2,917	0.21%
2022-Q3	1,288,705	2,944	0.23%
2022-Q4	1,087,807	2,450	0.23%
2023-Q1	1,254,327	3,166	0.25%
2023-Q2	1,230,225	3,673	0.30%
2023-Q3	1,145,450	3,719	0.32%
2023-Q4	976,992	3,231	0.33%
2024-Q1	1,222,059	4,206	0.34%
2024-Q2	1,178,524	5,121	0.43%
2024-Q3	1,163,001	5,161	0.44%
2024-Q4	1,022,498	5,040	0.49%
2025-Q1	932,377	5,524	0.59%
2025-Q2	838,935	6,915	0.82%
2025-Q3	754,013	7,963	1.06%
2025-Q4	464,101	5,581	1.20%

Note: This table presents summary statistics for AI-related review percentage change over the period from 2021 to 2025, on a quarterly basis. Year-Quarter identifies the calendar quarter of observation. *Total Reviews* is the total number of English-language employee reviews submitted on Glassdoor in that quarter. *AI-Related Reviews* is the number of reviews containing at least one AI-related keyword as defined in Appendix A. *Percentage (%)* is the percentage of employee reviews that contains AI-related comments.

Table 2: Employee AI Mention and Job Satisfaction

	(1)	(2)	(3)	(4)	(5)	(6)
Dep. Var=			*Job Satisfaction*			
	All Employees	All Employees	Employees from Firms with at least one AI related review	Employees from Firm-quarters with AI related reviews	Public Firm Employees	Public Firm Employees
Employee AI Mention	-0.134***	-0.048***	-0.135***	-0.158***	-0.271***	-0.302***
	(-15.33)	(-4.25)	(-15.47)	(-16.25)	(-15.13)	(-13.57)
*Employee AI Mention*PostGPT*		-0.119***				
		(-7.68)				
Seniority	0.040***	0.041***	0.031***	0.028***	0.032***	0.032***
	(43.98)	(43.99)	(18.40)	(8.13)	(12.45)	(9.32)
Size_Lag						-0.009
						(-0.34)
ROA_Lag						0.091
						(0.64)
TobinQ_Lag						0.041***
						(9.76)
Lev_Lag						-0.065
						(-1.11)
Return_Lag						-0.046***
						(-4.13)
R&D_Lag						-1.614***
						(-3.10)
Employment Status	Yes	Yes	Yes	Yes	Yes	Yes
Quarter Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Firm Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	21,894,847	21,894,847	10,742,010	4,531,105	5,691,021	3,904,975
R-squared	0.158	0.158	0.092	0.094	0.078	0.076

Note: This table examines the impact of AI-related review on employee satisfaction. The dependent variable, *Job Satisfaction*, measures an employee's overall satisfaction with the employer. Column (1) uses all English-language Glassdoor employee reviews between 2021 and 2025 as the sample. Column (2) presents a dynamic analysis of shifting employee sentiment regarding AI over the sample period. Column (3) restricts the sample to firms having at least one AI-related employee review between 2021 and 2025. Column (4) further restricts the sample to firm-quarters having at least one AI-related employee review between 2021 and 2025. Column (5) restricts the sample to Glassdoor reviews from employees at listed firms and excludes firm-level control variables. Column (6) restricts the sample to Glassdoor reviews from employees at listed firms and includes firm-level control variables. The independent variable, *Employee AI Mention*, is a binary indicator equal to 1 if the employee's review contains any of the AI-related keywords, and 0 otherwise. *PostGPT* is an indicator variable equal to 1 for reviews posted in 2023, 2024, or 2025, after the launch of ChatGPT. *Seniority* is the reviewer's self-reported job seniority level, ranging from 1 to 7. Employment Status includes regular, part-time, temporary, apprentice, contract, freelance, intern, per-diem, PhD, reserve, self-employment, seasonal, trainee, and unknown. Standard errors are clustered at the firm level. *t*-statistics are reported in parentheses. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

Table 3: Employee AI Sentiment and Firm Productivity

	(1)	(2)	(3)	(4)	(5)
Dep. Var=	*TFP_LP*	*TFP_LP*	*TFP_OP*	*TFP_OLS*	*TFP_LP*
Firm Employee AI Sentiment	0.014***	0.015***	0.017**	0.014**	
	(2.62)	(2.87)	(2.31)	(2.02)	
Firm Employee AI Sentiment GPT					0.010**
					(2.34)
Size		-0.065**	0.054	-0.023	-0.055**
		(-2.44)	(1.49)	(-0.65)	(-2.25)
Tangibility		-0.876***	-0.877***	-1.273***	-0.756***
		(-2.59)	(-3.34)	(-4.99)	(-2.73)
Sales_Growth		0.074**	0.180***	0.194***	0.077***
		(2.55)	(4.33)	(4.85)	(2.91)
Leverage		-0.070	-0.028	0.008	-0.069
		(-1.35)	(-0.34)	(0.10)	(-1.47)
Capex		-0.086	-0.637**	-0.618**	-0.222
		(-0.24)	(-2.49)	(-2.44)	(-0.88)
Management AI Sentiment		-0.014	-0.030	-0.033	-0.021
		(-0.61)	(-0.76)	(-0.87)	(-0.77)
Annual Job Satisfaction		0.024	-0.023	-0.042	0.028
		(1.18)	(-0.78)	(-1.33)	(1.33)
Year Fixed Effects	Yes	Yes	Yes	Yes	Yes
Firm Fixed Effects	Yes	Yes	Yes	Yes	Yes
Observations	1,994	1,983	1,983	1,987	2,393
R-squared	0.957	0.962	0.972	0.959	0.960

Notes: This table examines the relationship between employee AI sentiment and firm productivity. The sample is restricted to firm-years with at least one employee review mentioning AI. The dependent variable is total factor productivity (TFP). Columns (1) and (2) use *TFP_LP*, estimated following Levinsohn and Petrin (1999), which instruments for unobserved productivity shocks using intermediate inputs (cost of goods sold) to address simultaneity bias. Column (1) excludes control variables; Column (2) includes them. Column (3) uses *TFP_OP*, estimated following Olley and Pakes (1996), which uses capital expenditures as a proxy variable and additionally corrects for selection bias due to firm exit. Column (4) uses *TFP_OLS*, defined as the residual from an OLS regression of log sales on log capital and log labor. Column (5) replaces the Harvard-dictionary-based sentiment measure with GPT-generated sentiment. The key independent variable, *Firm Employee AI Sentiment*, is the average sentiment of employee AI related reviews, which are calculated using the Harvard IV-4 General Inquirer dictionary. *Firm Employee AI Sentiment GPT* is defined as the average sentiment of employee AI related reviews, where each AI-related employee comment is classified as 1 (positive), 0 (neutral), or -1 (negative) by ChatGPT. Control variables include *Size*, *Tangibility*, *Sales Growth*, *Leverage*, *Capex*, *Management AI Sentiment*, and *Annual Job Satisfaction*. All regressions include firm and year fixed effects. Standard errors are clustered at the firm level. t-statistics are reported in parentheses. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

Table 4: Determinants of Employee AI Sentiment

	(1)	(2)	(3)	(4)
Dep. Var=		*Employee AI Sentiment*		
AI Layoff	-0.059**	-0.069***	-0.055**	-0.093***
	(-2.57)	(-3.04)	(-2.29)	(-2.86)
Mgmt AI Expert	0.468***	0.548***	0.477***	-0.167
	(4.69)	(5.23)	(3.91)	(-0.40)
Management AI Sentiment	0.036	0.032	0.033	-0.015
	(1.06)	(0.95)	(0.77)	(-0.17)
AI Investment	-0.008	-0.016	-0.023	0.001
	(-0.68)	(-1.23)	(-1.50)	(0.06)
Seniority	0.021***	0.019***	0.018***	0.018***
	(3.68)	(3.37)	(3.08)	(2.73)
Leadership	0.034***	0.034***	0.031***	0.037***
	(3.59)	(3.61)	(3.13)	(3.64)
Career Opportunity	0.023**	0.024**	0.022**	0.019
	(2.17)	(2.25)	(2.03)	(1.59)
Employment Status	Yes	Yes	Yes	Yes
Quarter Fixed Effects		Yes	Yes	Yes
Industry Fixed Effects			Yes	
Firm Fixed Effects				Yes
Observations	9,419	9,419	8,601	8,851
R-squared	0.024	0.026	0.028	0.137

Note: This table examines the determinants of employee AI sentiment, using a sample of public firm employee reviews between 2021 and 2025. The dependent variable, Employee AI Sentiment is the sentiment of an individual employee's AI-related comment. AI Layoff is the natural logarithm of the cumulative number of AI-caused layoff announcements since 2021. Mgmt AI Expert is the percentage of a firm's directors and executives who possess AI-related skills, as identified from LinkedIn profiles, relative to the total number of directors and executives at the firm. Management AI Sentiment is the average executive AI sentiment constructed by averaging the sentiment of all AI-related conference call discussions during the fiscal year. AI Investment is the natural logarithm of the cumulative number of firm AI investment announcements since 2021. Seniority is the reviewer's self-reported job seniority level, ranging from 1 to 7. Leadership and Career Opportunity are the employee's ratings for senior leadership and career opportunities on Glassdoor, respectively, each ranging from 1 to 5. Employment status categories include regular, part-time, temporary, apprentice, contract, freelance, intern, per-diem, PhD, reserve, self-employed, seasonal, trainee, and unknown. All regressions include firm and time fixed effects. Standard errors are clustered at the firm level. t-statistics are reported in parentheses. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

Table 5: Management AI Sentiment and Firm Productivity

	(1)	(2)	(3)	(4)
Dep. Var=	*TFP_LP*	*TFP_LP*	*TFP_OP*	*TFP_OLS*
Management AI Mention	0.000			
	(0.04)			
Management AI Investment Mention	0.009			
	(1.46)			
Management AI Sentiment		0.012	0.019	0.018
		(1.01)	(1.40)	(1.32)
Size	-0.016	-0.022	0.036	-0.049**
	(-0.77)	(-1.16)	(1.60)	(-2.19)
Tangibility	-0.241***	-0.332***	-0.707***	-1.167***
	(-2.64)	(-3.17)	(-5.00)	(-8.98)
Sales_Growth	0.186***	0.183***	0.232***	0.233***
	(11.83)	(9.84)	(12.31)	(12.24)
Leverage	-0.021	-0.013	-0.013	-0.006
	(-0.53)	(-0.28)	(-0.25)	(-0.11)
Capex	-0.263***	-0.312***	-0.441***	-0.439***
	(-6.49)	(-6.27)	(-8.97)	(-8.91)
Year Fixed Effects	Yes	Yes	Yes	Yes
Firm Fixed Effects	Yes	Yes	Yes	Yes
Observations	12,725	10,242	10,242	10,264
R-squared	0.891	0.906	0.930	0.918

Note: This table examines the relationship between AI-related discussions by executives in earnings conference calls and firm productivity. The dependent variable in Column (1) and Column (2) is *TFP_LP*, the total factor productivity estimated using the Levinsohn-Petrin (1999) method, which uses intermediate inputs (cost of goods sold) as a proxy for unobserved productivity shocks to correct for simultaneity bias. The dependent variable in Column (3) is *TFP_OP*, estimated using the Olley-Pakes (1996) method, which uses capital expenditures as a proxy variable and additionally corrects for selection bias arising from firm exit. The dependent variable in Column (4) is *TFP_OLS*, estimated as the residual from an OLS regression of log sales on log capital and log labor. All TFP measures are estimated separately within each two-digit SIC industry and are demeaned by industry-year mean to remove time trends and industry-level shocks. The key independent variables capture executive discussions of AI from earnings conference calls. *Management AI Mention* is an indicator variable equal to 1 if executives mention AI-related content at least once in conference calls during the fiscal year. *Management AI Investment Mention* is an indicator variable equal to 1 if executives mention AI-related investments in at least one conference call during the fiscal year, and 0 otherwise. *Management AI Sentiment* is management AI sentiment constructed by averaging the sentiment of all AI-related conference call discussions during the fiscal year. All regressions include firm and year fixed effects. Standard errors are clustered at the firm level. t-statistics are reported in parentheses. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

Table 6: Job Displacement Impact of AI

Panel A: Firm-Role-Year Level Tests

	(1)	(2)	(3)	(4)	(5)	(6)
Dep. Var=	*Hire Rate*		*Turnover*		*Employee Growth*	
Ln_#AI	-0.044***	-0.039***	0.039***	0.043***	-0.199***	-0.192***
	(-83.11)	(-81.44)	(69.19)	(78.83)	(-114.11)	(-120.42)
*Ln_#AI*PostGPT*		-0.013***		-0.010***		-0.018***
		(-34.03)		(-39.47)		(-18.02)
Firm*Role FE	Yes	Yes	Yes	Yes	Yes	Yes
Quarterly FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	22,678,945	22,678,945	22,678,945	22,678,945	22,336,097	22,336,097
R-squared	0.257	0.258	0.237	0.238	0.129	0.129

Panel B: Cross-sectional Analysis Based on COVID-Period Employment Growth Rates

	(1)	(2)	(3)	(4)	(5)	(6)
Dep. Var=	*Hire Rate*		*Turnover*		*Employee Growth*	
	High COVID Hiring	Low COVID Hiring	High COVID Hiring	Low COVID Hiring	High COVID Hiring	Low COVID Hiring
Ln_#AI	-0.050***	-0.039***	0.038***	0.040***	-0.213***	-0.186***
	(-60.08)	(-63.01)	(46.09)	(51.89)	(-75.02)	(-91.31)
Firm*Role FE	Yes	Yes	Yes	Yes	Yes	Yes
Quarterly FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	8,888,736	13,661,938	8,888,736	13,661,938	8,696,448	13,511,985
R-squared	0.258	0.252	0.229	0.241	0.134	0.124

Panel C: Alternative AI Talent Measurement						
	(1)	(2)	(3)	(4)	(5)	(6)
Dep. Var=	*Hire Rate*		*Turnover*		*Employee Growth*	
Ln_#AI	-0.046***	-0.037***	0.022***	0.031***	-0.176***	-0.169***
	(-59.38)	(-48.55)	(33.96)	(47.21)	(-63.78)	(-60.18)
*Ln_#AI*PostGPT*		-0.016***		-0.017***		-0.012***
		(-27.06)		(-37.90)		(-7.76)
Firm*Role FE	Yes	Yes	Yes	Yes	Yes	Yes
Quarterly FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	22,678,945	22,678,945	22,678,945	22,678,945	22,336,097	22,336,097
R-squared	0.255	0.256	0.235	0.235	0.119	0.119

Panel D: Firm-Year Level Tests						
	(1)	(2)	(3)	(4)	(5)	(6)
Dep. Var=	*Hire Rate*		*Turnover*		*Employee Growth*	
Ln_#AI	-0.016***	-0.015***	0.036***	0.036***	-0.092***	-0.092***
	(-11.88)	(-11.41)	(37.55)	(38.06)	(-28.92)	(-28.71)
*Ln_#AI*PostGPT*		-0.002***		-0.002***		-0.002***
		(-13.83)		(-13.17)		(-4.63)
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
Quarterly FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	152,015	152,015	152,015	152,015	151,773	151,773
R-squared	0.516	0.517	0.486	0.487	0.274	0.274

Note: This table estimates the impact of AI talent accumulation on labor replacement and workforce dynamics. Panel A is structured as a firm-role-quarter panel. Panel B divides firms into high and low COVID hiring subsamples based on their employee growth between December 2019 and December 2021. Firms with employment growth exceeding 50% are classified as High COVID Hiring, and those below 50% as Low COVID Hiring. Panel C uses an alternative measure of AI skilled employees that are identified based on LinkedIn employee job titles only, while all other panels identify AI-skilled workers based on both job titles and job descriptions on LinkedIn. Panel D aggregates the firm- role-quarter observations to the firm-quarter level. *Hire Rate* is the average employee hire rate over the next two quarters. *Turnover* is the average employee turnover rate over the next two quarters. *Employee Growth* is defined as the average employee growth rate over the next two quarters. The key independent variable is the natural logarithm of the number of AI skilled employees in a firm (*Ln_#AI*). *PostGPT* is an indicator variable equal to 1 for periods from 2023 onward, following the public launch of ChatGPT. All regressions include firm-job role and calendar quarter fixed effects. Standard errors are clustered at the firm level. t-statistics are reported in parentheses. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

Table 7: AI Impact on Employee Hiring and Turnover

Panel A: Hiring Impact by industry

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	Dep. Var.=*Hire Rate*									
	Agri.	Mining	Constr.	Mfg.	Utilities	Trade	Transport.	Info.	Finance	Services
Ln_#AI	-0.044***	-0.039***	-0.042***	-0.043***	-0.046***	-0.041***	-0.043***	-0.052***	-0.044***	-0.046***
	(-8.22)	(-13.31)	(-12.61)	(-44.92)	(-21.67)	(-23.85)	(-15.07)	(-35.11)	(-35.88)	(-39.79)
Firm-Role FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Quarterly FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	21,872	406,873	166,174	6,087,676	725,156	1,366,369	511,190	2,022,408	2,818,622	1,906,958
R-squared	0.251	0.230	0.243	0.248	0.224	0.249	0.251	0.276	0.254	0.247

Panel B: Hiring Impact by Seniority, Gender and Race

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	Dep. Var.=*Hire Rate*									
	Rank & File	Mid Manager	Executive	Female	Male	Asian	Black	Hispanic	White	Native & Other
Ln_#AI	-0.054***	-0.043***	-0.018***	-0.041***	-0.059***	-0.037***	-0.020***	-0.020***	-0.057***	-0.001
	(-87.24)	(-60.42)	(-22.81)	(-62.33)	(-91.75)	(-45.37)	(-25.29)	(-26.06)	(-90.76)	(-0.71)
Firm-Role FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Quarterly FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	10,926,899	7,853,543	2,479,493	9,332,927	11,179,013	3,370,693	3,520,922	3,470,295	12,864,612	302,832
R-squared	0.280	0.244	0.239	0.261	0.259	0.273	0.258	0.260	0.258	0.241

Panel C: Turnover Impact by industry

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	Dep. Var.=*Turnover*									
	Agri.	Mining	Constr.	Mfg.	Utilities	Trade	Transport.	Info.	Finance	Services
Ln_#AI	0.023***	0.034***	0.039***	0.034***	0.032***	0.034***	0.036***	0.039***	0.032***	0.035***
	(3.73)	(13.44)	(7.62)	(41.85)	(19.87)	(16.99)	(9.71)	(31.13)	(24.84)	(30.06)
Firm-Role FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Quarterly FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	21,872	406,873	166,174	6,087,676	725,156	1,366,369	511,190	2,022,408	2,818,622	1,906,958
R-squared	0.233	0.221	0.219	0.227	0.214	0.240	0.236	0.236	0.230	0.227

Panel D: Turnover Impact by Seniority, Gender and Race

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	Dep. Var.=*Turnover*									
	Rank & File	Mid Manager	Executive	Female	Male	Asian	Black	Hispanic	White	Native & Other
Ln_#AI	0.024***	0.023***	0.014***	0.018***	0.027***	0.015***	0.009***	0.010***	0.026***	0.006***
	(39.80)	(40.77)	(20.80)	(33.34)	(46.27)	(23.71)	(15.35)	(17.14)	(45.62)	(4.76)
Firm-Role FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Quarterly FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	10,926,899	7,853,543	2,479,493	9,332,927	11,179,013	3,370,693	3,520,922	3,470,295	12,864,612	302,832
R-squared	0.230	0.195	0.190	0.214	0.211	0.226	0.214	0.213	0.210	0.221

Note: This figure presents cross-sectional estimates of how AI talent accumulation affects new hiring and employee turnover across industries, seniority levels, gender, and ethnicity groups. *Hire Rate* is the average employee hire rate over the next two quarters. *Turnover* is the average employee turnover rate over the next two quarters. *Ln_#AI* is the natural logarithm of the total number of AI employees in a given firm-role-quarter. Panels A and C report estimates across industry groups classified by two-digit NAICS codes. Industry abbreviations are as follows: Agri. = Agriculture, Constr. = Construction, Mfg. = Manufacturing, Transport. = Transportation, Info. = Information. Panels B and D report estimates by seniority, gender, and ethnicity. Seniority groups follow Revelio Labs' classification: Rank & File (seniority 1-3) comprises individual contributors and junior staff; Mid-Manager (seniority 4-5) comprises managers and senior managers; Executive (seniority 6-7) comprises directors and above. API refers to Asian and Pacific Islander, and Native & Other combines Native American and multiracial employees due to small group sizes. All regressions include firm-job role and calendar quarter fixed effects. Standard errors are clustered at the firm level. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

Table 8: Job Displacement Impact of Remote Work

Panel A: Overall Workforce Outcomes

	(1)	(2)	(3)	(4)	(5)	(6)
	Hire Rate	Turnover	Employee Growth	Hire Rate	Turnover	Employee Growth
Flex_Work%	0.026***	-0.001	0.059***			
	(2.85)	(-0.23)	(2.62)			
Fully_Remote%				0.030***	-0.001	0.062**
				(2.99)	(-0.19)	(2.45)
Partial_Remote%				0.019	0.046	-0.009
				(0.41)	(1.34)	(-0.07)
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
Quarterly FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	90,919	90,919	90,875	90,919	90,919	90,875
R-squared	0.605	0.537	0.335	0.605	0.537	0.335

Panel B: Hire Rate by Employee Seniority

	(1)	(2)	(3)	(4)	(5)	(6)
			Dep. Var.=*Hire Rate*			
	Rank & File	Mid Manager	Executive	Rank & File	Mid Manager	Executive
Flex_Work%	0.034*	0.044**	0.003			
	(1.82)	(2.55)	(0.15)			
Fully_Remote%				0.030	0.046**	0.012
				(1.55)	(2.45)	(0.46)
Partial_Remote%				0.052	0.111	0.083
				(0.54)	(1.24)	(0.74)
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
Quarterly FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	84,744	82,646	75,324	84,744	82,646	75,324
R-squared	0.486	0.408	0.268	0.486	0.408	0.268

Note: This table estimates the impact of remote work flexibility on hiring, turnover, and workforce growth using firm-quarter level data. Panel A reports results for the full workforce, examining hire rate, turnover, and employee growth as outcomes. *Hire Rate* is the average employee hire rate over the next two quarters. *Turnover* is the average employee turnover rate over the next two quarters. *Employee Growth* is defined as the average employee growth rate over the next two quarters. Panel B reports result separately for rank-and-file employees (seniority levels 1–3), mid-managers (levels 4–5), and top executives (levels 6–7), using *Hire Rate* as the outcome. The key independent variables capture work mode flexibility at the firm level: *Flex_Work%* is the percentage of full-time job postings that allow flexible work arrangements (fully or partially remote) since the beginning of 2021. *Fully_Remote%* is the percentage of full-time job postings that allow fully remote work since the beginning of 2021, and *Partial_Remote%* is the percentage of full-time job postings allowing partially remote work since the beginning of 2021. All three measures are constructed as firm-level cumulative ratios, where the numerator is the cumulative count of remote job postings since the beginning of 2021 and the denominator is the cumulative count of all full-time job postings since the beginning of 2021. As a result, these variables are time-invariant at the firm level, reflecting each firm's overall remote work posture across the sample window rather than quarter-to-quarter hiring decisions. All regressions include industry and calendar quarter fixed effects. Standard errors are clustered at the firm level. t-statistics are reported in parentheses. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

Table 9: Job Market Premium to Employees with AI Skills

Panel A: Salary Premium

	(1)	(2)
Dep. Var.=	*Salary*	
AI Job	11,575***	5,291***
	(8.05)	(9.13)
Role FE	Yes	
Firm-Role FE		Yes
Monthly FE	Yes	Yes
Observations	13,107,494	12,549,021
R-squared	0.573	0.775

Panel B: Job Market Mobility

	(1)	(2)	(3)	(4)
Dep. Var.=	*Promotion*	*Demotion*	*Parallel*	*Move to Large Firm*
AI Talent	0.016***	0.003	-0.018***	0.057***
	(7.76)	(1.39)	(-6.70)	(19.97)
Firm-Role FE	Yes	Yes	Yes	Yes
Left_Quarter FE	Yes	Yes	Yes	Yes
Observations	5,934,656	5,934,656	5,934,656	5,455,389
R-squared	0.129	0.150	0.151	0.154

Note: This table examines how AI skills are valued in the labor market. Panel A estimates the salary premium associated with AI-related skills based on salary information in job postings. The dependent variable is the average monthly advertised salary per job role (*Salary*). Column (1) includes role and monthly fixed effects; Column (2) replaces role fixed effects with firm-role fixed effects, absorbing all time-invariant differences within each firm-role pair. *AI Job* is an indicator for job postings requiring AI skills. Panel B examines how AI skills influence career outcomes at job transition, tracking individual U.S.-to-U.S. job moves from 2021 onward and comparing AI workers (*AI Talent = 1*) against non-AI workers (*AI Talent = 0*) leaving the same firm-role in the same quarter. The dependent variables in Columns (1) to (3) are binary indicators for *Promotion* (seniority increases), *Demotion* (seniority decreases), and *Parallel* (seniority unchanged), where seniority follows Revelio Labs' seven-level classification. The dependent variable in Column (4), *Move to Larger Firm*, is a binary indicator equal to 1 if the worker moved to a destination firm with 10,000 or more employees. All regressions include firm-role and left-quarter fixed effects. Standard errors are clustered at the firm level. t-statistics are reported in parentheses. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

Appendix A Variable Definition

Panel A: Dependent Variables	
Job Satisfaction	An employee's job satisfaction rating on a 1-5 scale as reported on Glassdoor. Data Source: *Glassdoor*.
Annual Job Satisfaction	Firm-level average employee satisfaction rating (1–5 scale) based on Glassdoor data for the fiscal year. Data Source: *Glassdoor*.
TFP_LP	Total factor productivity estimated using the Levinsohn-Petrin (1999) method which uses intermediate inputs (cost of goods sold) as a proxy for unobserved productivity shocks. Estimated separately within each two-digit SIC industry and demeaned by industry-year mean. Data Source: *Compustat*.
TFP_OP	Total factor productivity estimated using the Olley-Pakes (1996) method which uses capital expenditures as a proxy variable and additionally corrects for selection bias from firm exit. Estimated separately within each two-digit SIC industry and demeaned by industry-year mean. Data Source: *Compustat*.
TFP_OLS	Total factor productivity estimated as the residual from an OLS regression of log sales on log capital and log labor. Estimated separately within each two-digit SIC industry and demeaned by industry-year mean. Data Source: *Compustat*.
Hire Rate	Hire rate, which is defined as the average quarterly number of new hires divided by total headcount at the firm-role (or firm) level over the following two quarters. Data Source: *Revelio Labs*.
Turnover	Turnover rate, which is defined as the average quarterly number of employees who depart divided by total headcount at the firm-role (or firm) level over the following two quarters. Data Source: *Revelio Labs*.
Employee Growth	Employee growth rate, which is defined as the average quarterly net change in headcount scaled by current-quarter headcount over the following two quarters. Headcount is measured based on the number of a firm's employees on LinkedIn. Data Source: *Revelio Labs*.
Salary	Average monthly salary per firm-job role as disclosed in firm job postings. Data Source: *Revelio Labs*.
Promotion	Binary indicator for job promotions, which is set to 1 if the worker's seniority level increased at job transition and 0 otherwise. Data Source: *Revelio Labs*.
Demotion	Binary indicator for job demotions, which is set to 1 if the worker's seniority level decreased at job transition and 0 otherwise. Data Source: *Revelio Labs*.
Parallel	Binary indicator for same level job changes, which is set to 1 if the worker's seniority level remained unchanged at job transition and 0 otherwise. Data Source: *Revelio Labs*.
Move to Larger Firm	Binary indicator for moving to a large firm, which is set to 1 if the worker moved to a destination firm with 10000 or more employees and 0 otherwise. Data Source: *Revelio Labs*.

Panel B: Key Independent Variables

Employee AI Mention	Binary indicator for AI-related employee reviews, which is set to 1 if an employee review contains at least one AI-related keyword (as defined in Appendix B) and 0 otherwise. Data Source: *Glassdoor*.
Employee AI Sentiment	Employee AI sentiment, which is measured using a continuous polarity score ranging from -1 to 1, calculated as (positive word count minus negative word count) divided by (positive word count plus negative word count), where values closer to 1 indicate more positive sentiment and values closer to -1 indicate more negative sentiment. Data Source: *Glassdoor*.
Employee AI Sentiment GPT	Employee AI sentiment, which is alternatively measured as a categorical variable derived from the GPT model, where each AI-specific comment within a review is coded as 1 (positive), 0 (neutral), or -1 (negative) based on its sentiment. Data Source: *Glassdoor*.
Firm Employee AI Sentiment	The firm-year average of *Employee AI Sentiment*, aggregated across all individual reviews for a given firm and year. Source: *Glassdoor*.
Firm Employee AI Sentiment GPT	The firm-year average of *Employee AI Sentiment GPT*, aggregated across all individual reviews for a given firm and year. Source: *Glassdoor*.
Management AI Mention	Binary indicator for management discussion of AI issues during conference calls, which is set to 1 if executives mention AI-related content at least once in conference calls during the fiscal year and 0 otherwise. Data Source: *Capital IQ*.
Management AI Sentiment	Management AI sentiment, which is calculated as the mean HIV4-based polarity score across all AI-related executive discussions in conference calls during the fiscal year. A higher value indicates higher management AI sentiment. This variable is set to 0 if missing. Data Source: *Capital IQ*.
Management AI Investment Mention	Binary indicator for management discussion of AI inevstment during conference calls, which is set to 1 if executives mention both AI-related content and AI-related investments in at least one conference call during the fiscal year and 0 otherwise. Data Source: *Capital IQ*.
Mgmt AI Expert	Management AI expertise, which is calculated as the percentage of a firm's directors and executives who possess AI-related skills as identified from LinkedIn profiles. Data Source: *Revelio Labs*.
AI Investment	Natural logarithm of the cumulative number of firm AI investment announcements since 2021. Data Source: *RavenPack*.
AI Layoff	Natural logarithm of the cumulative number of AI-caused layoff announcements since 2021. Data Source: *RavenPack*.
Leadership	Employee satisfaction rating for senior leadership of the employer on Glassdoor, ranging from 1 to 5. Data Source: *Glassdoor*.
Career Opportunity	Employee satisfaction rating for career opportunities with the employer, ranging from 1 to 5. Data Source: *Glassdoor*.
Seniority	Individual self-reported seniority level on Glassdoor. Data Source: *Glassdoor*.
AI Job	Binary indicator for AI job postings, which is set to 1 if a job posting or job role contains AI-related keywords and 0 otherwise. Data Source: *Revelio Labs*.

AI Talent	Binary indicator for AI skilled workers, which equal to one if an employee's LinkedIn job title or profile contains AI-related keywords, and zero otherwise. Source: *Revelio Labs*.
Ln_#AI	Natural logarithm of the total number of AI skilled employees in a given firm-role-quarter (or a firm-quarter) plus one. Data Source: *Revelio Labs*.
PostGPT	Binary indicator for post-GPT period, which is set to 1 for observations after the launch of ChatGPT and 0 otherwise. Data Source: *Manual*.
Flex_Work%	The percentage of full-time job postings allowing flexible work arrangements (fully or partially remote) at the firm level, constructed as the cumulative count of flexible job postings divided by the cumulative count of all full-time job postings from 2021 Q1 to the current quarter. Time-invariant at the firm level. Data Source: *Revelio Labs*.
Fully_Remote%	The percentage of full-time job postings allowing fully remote work at the firm level, constructed as the cumulative count of fully remote job postings divided by the cumulative count of all full-time job postings from 2021 Q1 to the current quarter. Data Source: *Revelio Labs*.
Partial_Remote%	The percentage of full-time job postings allowing partially remote work at the firm level, constructed as the cumulative count of partially remote job postings divided by the cumulative count of all full-time job postings from 2021 Q1 to the current quarter. Data Source: *Revelio Labs*.
Panel C: Firm-Level Control Variables	
Size_Lag	Firm size from the last period, measured as the natural logarithm of total assets at the end of the prior quarter. Data Source: *Compustat*.
ROA_Lag	Return on assets from the last period, defined as net income from the prior quarter divided by total assets at the end of the prior quarter. Data Source: *Compustat*.
TobinQ_Lag	Tobin's Q from the last period, measured as the market value of equity plus book value of debt from the prior quarter divided by total assets at the end of the prior quarter. Data Source: *Compustat*.
Lev_Lag	Financial leverage ratio from the last period, defined as total debt from prior quarter divided by total assets at the end of the prior quarter. Data Source: *Compustat*.
Return_Lag	Stock return over the prior quarter. Data Source: *Compustat*.
R&D_Lag	Research and development expenditures during the prior quarter scaled by total assets at the end of the prior quarter. Data Source: *Compustat*.
Size	Firm size, measured as the natural logarithm of total assets at the end of the current year. Data Source: *Compustat*.
Tangibility	Asset tangibility, measured as ratio of net property plant and equipment to total assets at the end of the current year. Data Source: *Compustat*.
Sales_Growth	Annual growth rate of total sales revenue for the current year. Data Source: *Compustat*.
Capex	Capital expenditures scaled by total sales revenue from the current year. Data Source: *Compustat*.
Leverage	Financial leverage ratio, defined as total debt divided by total assets at the end of the current year. Data Source: *Compustat*.

Appendix B Additional Tables and Analysis

Table B1: Examples of AI-Related Review Under Each Category

Category	Example Review
AI Job Insecurity	Don't replace us with robots.
AI Job Insecurity	The account closed. AI defeated us humans.
AI Job Insecurity	The company is trying an automation shift that reduces full time hours and in general members of staff, this means staff having to be skilled and trained in every department.
AI Job Insecurity	This company has laid off everyone that worked for them and sell clients content written by ChatGPT.
AI Job Insecurity	Constant lay offs — I was laid off along with 90 other people after they began replacing us with AI.
AI Job Insecurity	Switched from being designer centered to AI-focused
AI Leadership	Even the team leaders don't know research. They just copy and paste from ChatGPT.
AI Leadership	There is absolutely room for AI throughout the company, but it should not be shoehorned in everywhere just to have it.
AI Leadership	Confused leaders, chasing trends, choosing AI to fill some sort of mythic gap.
AI Leadership	Care for your people more than you do for the robots! The leadership is evil and unethical. They mislead customers about the robot's performance and then blame employees when customers complain.
AI Leadership	They're replacing original, engaging learning content with AI-generated blandness, turning this place into an AI-first sweatshop.
AI Leadership	Untested adoption of AI in environment that needs human interaction.
AI Operation and Productivity	Work on broadening the scope of applications for our AI systems, to showcase the versatility of axiomatic approaches.
AI Operation and Productivity	Their AI tools have significantly improved our efficiency and productivity.
AI Operation and Productivity	Cutting-edge AI: utilizes advanced AI technology for efficient operations.
AI Operation and Productivity	Some really great AI productivity tools in place for Sales / CS.
AI Operation and Productivity	They are automating everything, therefore, everything is going a lot faster.
AI Operation and Productivity	Zero access to cloud or Llm
AI Operation and Productivity	This often times feels like you are working multiple jobs with how different the automation systems and mechanical systems are set up.
AI Operation and Productivity	Taking notes can sometimes feel overwhelming. I would appreciate it if Rula could incorporate AI tools to assist with note-taking.
AI UpSkill	Since it's a bank, most of the time you will get to work on legacy projects with old tech stack.
AI UpSkill	Can work on end to end solutions — chance to explore and learn new technologies and services.
AI UpSkill	Invest in Technical Upskilling or Advisory Support. Consider training, workshops, or bringing in technical advisors to bridge the gap.
AI UpSkill	Interesting work in the AI / ML space
AI UpSkill	Great datascience department! We have some really exciting and innovative projects we're working on. Fun work environment with lots of smart people!

AI UpSkill	Cool company and love working with a great team on advancing AI. Proud of the work we are doing and even getting a mention on BBC news.
AI UpSkill	With the AI boom, the company's business aligns well with current technological trends, indicating a strong potential for significant growth in the near future. This presents exciting opportunities for innovation and career development within the company.
AI UpSkill	Our team comes from some of the fastest-growing companies in Europe and some of the most innovative AI labs in the World.

Table B2: Listed Firm AI Layoff (Jan 2023 – Feb 2026) Analysis

Company	Layoff News Date	Does the Firm Have More AI Talent Than Peers?	Does the Firm Overhire?	Does the Firm Have Loss in Last Quarter?	Does the Firm Have Cash Constraint?	Classification
Accenture Plc	9/26/2025	Yes	Yes	No	No	AI Washing
Advanced Micro Devices	11/13/2024	Yes	Yes	No	No	AI Washing
Alphabet Inc	12/20/2024	Yes	No	No	No	AI Displacement
Alphabet Inc	10/2/2025	Yes	No	No	No	AI Displacement
Alphabet Inc	12/27/2023	Yes	No	No	No	AI Displacement
Amazon.Com Inc	11/17/2023	Yes	No	No	Yes	AI Investment
Amazon.Com Inc	4/4/2024	Yes	No	No	Yes	AI Investment
Amazon.Com Inc	10/27/2025	Yes	No	No	No	AI Displacement
Amazon.Com Inc	1/28/2026	Yes	No	No	No	AI Displacement
Amdocs	8/12/2025	Yes	No	No	No	AI Displacement
Angi Inc	1/7/2026	No	No	No	No	AI Washing
Apple Inc	8/28/2024	Yes	No	No	No	AI Displacement
ASML Holding	1/28/2026	Yes	Yes	No	No	AI Washing
Atlassian Corp	8/3/2025	Yes	No	Yes	No	AI Washing
Autodesk Inc	1/22/2026	Yes	No	No	No	AI Displacement
Autodesk Inc	2/28/2025	Yes	No	No	No	AI Displacement
Bank of America	6/14/2025	Yes	No	No	No	AI Displacement
Best Buy	4/10/2024	Yes	No	No	No	AI Displacement
Block Inc	2/26/2026	Yes	No	No	No	AI Displacement
Broadcom Inc	10/19/2025	Yes	Yes	No	No	AI Washing
Buzzfeed Inc	2/13/2025	No	No	No	Yes	AI Investment
Chegg Inc	10/28/2025	Yes	No	Yes	Yes	AI Washing
Chegg Inc	11/13/2024	Yes	No	Yes	No	AI Washing
Ciena Corp	9/4/2025	Yes	No	No	No	AI Displacement
Cisco Systems Inc	8/15/2024	Yes	No	No	No	AI Displacement
Cisco Systems Inc	8/22/2025	Yes	No	No	No	AI Displacement
ConocoPhillips	9/5/2025	Yes	No	No	No	AI Displacement

CrowdStrike Holdings	5/7/2025	Yes	Yes	Yes	No	AI Washing
CVS Health Corp	10/1/2024	Yes	No	No	No	AI Displacement
Dell Technologies Inc	8/6/2024	Yes	No	No	Yes	AI Investment
Dell Technologies Inc	2/6/2023	Yes	No	No	No	AI Displacement
Dow Inc	1/29/2026	Yes	No	Yes	Yes	AI Washing
Dropbox Inc	4/27/2023	Yes	No	No	No	AI Displacement
Dropbox Inc	10/30/2024	Yes	No	No	No	AI Displacement
Duolingo Inc	1/8/2024	Yes	Yes	No	No	AI Washing
eBay Inc	2/26/2026	Yes	Yes	No	No	AI Washing
Fiverr International Ltd	9/16/2025	Yes	No	No	No	AI Displacement
Goldman Sachs	10/14/2025	Yes	No	No	No	AI Displacement
HP Inc	11/26/2025	Yes	No	No	No	AI Displacement
Intel Corp	8/2/2024	Yes	No	Yes	Yes	AI Washing
Intl Business Machines	5/10/2023	Yes	No	No	No	AI Displacement
Intl Business Machines	9/20/2024	Yes	No	No	No	AI Displacement
Intl Business Machines	11/15/2025	Yes	No	No	No	AI Displacement
Intuit Inc	7/10/2024	Yes	No	No	No	AI Displacement
Intuit Inc	7/15/2025	Yes	No	No	No	AI Displacement
Mastercard Inc	8/21/2024	Yes	Yes	No	No	AI Washing
Meta Platforms Inc	3/14/2023	Yes	No	No	No	AI Displacement
Meta Platforms Inc	10/29/2025	Yes	No	No	No	AI Displacement
Meta Platforms Inc	1/12/2026	Yes	No	No	No	AI Displacement
Microsoft Corp	10/25/2023	Yes	No	No	No	AI Displacement
Microsoft Corp	6/4/2024	Yes	No	No	No	AI Displacement
Microsoft Corp	7/2/2025	Yes	No	No	No	AI Displacement
Microsoft Corp	1/7/2026	Yes	No	No	No	AI Displacement
Morgan Stanley	3/19/2025	Yes	No	No	No	AI Displacement
Netflix Inc	12/26/2024	No	No	No	No	AI Washing
Oracle Corp	8/19/2025	Yes	No	No	No	AI Displacement
Paycom Software Inc	10/1/2025	No	No	No	No	AI Washing
PayPal Holdings Inc	1/31/2024	Yes	No	No	No	AI Displacement
Peloton Interactive Inc	2/5/2026	No	No	Yes	Yes	AI Washing

Pinterest Inc	1/27/2026	Yes	No	No	No	AI Displacement
Rogers Communications	7/9/2025	No	No	No	Yes	AI Investment
Sabre Corp	2/20/2026	Yes	No	Yes	Yes	AI Washing
Salesforce Inc	9/1/2025	Yes	No	No	No	AI Displacement
Salesforce Inc	2/10/2026	Yes	No	No	No	AI Displacement
Shopify Inc	8/2/2023	No	No	Yes	No	AI Washing
Snap Inc	2/6/2024	Yes	No	Yes	No	AI Washing
Synopsys Inc	11/14/2025	No	No	No	No	AI Washing
United Parcel Service Inc	1/31/2024	Yes	No	No	No	AI Displacement
Upstart Holdings Inc	2/1/2023	Yes	Yes	Yes	Yes	AI Washing
Verizon Communications	11/17/2025	No	No	No	No	AI Washing
Walmart Inc	6/3/2025	No	No	No	No	AI Washing
Warner Music Group	3/29/2023	No	No	No	No	AI Washing
Wells Fargo & Co	12/9/2025	Yes	No	No	No	AI Displacement
Wells Fargo & Co	2/12/2026	Yes	No	No	No	AI Displacement
Workday Inc	2/5/2025	Yes	No	No	No	AI Displacement

Notes: This table classifies AI-related layoff announcements from January 2023 to February 2026 into three groups based on four firm-level indicators. The first indicator, "Does the Firm Have More AI Talent Than Peers?", measures whether the firm's average share of AI talent in its total workforce, computed by averaging quarterly AI headcount ratios from 2023 onwards, exceeds the industry average among all firms operating in the same 2-digit NAICS industry over the same period. "Yes" indicates the firm has a higher AI talent concentration than its industry peers; "No" indicates it falls below. The second indicator, "Does the Firm Overhire?", measures whether the firm's total employment grew faster than its industry peers between 2021 and the year immediately preceding the layoff announcement. "Yes" indicates above-peer employment growth; "No" indicates below-peer growth. The third indicator, "Does the Firm Have Loss in Last Quarter?", measures whether the firm reported negative net income in the quarter immediately preceding the layoff announcement. "Yes" indicates a net loss; "No" indicates a net profit. The fourth indicator, "Does the Firm Have Cash Constraints?", measures whether free cash flow, defined as operating cash flow minus capital expenditures, falls short of short-term debt at the fiscal year end preceding the layoff announcement. "Yes" indicates that free cash flow is insufficient to cover short-term debt; "No" indicates that it is. A layoff event is classified as "AI Investment" when the announcing firm simultaneously exhibits below-industry employment growth, a net profit in the prior quarter, and cash constraints at the preceding fiscal year end. A layoff event is classified as "AI Displacement" when the announcing firm simultaneously exhibits above-industry AI talent concentration, below-industry employment growth, a net profit in the prior quarter, and no cash constraints. All remaining events are classified as AI Washing, reflecting cases where the firm overhired, reported a loss prior to the layoff, or otherwise exhibits observable characteristics inconsistent with a credible AI-driven rationale for the workforce reduction.

Table B3: AI Job Posts and Salary Premium by month

Period	*#AI_Posts*	*#Total Posts*	*AI-Post %*	*AI Pay*	*No-AI Pay*	*Premium*
	(1)	(2)	(3) = (1)/(2)	(4)	(5)	(6)=(4)/(5)
2021m1	14,411	684,008	2.11%	103,454	62,113	66.56%
2021m2	16,149	710,870	2.27%	102,257	61,611	65.97%
2021m3	20,601	1,188,664	1.73%	90,619	50,875	78.12%
2021m4	23,798	1,413,955	1.68%	94,095	51,388	83.11%
2021m5	25,778	1,201,960	2.14%	97,054	56,936	70.46%
2021m6	17,461	850,848	2.05%	98,732	57,922	70.46%
2021m7	24,795	1,115,939	2.22%	94,507	57,938	63.12%
2021m8	59,194	1,876,368	3.15%	99,704	62,950	58.39%
2021m9	55,917	1,717,055	3.26%	98,537	62,593	57.42%
2021m10	77,018	2,418,638	3.18%	104,557	64,275	62.67%
2021m11	63,085	2,074,917	3.04%	101,623	63,133	60.97%
2021m12	67,368	1,958,275	3.44%	99,035	64,391	53.80%
2022m1	87,819	2,548,547	3.45%	102,027	67,210	51.80%
2022m2	93,768	2,573,073	3.64%	102,231	69,996	46.05%
2022m3	82,432	2,354,825	3.50%	101,741	69,852	45.65%
2022m4	62,151	2,075,836	2.99%	109,496	69,137	58.38%
2022m5	100,650	3,368,628	2.99%	107,372	70,204	52.94%
2022m6	70,321	2,698,203	2.61%	111,871	67,815	64.96%
2022m7	71,417	2,719,323	2.63%	113,992	68,163	67.23%
2022m8	69,238	2,505,521	2.76%	116,507	68,698	69.59%
2022m9	63,176	2,559,708	2.47%	112,729	66,381	69.82%
2022m10	59,613	2,170,220	2.75%	111,800	67,215	66.33%
2022m11	53,791	2,055,171	2.62%	108,279	65,258	65.93%
2022m12	45,547	1,823,997	2.50%	108,176	64,639	67.35%
2023m1	53,025	1,972,331	2.69%	106,209	65,899	61.17%
2023m2	31,334	1,271,192	2.46%	103,967	62,447	66.49%
2023m3	31,451	1,596,368	1.97%	110,050	62,105	77.20%
2023m4	27,558	1,325,066	2.08%	110,494	64,811	70.49%
2023m5	27,921	1,261,129	2.21%	104,456	62,803	66.32%
2023m6	35,578	1,466,246	2.43%	108,673	64,601	68.22%
2023m7	26,708	1,171,330	2.28%	105,395	61,386	71.69%
2023m8	34,963	1,525,141	2.29%	108,785	60,791	78.95%
2023m9	38,127	1,546,376	2.47%	113,064	61,725	83.17%
2023m10	39,808	1,556,635	2.56%	114,460	64,842	76.52%
2023m11	28,727	1,137,697	2.53%	112,321	61,723	81.98%
2023m12	28,763	1,124,901	2.56%	115,650	63,537	82.02%
2024m1	35,042	1,392,255	2.52%	113,456	64,173	76.80%
2024m2	30,222	1,124,456	2.69%	108,972	62,981	73.02%
2024m3	40,180	1,288,621	3.12%	101,421	62,391	62.56%
2024m4	36,645	1,373,967	2.67%	112,268	64,320	74.55%
2024m5	33,058	1,004,489	3.29%	117,369	70,010	67.65%

2024m6	37,017	1,217,126	3.04%	121,311	67,472	79.80%
2024m7	33,869	1,300,022	2.61%	119,676	66,780	79.21%
2024m8	36,721	1,297,031	2.83%	116,369	64,422	80.63%
2024m9	31,467	1,267,167	2.48%	113,137	59,280	90.85%
2024m10	36,636	1,323,835	2.77%	116,512	61,410	89.73%
2024m11	33,297	1,193,626	2.79%	117,344	64,161	82.89%
2024m12	31,236	1,301,095	2.40%	116,901	61,531	89.99%
2025m1	40,220	1,522,181	2.64%	120,297	63,417	89.69%
2025m2	52,049	1,489,629	3.49%	119,585	68,504	74.57%
2025m3	61,753	1,701,108	3.63%	122,339	68,459	78.70%
2025m4	92,991	2,552,263	3.64%	119,552	66,675	79.31%
2025m5	102,380	2,428,331	4.22%	118,673	68,507	73.23%
2025m6	109,443	2,719,725	4.02%	117,879	65,140	80.96%
2025m7	135,708	3,014,309	4.50%	117,492	67,309	74.56%
2025m8	139,150	2,684,886	5.18%	131,635	69,402	89.67%
2025m9	120,809	2,929,879	4.12%	124,566	64,657	92.66%
2025m10	110,144	2,559,893	4.30%	125,230	67,027	86.84%
2025m11	100,908	2,319,553	4.35%	128,331	73,157	75.42%
2025m12	94,017	1,840,508	5.11%	132,138	73,613	79.50%

Note: To ensure accuracy in identifying AI-integrated roles, this analysis includes only job posts with descriptions. This allows for keyword filtering to accurately identify roles that require or utilize AI tools. *#AI_Posts* is the number of job postings containing AI-related keywords. #Total Posts is the total number of job postings in that month. *AI-Post %* is the ratio of AI-related to total postings. *AI Pay* and *No-AI Pay* are the average advertised salaries for AI-related and non-AI-related postings, respectively. Premium is the ratio of AI *Pay* to *No-AI Pay*, expressed as a percentage above the non-AI average.

Table B4: Top 100 Expanding Job Roles by Employment Share, 2021-2025

	Employment Share (% of Sample)					
Jon Position	2021	2022	2023	2024	2025	*Δ/21*
Spring	0.056%	0.094%	0.128%	0.152%	0.176%	215%
Software Designer	0.158%	0.288%	0.363%	0.397%	0.430%	172%
Resident	0.054%	0.091%	0.115%	0.127%	0.142%	161%
Teaching	0.147%	0.214%	0.286%	0.332%	0.372%	154%
Analyst Programmer	0.000%	0.001%	0.001%	0.001%	0.001%	135%
Technical Coordinator	0.027%	0.043%	0.054%	0.059%	0.062%	130%
Treasurer	0.037%	0.049%	0.064%	0.074%	0.083%	123%
Purchaser	0.009%	0.015%	0.018%	0.020%	0.020%	122%
Lecturer	0.007%	0.011%	0.014%	0.015%	0.016%	119%
Software Programmer	0.060%	0.100%	0.119%	0.124%	0.128%	115%
Financial Adviser	0.020%	0.027%	0.035%	0.039%	0.041%	111%
Doctor	0.013%	0.019%	0.023%	0.025%	0.027%	111%
Tax Preparer	0.293%	0.417%	0.525%	0.583%	0.616%	110%
Public Relations	0.074%	0.094%	0.116%	0.134%	0.149%	101%
Industrial Engineering	0.019%	0.027%	0.033%	0.037%	0.037%	97%
Systems Programmer	0.013%	0.021%	0.024%	0.025%	0.026%	97%
Substitute Teacher	0.221%	0.292%	0.355%	0.396%	0.430%	95%
Thesis Student	0.001%	0.001%	0.001%	0.001%	0.001%	90%
Site	0.113%	0.160%	0.187%	0.202%	0.215%	89%
Student Worker	0.148%	0.224%	0.255%	0.265%	0.278%	88%
Ambassador	0.100%	0.119%	0.148%	0.170%	0.188%	87%
Day Job	0.001%	0.002%	0.002%	0.002%	0.002%	87%
Bank Officer	0.011%	0.015%	0.018%	0.020%	0.021%	87%
Electronics Engineer	0.101%	0.132%	0.165%	0.181%	0.189%	87%
Marketing Sales	0.041%	0.054%	0.063%	0.070%	0.075%	84%
Technical Officer	0.008%	0.010%	0.012%	0.013%	0.014%	82%
Safety Officer	0.020%	0.028%	0.033%	0.035%	0.036%	82%
Captain	0.122%	0.146%	0.181%	0.210%	0.222%	81%
Professor	0.044%	0.054%	0.064%	0.072%	0.079%	80%
Mechanical	0.082%	0.099%	0.123%	0.139%	0.147%	80%
Software Engg	0.002%	0.004%	0.004%	0.004%	0.004%	77%
Computer Scientist	0.037%	0.045%	0.053%	0.060%	0.065%	77%
Relationship Officer	0.006%	0.009%	0.010%	0.010%	0.010%	76%
Advertising	0.051%	0.067%	0.075%	0.082%	0.088%	73%
Computer Programmer	0.023%	0.030%	0.036%	0.038%	0.039%	73%
Real Estate Agent	0.095%	0.123%	0.143%	0.156%	0.164%	73%
Credit Officer	0.018%	0.026%	0.029%	0.030%	0.031%	73%
Promoter	0.039%	0.042%	0.051%	0.060%	0.066%	72%
Economist	0.034%	0.043%	0.050%	0.054%	0.058%	69%
Civil Engineer	0.073%	0.087%	0.104%	0.117%	0.124%	69%
Technique	0.002%	0.003%	0.003%	0.004%	0.004%	68%

Technical Sales Engineer	0.014%	0.018%	0.021%	0.022%	0.023%	67%
Laboratory	0.076%	0.087%	0.105%	0.119%	0.127%	67%
Computer Engineer	0.060%	0.072%	0.084%	0.094%	0.100%	67%
Computer	0.028%	0.036%	0.042%	0.045%	0.047%	67%
Comercial	0.001%	0.002%	0.002%	0.002%	0.002%	66%
Secretary	0.049%	0.059%	0.070%	0.077%	0.082%	65%
Document Controller	0.007%	0.009%	0.011%	0.011%	0.011%	64%
Statistician	0.048%	0.062%	0.070%	0.075%	0.078%	64%
Computer Technician	0.168%	0.225%	0.258%	0.268%	0.274%	63%
Teacher	0.148%	0.176%	0.206%	0.227%	0.241%	63%
Graphic Design	0.099%	0.133%	0.149%	0.156%	0.161%	63%
Pilot	0.060%	0.076%	0.089%	0.096%	0.098%	62%
Chemical Engineer	0.042%	0.050%	0.059%	0.066%	0.067%	62%
Engineer	0.148%	0.190%	0.216%	0.229%	0.239%	61%
Salesman	0.090%	0.116%	0.130%	0.137%	0.145%	61%
Purchasing Agent	0.039%	0.050%	0.058%	0.061%	0.063%	60%
Aircraft Technician	0.012%	0.017%	0.019%	0.019%	0.020%	60%
Flight Attendant	0.122%	0.149%	0.180%	0.191%	0.196%	60%
Actor	0.040%	0.053%	0.058%	0.061%	0.064%	60%
Research Engineer	0.166%	0.195%	0.220%	0.245%	0.265%	60%
Nursing	0.050%	0.062%	0.072%	0.077%	0.080%	59%
Animator	0.022%	0.030%	0.033%	0.034%	0.036%	59%
Mechanical Engineering	0.269%	0.348%	0.405%	0.430%	0.426%	58%
Finance Officer	0.005%	0.007%	0.008%	0.008%	0.008%	58%
Photographer	0.090%	0.116%	0.130%	0.136%	0.142%	57%
Healthcare Representative	0.184%	0.233%	0.260%	0.275%	0.289%	57%
Information Systems	0.012%	0.015%	0.018%	0.018%	0.019%	56%
Human Resources Officer	0.016%	0.021%	0.024%	0.024%	0.025%	56%
Marketing Representative	0.169%	0.199%	0.225%	0.246%	0.264%	56%
Quality Inspector	0.064%	0.081%	0.093%	0.099%	0.100%	55%
Counselor	0.072%	0.083%	0.096%	0.105%	0.112%	55%
Security Guard	0.034%	0.040%	0.046%	0.051%	0.053%	55%
Aircraft Maintenance Technician	0.035%	0.044%	0.051%	0.053%	0.055%	55%
Athlete	0.031%	0.039%	0.043%	0.045%	0.048%	55%
Financial Service	0.025%	0.030%	0.033%	0.036%	0.038%	55%
Medical Sales Representative	0.068%	0.083%	0.093%	0.099%	0.105%	55%
Systems Consultant	0.043%	0.056%	0.063%	0.065%	0.066%	55%
R&D Engineer	0.061%	0.072%	0.082%	0.090%	0.095%	55%
Foreman	0.085%	0.098%	0.114%	0.125%	0.131%	54%
Researcher	0.054%	0.059%	0.066%	0.075%	0.083%	53%
Station	0.047%	0.063%	0.068%	0.070%	0.072%	53%
Event	0.018%	0.022%	0.024%	0.027%	0.028%	52%
Emergency Medical Technician	0.113%	0.135%	0.155%	0.165%	0.172%	51%
Banking Officer	0.008%	0.010%	0.010%	0.011%	0.012%	51%
Marketing Research	0.051%	0.064%	0.071%	0.075%	0.077%	51%

Machine Engineer	0.153%	0.182%	0.199%	0.215%	0.232%	51%
Press	0.019%	0.025%	0.026%	0.027%	0.028%	50%
Attorney	0.156%	0.169%	0.192%	0.216%	0.234%	50%
Investment Representative	0.020%	0.024%	0.028%	0.029%	0.030%	50%
Bank Teller	0.234%	0.284%	0.324%	0.346%	0.352%	50%
Accounting Finance	0.131%	0.151%	0.173%	0.188%	0.196%	50%
Material	0.023%	0.025%	0.030%	0.033%	0.034%	50%
Research	0.139%	0.173%	0.191%	0.201%	0.208%	50%
Media Relations	0.065%	0.084%	0.091%	0.094%	0.097%	49%
Counter	0.021%	0.024%	0.028%	0.030%	0.032%	49%
Conseiller Commercial	0.000%	0.000%	0.000%	0.000%	0.000%	49%
Electronic Engineer	0.010%	0.012%	0.013%	0.014%	0.015%	49%
Electrical Engineering	0.070%	0.085%	0.098%	0.104%	0.104%	49%
Accounts Officer	0.002%	0.003%	0.003%	0.003%	0.003%	49%

Top 100 Contracting Job Roles by Employment Share, 2021-2025

Jon Position	*Employment Share (% of Sample)*					
	2021	2022	2023	t2024	2025	*Δ/21*
General Ledger Accountant	0.0004%	0.0003%	0.0001%	0.0001%	0.0001%	-75%
Financial Planning & Analysis	0.006%	0.004%	0.003%	0.002%	0.002%	-69%
Media Planner	0.012%	0.007%	0.005%	0.004%	0.004%	-67%
Mortgage Loan Processor	0.021%	0.015%	0.009%	0.008%	0.007%	-66%
Client Solutions	0.084%	0.054%	0.039%	0.032%	0.029%	-66%
Digital Account	0.027%	0.018%	0.013%	0.011%	0.010%	-65%
Sales Development	0.060%	0.040%	0.030%	0.025%	0.022%	-63%
Demand Planning	0.035%	0.023%	0.017%	0.015%	0.013%	-62%
Supply Planner	0.014%	0.009%	0.007%	0.006%	0.005%	-61%
Merchandise Planner	0.033%	0.022%	0.017%	0.014%	0.013%	-60%
Clinical Trial	0.040%	0.029%	0.023%	0.019%	0.017%	-58%
Barista	0.270%	0.171%	0.135%	0.120%	0.115%	-57%
Business Banking Relationship	0.005%	0.004%	0.003%	0.002%	0.002%	-57%
Treasury Analyst	0.020%	0.015%	0.011%	0.009%	0.009%	-57%
Mobile	0.016%	0.010%	0.008%	0.007%	0.007%	-56%
Cloud Architect	0.038%	0.027%	0.021%	0.018%	0.017%	-55%
Market Sales	0.143%	0.104%	0.082%	0.071%	0.065%	-55%
Fulfillment	0.086%	0.056%	0.045%	0.041%	0.039%	-55%
Customer Marketing	0.032%	0.025%	0.020%	0.017%	0.015%	-54%
Digital Product	0.100%	0.074%	0.059%	0.051%	0.046%	-54%
Financial Analyst Fp A	0.018%	0.014%	0.011%	0.009%	0.008%	-53%
Specialty Sales Representative	0.209%	0.160%	0.126%	0.108%	0.098%	-53%
Finance Analyst	0.091%	0.064%	0.052%	0.046%	0.043%	-52%
Merchant	0.019%	0.014%	0.011%	0.010%	0.009%	-52%
Project Scrum	0.136%	0.103%	0.082%	0.071%	0.066%	-52%
Commercial Account	0.011%	0.009%	0.007%	0.006%	0.005%	-52%
Account Delivery	0.003%	0.002%	0.002%	0.002%	0.001%	-52%

Retail Operations	0.007%	0.005%	0.004%	0.004%	0.003%	-52%
Network Consulting Engineer	0.008%	0.006%	0.005%	0.004%	0.004%	-52%
Financial Planning Analyst	0.053%	0.041%	0.033%	0.029%	0.026%	-51%
Brand Marketing	0.094%	0.069%	0.056%	0.050%	0.046%	-51%
Sales Trainer	0.010%	0.007%	0.005%	0.005%	0.005%	-51%
Business Planning Analyst	0.023%	0.017%	0.014%	0.012%	0.011%	-51%
Customer Development	0.039%	0.029%	0.024%	0.021%	0.019%	-51%
Desktop Support Analyst	0.004%	0.003%	0.002%	0.002%	0.002%	-51%
Lean Sigma Belt	0.014%	0.012%	0.009%	0.008%	0.007%	-50%
Revenue Accountant	0.011%	0.008%	0.007%	0.006%	0.005%	-50%
Beauty	0.068%	0.046%	0.039%	0.036%	0.034%	-50%
UX Designer	0.095%	0.073%	0.060%	0.052%	0.048%	-50%
Pharmacovigilance	0.020%	0.015%	0.012%	0.011%	0.010%	-50%
Retail Account	0.004%	0.004%	0.003%	0.002%	0.002%	-49%
Customer Engagement	0.023%	0.018%	0.014%	0.013%	0.012%	-49%
Controller	0.047%	0.036%	0.029%	0.026%	0.024%	-49%
Key Account	0.053%	0.042%	0.034%	0.030%	0.027%	-49%
Technical Test	0.002%	0.002%	0.002%	0.001%	0.001%	-48%
Sales Support	0.013%	0.010%	0.008%	0.007%	0.007%	-47%
Teller	0.081%	0.059%	0.049%	0.045%	0.043%	-47%
Desktop Engineer	0.002%	0.002%	0.002%	0.001%	0.001%	-47%
Commercial Operations	0.004%	0.003%	0.002%	0.002%	0.002%	-47%
Business Development Representative	0.117%	0.093%	0.075%	0.067%	0.062%	-47%
Media Buyer	0.008%	0.006%	0.005%	0.005%	0.004%	-47%
Keyholder	0.138%	0.099%	0.083%	0.076%	0.073%	-47%
Ecommerce	0.027%	0.021%	0.017%	0.015%	0.014%	-47%
Account Development	0.184%	0.139%	0.116%	0.107%	0.098%	-47%
Resourcing	0.001%	0.001%	0.001%	0.001%	0.001%	-47%
Customer Sales	0.009%	0.007%	0.006%	0.005%	0.005%	-47%
Supply Chain Planning	0.117%	0.091%	0.075%	0.068%	0.063%	-46%
Supply Planning	0.005%	0.004%	0.004%	0.003%	0.003%	-46%
Talent Development	0.086%	0.072%	0.060%	0.052%	0.046%	-46%
Escalation	0.005%	0.004%	0.003%	0.003%	0.003%	-46%
Loan Processor	0.072%	0.059%	0.044%	0.040%	0.039%	-46%
Transformation	0.030%	0.023%	0.019%	0.017%	0.016%	-46%
Project Delivery	0.017%	0.013%	0.011%	0.010%	0.009%	-46%
Finance Operations	0.011%	0.008%	0.007%	0.006%	0.006%	-46%
Genius	0.031%	0.024%	0.021%	0.018%	0.017%	-45%
Business Operations Analyst	0.078%	0.064%	0.053%	0.046%	0.043%	-45%
Devops	0.004%	0.003%	0.003%	0.002%	0.002%	-45%
Exploration Geologist	0.002%	0.002%	0.001%	0.001%	0.001%	-45%
Pmo Analyst	0.002%	0.002%	0.001%	0.001%	0.001%	-45%
Medical Representative	0.012%	0.009%	0.008%	0.007%	0.007%	-44%
Business Process	0.025%	0.020%	0.017%	0.015%	0.014%	-44%

HR Operations	0.012%	0.010%	0.008%	0.007%	0.007%	-44%
Supplier Quality	0.007%	0.006%	0.005%	0.004%	0.004%	-44%
Retail Banking	0.001%	0.001%	0.001%	0.001%	0.001%	-43%
Operating Officer	0.044%	0.035%	0.030%	0.027%	0.025%	-43%
Business Transformation	0.064%	0.056%	0.046%	0.040%	0.036%	-43%
Customer Solutions	0.010%	0.007%	0.006%	0.006%	0.006%	-43%
Retail Sales Representative	0.021%	0.016%	0.013%	0.012%	0.012%	-43%
Legal Counsel	0.017%	0.013%	0.012%	0.010%	0.010%	-43%
Inside Sales Representative	0.078%	0.062%	0.053%	0.047%	0.045%	-42%
Integration	0.008%	0.006%	0.005%	0.005%	0.005%	-42%
Sales Operations Analyst	0.051%	0.044%	0.037%	0.033%	0.030%	-42%
Analytics	0.123%	0.094%	0.081%	0.075%	0.071%	-42%
Presales	0.002%	0.002%	0.001%	0.001%	0.001%	-42%
Mortgage Underwriter	0.018%	0.015%	0.012%	0.011%	0.010%	-42%
Grocery	0.047%	0.034%	0.030%	0.028%	0.028%	-42%
Marketing Project	0.044%	0.035%	0.030%	0.027%	0.026%	-42%
Development Coordinator	0.021%	0.017%	0.015%	0.013%	0.012%	-41%
Account Coordinator	0.072%	0.062%	0.051%	0.046%	0.042%	-41%
Client Account	0.011%	0.009%	0.008%	0.007%	0.007%	-41%
Ledger Accountant	0.009%	0.007%	0.006%	0.005%	0.005%	-41%
Human Resources Representative	0.017%	0.015%	0.013%	0.011%	0.010%	-41%
Project Support	0.002%	0.001%	0.001%	0.001%	0.001%	-41%
Digital Project	0.020%	0.017%	0.014%	0.013%	0.012%	-41%
Customer Success	0.181%	0.155%	0.129%	0.115%	0.107%	-41%
Drug Safety	0.012%	0.011%	0.009%	0.008%	0.007%	-41%
Application Support	0.009%	0.007%	0.006%	0.006%	0.005%	-40%
Operations Project	0.014%	0.013%	0.011%	0.009%	0.008%	-40%
Product Sales	0.017%	0.013%	0.011%	0.010%	0.010%	-40%
Information Security	0.027%	0.021%	0.018%	0.017%	0.016%	-40%
Vendor	0.009%	0.007%	0.007%	0.006%	0.006%	-40%

Note: This table presents the 100 most expanding and 100 most contracting job positions out of 1,500 total positions, ranked by the relative change in employment share between 2021 and 2025. The sample comprises employees with LinkedIn profiles across all listed firms. *Employment share* is calculated annually as the average monthly headcount in each position as a percentage of total sample employment. *Δ/2021* is defined as (Share_2025 - Share_2021) / Share_2021 × 100, expressed as a percentage, where positive (negative) values indicate expanding (contracting) job positions.

Appendix C AI-Related Keyword Identification Methodology

To identify AI-related employee reviews and job postings, we developed a keyword-based classification approach using regular expression (regex) pattern matching. A review or job posting was classified as "AI-related" if it contained at least one match from a curated keyword list. All matching was performed with case-insensitive flags, so that capitalization variants (e.g., "AI," "ai," "Ai") were captured uniformly.

Several design principles guided keyword construction. Short acronyms such as "AI," "NLP," and "LLM" were matched using strict word-boundary anchors (e.g., \bAI\b) to prevent false positives from partial matches within common words such as "again" or "fair." Ambiguous terms were handled with specificity constraints: "ML" was matched only within phrases such as *ML model*, and "transformer" only when followed by *model* or *architecture*, to distinguish AI-specific usage from unrelated references. For terms with inconsistent formatting, multiple surface forms were included (e.g., *chatgpt*, *chat gpt*, *chat-gpt*; *gen ai*, *gen-ai*, *generative ai*). Product names such as *Claude* and *copilot* were included as standalone matches, accepting a small false-positive risk given their growing prevalence as AI tools in workplace discourse.

Conference call transcripts differ in character from employee reviews: they are directed at a general audience and tend to express AI-related content in broader, less role-specific terms. We therefore applied a separate keyword identification strategy for this corpus, using GPT to read a sample of 2025 conference call files and extract an additional set of AI-relevant terms, reported in Table C2.

Table C1 presents the complete keyword list organized by category. "Full boundary" indicates exact token matching; "Prefix boundary" allows trailing characters (e.g., matching both *neural network* and *neural networks*). To construct this keyword list, we first extracted 200,000 job descriptions and applied GPT to identify AI-related postings, which yielded an initial candidate set of more than 1,600 unique terms. We then manually reviewed, consolidated, and refined this candidate set, retaining only the most precise and non-redundant terms to produce the final keyword list presented below.

Table C1: AI-Related Keywords for Employee Review/LinkedIn Job Classification

Category	Keyword	Regex Pattern
Part-of-Word Fragments	ai+	ai\+
Part-of-Word Fragments	ai-	ai-
Part-of-Word Fragments	ai/	ai/
Part-of-Word Fragments	gpt-	gpt-
Part-of-Word Fragments	llm-	llm-
Part-of-Word Fragments	ml-	ml-
Part-of-Word Fragments	ml/	ml/
Part-of-Word Fragments	robot-	robot-
Part-of-Word Fragments	embedding-	embedding-
Part-of-Word Fragments	plc/	plc/
Core AI Terms	ai	\bai\b

Core AI Terms	a.i	\ba\.i\b
Core AI Terms	a.i.	\ba\.i\.\b
Core AI Terms	artificial intelligence	\bartificial intelligence\b
Core AI Terms	artificialintelligence	\bartificialintelligence\b
Core AI Terms	intelligence artificielle	\bintelligence artificielle\b
Core AI Terms	artificial general intelligence	\bartificial general intelligence\b
Core AI Terms	agi	\bagi\b
Core AI Terms	applied intelligence	\bapplied intelligence\b
Core AI Terms	augmented intelligence	\baugmented intelligence\b
Core AI Terms	computational intelligence	\bcomputational intelligence\b
Core AI Terms	machine intelligence	\bmachine intelligence\b
Core AI Terms	cognitive	\bcognitive\b
Core AI Terms	aiaas	\baiaas\b
Core AI Terms	aiml	\baiml\b
Core AI Terms	nextgenai	\bnextgenai\b
Core AI Terms	apple intelligence	\bapple intelligence\b
Generative AI	generativeai	\bgenerativeai\b
Generative AI	genai	\bgenai\b
Generative AI	gen-ai	\bgen-ai\b
Generative AI	generative design	\bgenerative design\b
Generative AI	generative media	\bgenerative media\b
Generative AI	generative models	\bgenerative models\b
Machine Learning	machine learning	\bmachine learning\b
Machine Learning	ml	\bml\b
Machine Learning	machine-learning	\bmachine-learning\b
Machine Learning	machine learning/ai	\bmachine learning/ai\b
Machine Learning	dl/ml	\bdl/ml\b
Machine Learning	deep learning	\bdeep learning\b
Machine Learning	deeplearning	\bdeeplearning\b
Machine Learning	supervised learning	\bsupervised learning\b
Machine Learning	supervised/unsupervised learning	\bsupervised/unsupervised learning\b
Machine Learning	supervised and unsupervised modeling	\bsupervised and unsupervised modeling\b
Machine Learning	supervised and unsupervised training	\bsupervised and unsupervised training\b
Machine Learning	supervised data	\bsupervised data\b
Machine Learning	unsupervised learning	\bunsupervised learning\b
Machine Learning	self-supervised learning	\bself-supervised learning\b
Machine Learning	self-supervised techniques	\bself-supervised techniques\b
Machine Learning	reinforcement learning	\breinforcement learning\b
Machine Learning	reinforcement-learning	\breinforcement-learning\b
Machine Learning	rlhf	\brlhf\b
Machine Learning	imitation learning	\bimitation learning\b
Machine Learning	active learning	\bactive learning\b
Machine Learning	few-shot learning	\bfew-shot learning\b

Machine Learning	transfer learning	\btransfer learning\b
Machine Learning	incremental learning	\bincremental learning\b
Machine Learning	real-time learning	\breal-time learning\b
Machine Learning	representation learning	\brepresentation learning\b
Machine Learning	statistical learning	\bstatistical learning\b
Machine Learning	statistical learning theory	\bstatistical learning theory\b
Machine Learning	graph-based learning	\bgraph-based learning\b
Machine Learning	multi-armed bandits	\bmulti-armed bandits\b
Machine Learning	exploration/exploitation tradeoffs	\bexploration/exploitation tradeoffs\b
Machine Learning	algorithms	\balgorithms\b
Machine Learning	fuzzy logic	\bfuzzy logic\b
Machine Learning	finite state automata	\bfinite state automata\b
Machine Learning	expert systems	\bexpert systems\b
Machine Learning	approximate inference	\bapproximate inference\b
Machine Learning	inference engine	\binference engine\b
Machine Learning	mathematical optimization	\bmathematical optimization\b
Machine Learning	optimization models	\boptimization models\b
Machine Learning	optimization techniques	\boptimization techniques\b
Machine Learning	probabilistic programming	\bprobabilistic programming\b
Machine Learning	graphical models	\bgraphical models\b
Machine Learning	mixture models	\bmixture models\b
Machine Learning	causal models	\bcausal models\b
Machine Learning	bayesian	\bbayesian\b
Neural Networks & Architectures	neural network	\bneural network\b
Neural Networks & Architectures	neural networks	\bneural networks\b
Neural Networks & Architectures	neural models	\bneural models\b
Neural Networks & Architectures	dnn	\bdnn\b
Neural Networks & Architectures	rnn	\brnn\b
Neural Networks & Architectures	rnns	\brnns\b
Neural Networks & Architectures	lstm	\blstm\b
Neural Networks & Architectures	lstms	\blstms\b
Neural Networks & Architectures	gnn	\bgnn\b
Neural Networks & Architectures	gnns	\bgnns\b
Neural Networks & Architectures	spiking neuronal network	\bspiking neuronal network\b
Neural Networks & Architectures	snns	\bsnns\b
Neural Networks & Architectures	transformer models	\btransformer models\b
Neural Networks & Architectures	transformer-based approaches	\btransformer-based approaches\b
Neural Networks & Architectures	auto-encoding models	\bauto-encoding models\b
Neural Networks & Architectures	auto-regressive models	\bauto-regressive models\b
Neural Networks & Architectures	variational autoencoders	\bvariational autoencoders\b
Neural Networks & Architectures	gans	\bgans\b
Neural Networks & Architectures	diffusion models	\bdiffusion models\b
Neural Networks & Architectures	graph-based models	\bgraph-based models\b
LLMs & Foundation Models	llm	\bllm\b
LLMs & Foundation Models	llms	\bllms\b

LLMs & Foundation Models	language model	\blanguage model\b
LLMs & Foundation Models	foundation models	\bfoundation models\b
LLMs & Foundation Models	foundational models	\bfoundational models\b
LLMs & Foundation Models	frontier models	\bfrontier models\b
LLMs & Foundation Models	frontier model capabilities	\bfrontier model capabilities\b
LLMs & Foundation Models	large-scale models	\blarge-scale models\b
LLMs & Foundation Models	large biological models	\blarge biological models\b
LLMs & Foundation Models	pre-trained models	\bpre-trained models\b
LLMs & Foundation Models	reasoning models	\breasoning models\b
LLMs & Foundation Models	vision models	\bvision models\b
LLMs & Foundation Models	video models	\bvideo models\b
LLMs & Foundation Models	gpt	\bgpt\b
LLMs & Foundation Models	chatgpt	\bchatgpt\b
LLMs & Foundation Models	claude	\bclaude\b
LLMs & Foundation Models	gemini	\bgemini\b
LLMs & Foundation Models	google gemini	\bgoogle gemini\b
LLMs & Foundation Models	bard	\bbard\b
LLMs & Foundation Models	llama	\bllama\b
LLMs & Foundation Models	mistral	\bmistral\b
LLMs & Foundation Models	deepseek	\bdeepseek\b
LLMs & Foundation Models	bert	\bbert\b
LLMs & Foundation Models	albert.ai	\balbert\.ai\b
LLMs & Foundation Models	t5	\bt5\b
LLMs & Foundation Models	blip	\bblip\b
LLMs & Foundation Models	albef	\balbef\b
LLMs & Foundation Models	eva-enormous	\beva-enormous\b
LLMs & Foundation Models	alphafold	\balphafold\b
LLMs & Foundation Models	openfold	\bopenfold\b
LLMs & Foundation Models	medpalm	\bmedpalm\b
LLMs & Foundation Models	multimodal	\bmultimodal\b
LLMs & Foundation Models	multi-modal	\bmulti-modal\b
LLMs & Foundation Models	multi-modal understanding	\bmulti-modal understanding\b
LLMs & Foundation Models	vlms	\bvlms\b
LLMs & Foundation Models	vllm	\bvllm\b
LLMs & Foundation Models	vllms	\bvllms\b
Generative AI Tools & Platforms	stable diffusion	\bstable diffusion\b
Generative AI Tools & Platforms	dalle	\bdalle\b
Generative AI Tools & Platforms	midjourney	\bmidjourney\b
Generative AI Tools & Platforms	midjourney.com	\bmidjourney\.com\b
Generative AI Tools & Platforms	sora	\bsora\b
Generative AI Tools & Platforms	veo 3	\bveo 3\b
Generative AI Tools & Platforms	image/video generation platforms	\bimage/video generation platforms\b
Generative AI Tools & Platforms	heygen	\bheygen\b
Generative AI Tools & Platforms	synthesia	\bsynthesia\b
Generative AI Tools & Platforms	synthflow	\bsynthflow\b

Generative AI Tools & Platforms	pika	\bpika\b
Generative AI Tools & Platforms	lovart	\blovart\b
Generative AI Tools & Platforms	creatify	\bcreatify\b
Generative AI Tools & Platforms	d-id	\bd-id\b
Generative AI Tools & Platforms	arcads	\barcads\b
Generative AI Tools & Platforms	jasper art	\bjasper art\b
Generative AI Tools & Platforms	jasper video	\bjasper video\b
Generative AI Tools & Platforms	canva magic studio	\bcanva magic studio\b
Generative AI Tools & Platforms	canva pro	\bcanva pro\b
Generative AI Tools & Platforms	adobe sensei	\badobe sensei\b
Generative AI Tools & Platforms	content generation	\bcontent generation\b
Generative AI Tools & Platforms	image generation	\bimage generation\b
Generative AI Tools & Platforms	video generation	\bvideo generation\b
Generative AI Tools & Platforms	text-to-video applications	\btext-to-video applications\b
Prompt Engineering	prompt engineering	\bprompt engineering\b
Prompt Engineering	prompt engineer	\bprompt engineer\b
Prompt Engineering	prompt engineers	\bprompt engineers\b
Prompt Engineering	prompting	\bprompting\b
Prompt Engineering	prompting strategies	\bprompting strategies\b
Prompt Engineering	prompts	\bprompts\b
Prompt Engineering	prompt design	\bprompt design\b
Prompt Engineering	design prompts	\bdesign prompts\b
Prompt Engineering	custom prompting	\bcustom prompting\b
Prompt Engineering	prompt chaining	\bprompt chaining\b
Prompt Engineering	prompt libraries	\bprompt libraries\b
Prompt Engineering	prompt writing	\bprompt writing\b
Prompt Engineering	prompt injection	\bprompt injection\b
Prompt Engineering	prompt-based labeling	\bprompt-based labeling\b
Prompt Engineering	training prompts	\btraining prompts\b
Prompt Engineering	chain of thought	\bchain of thought\b
Prompt Engineering	chain of reasoning	\bchain of reasoning\b
Prompt Engineering	chain-of-thought reasoning	\bchain-of-thought reasoning\b
Prompt Engineering	multi-step reasoning	\bmulti-step reasoning\b
Prompt Engineering	structured outputs	\bstructured outputs\b
Prompt Engineering	function calling	\bfunction calling\b
Prompt Engineering	instruction tuning	\binstruction tuning\b
Agentic AI & Agent Frameworks	agentic	\bagentic\b
Agentic AI & Agent Frameworks	agent framework	\bagent framework\b
Agentic AI & Agent Frameworks	agent frameworks	\bagent frameworks\b
Agentic AI & Agent Frameworks	agent reasoning patterns	\bagent reasoning patterns\b
Agentic AI & Agent Frameworks	agent workflows	\bagent workflows\b
Agentic AI & Agent Frameworks	agent-based systems	\bagent-based systems\b
Agentic AI & Agent Frameworks	orchestration	\borchestration\b
Agentic AI & Agent Frameworks	multi-agent	\bmulti-agent\b
Agentic AI & Agent Frameworks	react agents	\breact agents\b

Agentic AI & Agent Frameworks	graph agents	\bgraph agents\b
Agentic AI & Agent Frameworks	intelligent agents	\bintelligent agents\b
Agentic AI & Agent Frameworks	conversational agents	\bconversational agents\b
Agentic AI & Agent Frameworks	agentops	\bagentops\b
Agentic AI & Agent Frameworks	agentforce	\bagentforce\b
Agentic AI & Agent Frameworks	langchain	\blangchain\b
Agentic AI & Agent Frameworks	lang chain	\blang chain\b
Agentic AI & Agent Frameworks	langgraph	\blanggraph\b
Agentic AI & Agent Frameworks	llamaindex	\bllamaindex\b
Agentic AI & Agent Frameworks	autogen	\bautogen\b
Agentic AI & Agent Frameworks	crewai	\bcrewai\b
Agentic AI & Agent Frameworks	dspy	\bdspy\b
Agentic AI & Agent Frameworks	google adk	\bgoogle adk\b
Agentic AI & Agent Frameworks	mcp	\bmcp\b
Agentic AI & Agent Frameworks	mcps	\bmcps\b
Agentic AI & Agent Frameworks	model context protocol	\bmodel context protocol\b
Agentic AI & Agent Frameworks	bedrock agents	\bbedrock agents\b
Agentic AI & Agent Frameworks	human-in-the-loop	\bhuman-in-the-loop\b
RAG & Retrieval Systems	rag	\brag\b
RAG & Retrieval Systems	retrieval augmented generation	\bretrieval augmented generation\b
RAG & Retrieval Systems	retrieval-augmented generation	\bretrieval-augmented generation\b
RAG & Retrieval Systems	rag-based solutions	\brag-based solutions\b
RAG & Retrieval Systems	ragas	\bragas\b
RAG & Retrieval Systems	hybrid retrieval	\bhybrid retrieval\b
RAG & Retrieval Systems	hybrid search	\bhybrid search\b
RAG & Retrieval Systems	retrieval pipelines	\bretrieval pipelines\b
RAG & Retrieval Systems	retrieval and grounding workflows	\bretrieval and grounding workflows\b
RAG & Retrieval Systems	context construction	\bcontext construction\b
RAG & Retrieval Systems	context modeling	\bcontext modeling\b
RAG & Retrieval Systems	context pipelines	\bcontext pipelines\b
RAG & Retrieval Systems	graph-based retrieval	\bgraph-based retrieval\b
RAG & Retrieval Systems	chunking strategies	\bchunking strategies\b
RAG & Retrieval Systems	safe retrieval	\bsafe retrieval\b
RAG & Retrieval Systems	reranking	\breranking\b
RAG & Retrieval Systems	embedding	\bembedding\b
RAG & Retrieval Systems	embeddings	\bembeddings\b
RAG & Retrieval Systems	sentence-transformers	\bsentence-transformers\b
RAG & Retrieval Systems	information retrieval	\binformation retrieval\b
RAG & Retrieval Systems	intelligent search	\bintelligent search\b
RAG & Retrieval Systems	conversational search	\bconversational search\b
RAG & Retrieval Systems	semantic	\bsemantic\b
RAG & Retrieval Systems	bedrock knowledge base	\bbedrock knowledge base\b
RAG & Retrieval Systems	bedrock knowledge bases	\bbedrock knowledge bases\b
RAG & Retrieval Systems	haystack	\bhaystack\b

Vector Databases	vector database	\bvector database\b
Vector Databases	vector databases	\bvector databases\b
Vector Databases	vector db	\bvector db\b
Vector Databases	vector dbs	\bvector dbs\b
Vector Databases	vector search	\bvector search\b
Vector Databases	vector services	\bvector services\b
Vector Databases	vector stores	\bvector stores\b
Vector Databases	vectordbs	\bvectordbs\b
Vector Databases	pinecone	\bpinecone\b
Vector Databases	chroma	\bchroma\b
Vector Databases	weaviate	\bweaviate\b
Vector Databases	milvus	\bmilvus\b
Vector Databases	pgvector	\bpgvector\b
Vector Databases	qdrant	\bqdrant\b
Vector Databases	faiss	\bfaiss\b
NLP & Language Technology	nlp	\bnlp\b
NLP & Language Technology	nlp-	\bnlp-
NLP & Language Technology	natural language	\bnatural language\b
NLP & Language Technology	language technology	\blanguage technology\b
NLP & Language Technology	language engineer	\blanguage engineer\b
NLP & Language Technology	language understanding	\blanguage understanding\b
NLP & Language Technology	text analytics	\btext analytics\b
NLP & Language Technology	text mining	\btext mining\b
NLP & Language Technology	text classification	\btext classification\b
NLP & Language Technology	text extraction	\btext extraction\b
NLP & Language Technology	text generation	\btext generation\b
NLP & Language Technology	text summarization	\btext summarization\b
NLP & Language Technology	text understanding	\btext understanding\b
NLP & Language Technology	text and voice processing	\btext and voice processing\b
NLP & Language Technology	information extraction	\binformation extraction\b
NLP & Language Technology	ner	\bner\b
NLP & Language Technology	sentiment analysis	\bsentiment analysis\b
NLP & Language Technology	topic modeling	\btopic modeling\b
NLP & Language Technology	question answering	\bquestion answering\b
NLP & Language Technology	question understanding	\bquestion understanding\b
NLP & Language Technology	knowledge answering	\bknowledge answering\b
NLP & Language Technology	answer generation	\banswer generation\b
NLP & Language Technology	machine translation	\bmachine translation\b
NLP & Language Technology	computer-assistance translation	\bcomputer-assistance translation\b
NLP & Language Technology	multilinguality	\bmultilinguality\b
NLP & Language Technology	tokenization	\btokenization\b
NLP & Language Technology	lemmatization	\blemmatization\b
NLP & Language Technology	intent recognition	\bintent recognition\b
NLP & Language Technology	nl2sql	\bnl2sql\b
NLP & Language Technology	document classification	\bdocument classification\b

NLP & Language Technology	document understanding	\bdocument understanding\b
NLP & Language Technology	document imaging	\bdocument imaging\b
NLP & Language Technology	ocr	\bocr\b
NLP & Language Technology	optical character recognition	\boptical character recognition\b
NLP & Language Technology	intelligent document processing	\bintelligent document processing\b
NLP & Language Technology	corenlp	\bcorenlp\b
NLP & Language Technology	nltk	\bnltk\b
NLP & Language Technology	gensim	\bgensim\b
NLP & Language Technology	fasttext	\bfasttext\b
NLP & Language Technology	torchtext	\btorchtext\b
NLP & Language Technology	pycld2	\bpycld2\b
NLP & Language Technology	lucene	\blucene\b
NLP & Language Technology	solr	\bsolr\b
Speech & Audio	speech recognition	\bspeech recognition\b
Speech & Audio	speech analytics	\bspeech analytics\b
Speech & Audio	speech apis	\bspeech apis\b
Speech & Audio	speech processing	\bspeech processing\b
Speech & Audio	speech to text	\bspeech to text\b
Speech & Audio	speech-to-text	\bspeech-to-text\b
Speech & Audio	speech-to-speech	\bspeech-to-speech\b
Speech & Audio	real-time speech processing	\breal-time speech processing\b
Speech & Audio	audio/speech recognition	\baudio/speech recognition\b
Speech & Audio	asr	\basr\b
Speech & Audio	text to speech	\btext to speech\b
Speech & Audio	tts	\btts\b
Speech & Audio	text-to-call	\btext-to-call\b
Speech & Audio	voice recognition	\bvoice recognition\b
Speech & Audio	voice assistant	\bvoice assistant\b
Speech & Audio	voice assistants	\bvoice assistants\b
Speech & Audio	voice bots	\bvoice bots\b
Speech & Audio	voice control	\bvoice control\b
Speech & Audio	voice selection software	\bvoice selection software\b
Speech & Audio	voice transcribing	\bvoice transcribing\b
Speech & Audio	voice-activated equipment	\bvoice-activated equipment\b
Speech & Audio	voice-controlled system	\bvoice-controlled system\b
Speech & Audio	personal voice assistant	\bpersonal voice assistant\b
Speech & Audio	auto-transcriptions	\bauto-transcriptions\b
Speech & Audio	alexa	\balexa\b
Speech & Audio	vapi	\bvapi\b
Speech & Audio	dragon dictation	\bdragon dictation\b
Speech & Audio	dragon diction	\bdragon diction\b
Speech & Audio	talktype	\btalktype\b
Speech & Audio	spellex dictation	\bspellex dictation\b
Speech & Audio	lightkey	\blightkey\b
Speech & Audio	claroread	\bclaroread\b

Speech & Audio	caption.ed	\bcaption\.ed\b
Speech & Audio	call-to-text	\bcall-to-text\b
Speech & Audio	instrument recognition	\binstrument recognition\b
Speech & Audio	signal classification	\bsignal classification\b
Computer Vision & Image Processing	computer vision	\bcomputer vision\b
Computer Vision & Image Processing	computer-vision	\bcomputer-vision\b
Computer Vision & Image Processing	image analysis	\bimage analysis\b
Computer Vision & Image Processing	image and video analytics	\bimage and video analytics\b
Computer Vision & Image Processing	image classification	\bimage classification\b
Computer Vision & Image Processing	image comparison	\bimage comparison\b
Computer Vision & Image Processing	image labeling	\bimage labeling\b
Computer Vision & Image Processing	image processing	\bimage processing\b
Computer Vision & Image Processing	image recognition	\bimage recognition\b
Computer Vision & Image Processing	image registration	\bimage registration\b
Computer Vision & Image Processing	image segmentation	\bimage segmentation\b
Computer Vision & Image Processing	image-guidance technology	\bimage-guidance technology\b
Computer Vision & Image Processing	imaging recognition	\bimaging recognition\b
Computer Vision & Image Processing	imagenet	\bimagenet\b
Computer Vision & Image Processing	object classification	\bobject classification\b
Computer Vision & Image Processing	object detection	\bobject detection\b
Computer Vision & Image Processing	object recognition	\bobject recognition\b
Computer Vision & Image Processing	object tracking	\bobject tracking\b
Computer Vision & Image Processing	object-tracking	\bobject-tracking\b
Computer Vision & Image Processing	face recognition	\bface recognition\b
Computer Vision & Image Processing	facial recognition	\bfacial recognition\b
Computer Vision & Image Processing	gesture recognition	\bgesture recognition\b
Computer Vision & Image Processing	activity recognition	\bactivity recognition\b
Computer Vision & Image Processing	human movement recognition	\bhuman movement recognition\b
Computer Vision & Image Processing	liveness detection	\bliveness detection\b

Computer Vision & Image Processing	pattern recognition	\bpattern recognition\b
Computer Vision & Image Processing	recognition technology	\brecognition technology\b
Computer Vision & Image Processing	scene understanding	\bscene understanding\b
Computer Vision & Image Processing	visual target tracking	\bvisual target tracking\b
Computer Vision & Image Processing	embedded vision	\bembedded vision\b
Computer Vision & Image Processing	vision application	\bvision application\b
Computer Vision & Image Processing	vision inspection systems	\bvision inspection systems\b
Computer Vision & Image Processing	vision libraries	\bvision libraries\b
Computer Vision & Image Processing	vision software	\bvision software\b
Computer Vision & Image Processing	vision system	\bvision system\b
Computer Vision & Image Processing	vision systems	\bvision systems\b
Computer Vision & Image Processing	intelligent visions systems	\bintelligent visions systems\b
Computer Vision & Image Processing	multiple view geometry	\bmultiple view geometry\b
Computer Vision & Image Processing	point cloud processing	\bpoint cloud processing\b
Computer Vision & Image Processing	neural radiance fields	\bneural radiance fields\b
Computer Vision & Image Processing	sfm/slam	\bsfm/slam\b
Computer Vision & Image Processing	3d	\b3d\b
Computer Vision & Image Processing	opencv	\bopencv\b
Computer Vision & Image Processing	open-cv	\bopen-cv\b
Computer Vision & Image Processing	halcon	\bhalcon\b
Computer Vision & Image Processing	matrox mil	\bmatrox mil\b
Computer Vision & Image Processing	halide	\bhalide\b
Computer Vision & Image Processing	cognex	\bcognex\b
Computer Vision & Image Processing	openvino	\bopenvino\b
Computer Vision & Image Processing	deepstream	\bdeepstream\b
Computer Vision & Image Processing	tao toolkit	\btao toolkit\b
Computer Vision & Image Processing	medical image analysis	\bmedical image analysis\b
Computer Vision & Image Processing	medical imaging	\bmedical imaging\b

Computer Vision & Image Processing	av perception	\bav perception\b
Video Analytics & Generation	video analysis	\bvideo analysis\b
Video Analytics & Generation	video analytics	\bvideo analytics\b
Video Analytics & Generation	video processing	\bvideo processing\b
Video Analytics & Generation	video tagging	\bvideo tagging\b
Video Analytics & Generation	video understanding	\bvideo understanding\b
Conversational AI & Chatbots	chatbot	\bchatbot\b
Conversational AI & Chatbots	chatbots	\bchatbots\b
Conversational AI & Chatbots	chat bot	\bchat bot\b
Conversational AI & Chatbots	chat bots	\bchat bots\b
Conversational AI & Chatbots	chat-bots	\bchat-bots\b
Conversational AI & Chatbots	aichat	\baichat\b
Conversational AI & Chatbots	bots	\bbots\b
Conversational AI & Chatbots	bot interactions	\bbot interactions\b
Conversational AI & Chatbots	bot responses	\bbot responses\b
Conversational AI & Chatbots	chat completion	\bchat completion\b
Conversational AI & Chatbots	chat-based tutoring	\bchat-based tutoring\b
Conversational AI & Chatbots	conversational assistant	\bconversational assistant\b
Conversational AI & Chatbots	conversational assistants	\bconversational assistants\b
Conversational AI & Chatbots	conversational design	\bconversational design\b
Conversational AI & Chatbots	conversational designer	\bconversational designer\b
Conversational AI & Chatbots	conversational intelligence	\bconversational intelligence\b
Conversational AI & Chatbots	conversational interfaces	\bconversational interfaces\b
Conversational AI & Chatbots	conversational recruiting assistant	\bconversational recruiting assistant\b
Conversational AI & Chatbots	conversational software	\bconversational software\b
Conversational AI & Chatbots	conversational user interfaces	\bconversational user interfaces\b
Conversational AI & Chatbots	conversation design	\bconversation design\b
Conversational AI & Chatbots	conversation engines	\bconversation engines\b
Conversational AI & Chatbots	conversation intelligence	\bconversation intelligence\b
Conversational AI & Chatbots	conversation memory	\bconversation memory\b
Conversational AI & Chatbots	dialogue system	\bdialogue system\b
Conversational AI & Chatbots	dialogue systems	\bdialogue systems\b
Conversational AI & Chatbots	digital assistant	\bdigital assistant\b
Conversational AI & Chatbots	virtual agent	\bvirtual agent\b
Conversational AI & Chatbots	virtual agents	\bvirtual agents\b
Conversational AI & Chatbots	virtual assistant	\bvirtual assistant\b
Conversational AI & Chatbots	virtual assistants	\bvirtual assistants\b
Conversational AI & Chatbots	multi-turn memory	\bmulti-turn memory\b
Conversational AI & Chatbots	response generation	\bresponse generation\b
Conversational AI & Chatbots	dialogflow	\bdialogflow\b
Conversational AI & Chatbots	amazon lex	\bamazon lex\b
Conversational AI & Chatbots	oracle digital assistant	\boracle digital assistant\b
Conversational AI & Chatbots	power virtual agent	\bpower virtual agent\b

Conversational AI & Chatbots	power virtual agents	\bpower virtual agents\b
Conversational AI & Chatbots	einstein bots	\beinstein bots\b
Conversational AI & Chatbots	service cloud bots	\bservice cloud bots\b
Conversational AI & Chatbots	google ccai	\bgoogle ccai\b
Conversational AI & Chatbots	ibm watson	\bibm watson\b
Recommendation & Personalisation	recommendation algorithm	\brecommendation algorithm\b
Recommendation & Personalisation	recommendation engine	\brecommendation engine\b
Recommendation & Personalisation	recommendation engines	\brecommendation engines\b
Recommendation & Personalisation	recommendation systems	\brecommendation systems\b
Recommendation & Personalisation	recommendations platform	\brecommendations platform\b
Recommendation & Personalisation	recommender engine	\brecommender engine\b
Recommendation & Personalisation	recommender system	\brecommender system\b
Recommendation & Personalisation	recommender systems	\brecommender systems\b
Recommendation & Personalisation	content recommendation systems	\bcontent recommendation systems\b
Recommendation & Personalisation	collaborative filtering	\bcollaborative filtering\b
Recommendation & Personalisation	personalization	\bpersonalization\b
Recommendation & Personalisation	personalized ranking	\bpersonalized ranking\b
Recommendation & Personalisation	personalized recommendation	\bpersonalized recommendation\b
Recommendation & Personalisation	product recommendation	\bproduct recommendation\b
Recommendation & Personalisation	customer segmentation	\bcustomer segmentation\b
Recommendation & Personalisation	aws personalize	\baws personalize\b
Recommendation & Personalisation	predictive engagement	\bpredictive engagement\b
Recommendation & Personalisation	product categorization	\bproduct categorization\b
Data Science & Analytics	data science	\bdata science\b
Data Science & Analytics	data-science	\bdata-science\b
Data Science & Analytics	datascience	\bdatascience\b
Data Science & Analytics	data scientist	\bdata scientist\b
Data Science & Analytics	data scientists	\bdata scientists\b
Data Science & Analytics	datascientist	\bdatascientist\b
Data Science & Analytics	data privacy scientist	\bdata privacy scientist\b
Data Science & Analytics	data analysts	\bdata analysts\b
Data Science & Analytics	data analysis	\bdata analysis\b
Data Science & Analytics	data analytics	\bdata analytics\b
Data Science & Analytics	data-driven insights	\bdata-driven insights\b
Data Science & Analytics	analytics	\banalytics\b
Data Science & Analytics	advanced analytics	\badvanced analytics\b
Data Science & Analytics	advanced modeling	\badvanced modeling\b
Data Science & Analytics	business intelligence	\bbusiness intelligence\b
Data Science & Analytics	big data	\bbig data\b
Data Science & Analytics	statistical analysis	\bstatistical analysis\b
Data Science & Analytics	statistical methods	\bstatistical methods\b
Data Science & Analytics	statistical modeling	\bstatistical modeling\b
Data Science & Analytics	statistical modelling	\bstatistical modelling\b
Data Science & Analytics	statistical models	\bstatistical models\b
Data Science & Analytics	statistical shape modeling	\bstatistical shape modeling\b

Data Science & Analytics	quantitative analytical	\bquantitative analytical\b
Data Science & Analytics	predictive analysis	\bpredictive analysis\b
Data Science & Analytics	predictive analytics	\bpredictive analytics\b
Data Science & Analytics	predictive and prescriptive analytics	\bpredictive and prescriptive analytics\b
Data Science & Analytics	predictive modeling	\bpredictive modeling\b
Data Science & Analytics	predictive modelling	\bpredictive modelling\b
Data Science & Analytics	predictive models	\bpredictive models\b
Data Science & Analytics	prediction systems	\bprediction systems\b
Data Science & Analytics	prescriptive models	\bprescriptive models\b
Data Science & Analytics	forecasting	\bforecasting\b
Data Science & Analytics	forecasting models	\bforecasting models\b
Data Science & Analytics	supply-demand forecasting	\bsupply-demand forecasting\b
Data Science & Analytics	survival analysis	\bsurvival analysis\b
Data Science & Analytics	time series	\btime series\b
Data Science & Analytics	time-series	\btime-series\b
Data Science & Analytics	anomaly detection	\banomaly detection\b
Data Science & Analytics	graph analytics	\bgraph analytics\b
Data Science & Analytics	graph methods	\bgraph methods\b
Data Science & Analytics	graph models	\bgraph models\b
Data Science & Analytics	propensity modelling	\bpropensity modelling\b
Data Science & Analytics	propensity models	\bpropensity models\b
Data Science & Analytics	data mining	\bdata mining\b
Data Science & Analytics	data-mining	\bdata-mining\b
Data Science & Analytics	knowledge discovery	\bknowledge discovery\b
Data Science & Analytics	complex event processing	\bcomplex event processing\b
Data Science & Analytics	computational advertising	\bcomputational advertising\b
Data Science & Analytics	ctr	\bctr\b
Data Science & Analytics	cvr	\bcvr\b
ML Training, Fine-Tuning & Optimisation	training data	\btraining data\b
ML Training, Fine-Tuning & Optimisation	training and inference	\btraining and inference\b
ML Training, Fine-Tuning & Optimisation	model training	\bmodel training\b
ML Training, Fine-Tuning & Optimisation	distributed training	\bdistributed training\b
ML Training, Fine-Tuning & Optimisation	multi-node training	\bmulti-node training\b
ML Training, Fine-Tuning & Optimisation	multi-gpu	\bmulti-gpu\b
ML Training, Fine-Tuning & Optimisation	mixed precision	\bmixed precision\b
ML Training, Fine-Tuning & Optimisation	pretraining	\bpretraining\b
ML Training, Fine-Tuning & Optimisation	fine tuning	\bfine tuning\b
ML Training, Fine-Tuning & Optimisation	fine-tune	\bfine-tune\b

ML Training, Fine-Tuning & Optimisation	fine-tuning	\bfine-tuning\b
ML Training, Fine-Tuning & Optimisation	finetune	\bfinetune\b
ML Training, Fine-Tuning & Optimisation	preference alignment	\bpreference alignment\b
ML Training, Fine-Tuning & Optimisation	hyperparameter tuning	\bhyperparameter tuning\b
ML Training, Fine-Tuning & Optimisation	hyperparameters	\bhyperparameters\b
ML Training, Fine-Tuning & Optimisation	automl	\bautoml\b
ML Training, Fine-Tuning & Optimisation	feature engineering	\bfeature engineering\b
ML Training, Fine-Tuning & Optimisation	feature extraction	\bfeature extraction\b
ML Training, Fine-Tuning & Optimisation	feature selection	\bfeature selection\b
ML Training, Fine-Tuning & Optimisation	feature store	\bfeature store\b
ML Training, Fine-Tuning & Optimisation	data augmentation	\bdata augmentation\b
ML Training, Fine-Tuning & Optimisation	synthetic datasets	\bsynthetic datasets\b
ML Training, Fine-Tuning & Optimisation	retraining	\bretraining\b
ML Training, Fine-Tuning & Optimisation	tuning models	\btuning models\b
ML Training, Fine-Tuning & Optimisation	model compression	\bmodel compression\b
ML Training, Fine-Tuning & Optimisation	model quantization	\bmodel quantization\b
ML Training, Fine-Tuning & Optimisation	quantization	\bquantization\b
ML Training, Fine-Tuning & Optimisation	model optimization	\bmodel optimization\b
MLOps & Model Lifecycle	mlops	\bmlops\b
MLOps & Model Lifecycle	llmops	\bllmops\b
MLOps & Model Lifecycle	dataops	\bdataops\b
MLOps & Model Lifecycle	aiops	\baiops\b
MLOps & Model Lifecycle	model deployment	\bmodel deployment\b
MLOps & Model Lifecycle	models deployment	\bmodels deployment\b
MLOps & Model Lifecycle	model serving	\bmodel serving\b
MLOps & Model Lifecycle	model-serving apis	\bmodel-serving apis\b
MLOps & Model Lifecycle	serving patterns	\bserving patterns\b
MLOps & Model Lifecycle	model inference	\bmodel inference\b
MLOps & Model Lifecycle	edge deployment	\bedge deployment\b
MLOps & Model Lifecycle	model monitoring	\bmodel monitoring\b
MLOps & Model Lifecycle	model evaluation	\bmodel evaluation\b
MLOps & Model Lifecycle	model validation	\bmodel validation\b
MLOps & Model Lifecycle	model governance	\bmodel governance\b
MLOps & Model Lifecycle	model registry	\bmodel registry\b

MLOps & Model Lifecycle	model management	\bmodel management\b
MLOps & Model Lifecycle	model life cycle management	\bmodel life cycle management\b
MLOps & Model Lifecycle	model operations	\bmodel operations\b
MLOps & Model Lifecycle	model platform	\bmodel platform\b
MLOps & Model Lifecycle	model development	\bmodel development\b
MLOps & Model Lifecycle	model implementation	\bmodel implementation\b
MLOps & Model Lifecycle	model selection	\bmodel selection\b
MLOps & Model Lifecycle	model reliability	\bmodel reliability\b
MLOps & Model Lifecycle	model robustness	\bmodel robustness\b
MLOps & Model Lifecycle	experiment tracking	\bexperiment tracking\b
MLOps & Model Lifecycle	drift detection	\bdrift detection\b
MLOps & Model Lifecycle	drift monitoring	\bdrift monitoring\b
MLOps & Model Lifecycle	drift/quality monitoring	\bdrift/quality monitoring\b
MLOps & Model Lifecycle	data drift	\bdata drift\b
MLOps & Model Lifecycle	model drift	\bmodel drift\b
MLOps & Model Lifecycle	uncertainty quantification	\buncertainty quantification\b
MLOps & Model Lifecycle	bias handling	\bbias handling\b
MLOps & Model Lifecycle	bias mitigation	\bbias mitigation\b
MLOps & Model Lifecycle	bias testing	\bbias testing\b
MLOps & Model Lifecycle	red-teaming	\bred-teaming\b
MLOps & Model Lifecycle	adversarial attacks	\badversarial attacks\b
MLOps & Model Lifecycle	model poisoning	\bmodel poisoning\b
MLOps & Model Lifecycle	hallucination	\bhallucination\b
MLOps & Model Lifecycle	guardrails	\bguardrails\b
MLOps & Model Lifecycle	robust intelligence	\brobust intelligence\b
MLOps & Model Lifecycle	concept explainability	\bconcept explainability\b
MLOps & Model Lifecycle	shap	\bshap\b
MLOps & Model Lifecycle	real-time monitoring	\breal-time monitoring\b
MLOps & Model Lifecycle	condition monitoring	\bcondition monitoring\b
Data Engineering & Infrastructure	data engineer	\bdata engineer\b
Data Engineering & Infrastructure	data engineering	\bdata engineering\b
Data Engineering & Infrastructure	data platform	\bdata platform\b
Data Engineering & Infrastructure	data preprocessing	\bdata preprocessing\b
Data Engineering & Infrastructure	data cleaning	\bdata cleaning\b
Data Engineering & Infrastructure	data annotation	\bdata annotation\b
Data Engineering & Infrastructure	data labeling	\bdata labeling\b
Data Engineering & Infrastructure	data labelling	\bdata labelling\b
Data Engineering & Infrastructure	data collection and labeling	\bdata collection and labeling\b
Data Engineering & Infrastructure	annotation	\bannotation\b
Data Engineering & Infrastructure	data acquisition systems	\bdata acquisition systems\b
Data Engineering & Infrastructure	data fusion	\bdata fusion\b
Data Engineering & Infrastructure	data lineage	\bdata lineage\b
Data Engineering & Infrastructure	data governance	\bdata governance\b
Data Engineering & Infrastructure	data bricks	\bdata bricks\b
Data Engineering & Infrastructure	databricks	\bdatabricks\b

Data Engineering & Infrastructure	dataiku	\bdataiku\b
Data Engineering & Infrastructure	datarobot	\bdatarobot\b
Data Engineering & Infrastructure	process mining	\bprocess mining\b
Data Engineering & Infrastructure	task mining	\btask mining\b
Data Engineering & Infrastructure	sql	\bsql\b
Data Engineering & Infrastructure	sparql	\bsparql\b
Data Engineering & Infrastructure	pyspark	\bpyspark\b
Data Engineering & Infrastructure	spark-mllib	\bspark-mllib\b
Data Engineering & Infrastructure	sparkml	\bsparkml\b
Data Engineering & Infrastructure	apache mahout	\bapache mahout\b
Data Engineering & Infrastructure	kedro	\bkedro\b
Data Engineering & Infrastructure	kubeflow	\bkubeflow\b
Data Engineering & Infrastructure	kubeflow pipelines	\bkubeflow pipelines\b
Data Engineering & Infrastructure	mlflow	\bmlflow\b
Data Engineering & Infrastructure	mleap	\bmleap\b
Data Engineering & Infrastructure	tensorflow extended	\btensorflow extended\b
Data Engineering & Infrastructure	tfx	\btfx\b
Data Engineering & Infrastructure	cnvrg	\bcnvrg\b
Data Engineering & Infrastructure	biometric data	\bbiometric data\b
ML Frameworks & Libraries	tensorflow	\btensorflow\b
ML Frameworks & Libraries	tensor flow	\btensor flow\b
ML Frameworks & Libraries	tensorflow/	\btensorflow/
ML Frameworks & Libraries	tflite	\btflite\b
ML Frameworks & Libraries	pytorch	\bpytorch\b
ML Frameworks & Libraries	keras	\bkeras\b
ML Frameworks & Libraries	scikit	\bscikit\b
ML Frameworks & Libraries	scikit-	\bscikit-
ML Frameworks & Libraries	sci-kit learn	\bsci-kit learn\b
ML Frameworks & Libraries	sci-kit-learn	\bsci-kit-learn\b
ML Frameworks & Libraries	caffe	\bcaffe\b
ML Frameworks & Libraries	caffe2	\bcaffe2\b
ML Frameworks & Libraries	mxnet	\bmxnet\b
ML Frameworks & Libraries	theano	\btheano\b
ML Frameworks & Libraries	fastai	\bfastai\b
ML Frameworks & Libraries	xgboost	\bxgboost\b
ML Frameworks & Libraries	xg boost	\bxg boost\b
ML Frameworks & Libraries	light gbm	\blight gbm\b
ML Frameworks & Libraries	weka	\bweka\b
ML Frameworks & Libraries	libsvm	\blibsvm\b
ML Frameworks & Libraries	vowpal wabbit	\bvowpal wabbit\b
ML Frameworks & Libraries	cluto	\bcluto\b
ML Frameworks & Libraries	pymc	\bpymc\b
ML Frameworks & Libraries	nengo	\bnengo\b
ML Frameworks & Libraries	deepspeed	\bdeepspeed\b
ML Frameworks & Libraries	openllm	\bopenllm\b

ML Frameworks & Libraries	ollama	\bollama\b
ML Frameworks & Libraries	bentoml	\bbentoml\b
ML Frameworks & Libraries	onnx	\bonnx\b
ML Frameworks & Libraries	tensorrt	\btensorrt\b
ML Frameworks & Libraries	sglang	\bsglang\b
ML Frameworks & Libraries	langfuse	\blangfuse\b
ML Frameworks & Libraries	trulens	\btrulens\b
ML Frameworks & Libraries	arize phoenix	\barize phoenix\b
ML Frameworks & Libraries	python	\bpython\b
ML Frameworks & Libraries	matlab	\bmatlab\b
ML Frameworks & Libraries	matlab simulink	\bmatlab simulink\b
ML Frameworks & Libraries	hugging face	\bhugging face\b
ML Frameworks & Libraries	huggingface	\bhuggingface\b
GPU & HPC Computing	gpu	\bgpu\b
GPU & HPC Computing	gpus	\bgpus\b
GPU & HPC Computing	gpu applications	\bgpu applications\b
GPU & HPC Computing	gpu computing	\bgpu computing\b
GPU & HPC Computing	gpnpu	\bgpnpu\b
GPU & HPC Computing	npus	\bnpus\b
GPU & HPC Computing	neural processing unit	\bneural processing unit\b
GPU & HPC Computing	cuda	\bcuda\b
GPU & HPC Computing	cudnn	\bcudnn\b
GPU & HPC Computing	rocm	\brocm\b
GPU & HPC Computing	cloud tpu	\bcloud tpu\b
GPU & HPC Computing	hpc	\bhpc\b
GPU & HPC Computing	high performance computing	\bhigh performance computing\b
Cloud AI Platforms & Services	cloud	\bcloud\b
Cloud AI Platforms & Services	azure	\bazure\b
Cloud AI Platforms & Services	amazon bedrock	\bamazon bedrock\b
Cloud AI Platforms & Services	aws bedrock	\baws bedrock\b
Cloud AI Platforms & Services	bedrock	\bbedrock\b
Cloud AI Platforms & Services	amazon sagemaker	\bamazon sagemaker\b
Cloud AI Platforms & Services	aws sagemaker	\baws sagemaker\b
Cloud AI Platforms & Services	sagemaker	\bsagemaker\b
Cloud AI Platforms & Services	sage maker	\bsage maker\b
Cloud AI Platforms & Services	amazon textract	\bamazon textract\b
Cloud AI Platforms & Services	textract	\btextract\b
Cloud AI Platforms & Services	amazon dash cart	\bamazon dash cart\b
Cloud AI Platforms & Services	aws codewhisper	\baws codewhisper\b
Cloud AI Platforms & Services	google sge	\bgoogle sge\b
Cloud AI Platforms & Services	ibm odm	\bibm odm\b
Cloud AI Platforms & Services	microsoft dax	\bmicrosoft dax\b
Cloud AI Platforms & Services	microsoft power platform	\bmicrosoft power platform\b
Cloud AI Platforms & Services	power apps	\bpower apps\b
Cloud AI Platforms & Services	power platform	\bpower platform\b

Cloud AI Platforms & Services	einstein 1 platform	\beinstein 1 platform\b
Cloud AI Platforms & Services	metropolis microservices	\bmetropolis microservices\b
Cloud AI Platforms & Services	ray.io	\bray\.io\b
Cloud AI Platforms & Services	openai	\bopenai\b
Cloud AI Platforms & Services	deepmind	\bdeepmind\b
Cloud AI Platforms & Services	cortexai	\bcortexai\b
Cloud AI Platforms & Services	hyperscience	\bhyperscience\b
Cloud AI Platforms & Services	a360	\ba360\b
Robotics & Autonomous Systems	robot	\brobot\b
Robotics & Autonomous Systems	robotic	\brobotic\b
Robotics & Autonomous Systems	robotisation	\brobotisation\b
Robotics & Autonomous Systems	robots	\brobots\b
Robotics & Autonomous Systems	cobots	\bcobots\b
Robotics & Autonomous Systems	cobotics	\bcobotics\b
Robotics & Autonomous Systems	ro/cobots	\bro/cobots\b
Robotics & Autonomous Systems	autonomous	\bautonomous\b
Robotics & Autonomous Systems	autonomy	\bautonomy\b
Robotics & Autonomous Systems	inverse kinematic	\binverse kinematic\b
Robotics & Autonomous Systems	end-of-arm tooling	\bend-of-arm tooling\b
Robotics & Autonomous Systems	end of arm tooling	\bend of arm tooling\b
Robotics & Autonomous Systems	servo	\bservo\b
Robotics & Autonomous Systems	servo driven manipulators	\bservo driven manipulators\b
Robotics & Autonomous Systems	fanuc	\bfanuc\b
Robotics & Autonomous Systems	kuka	\bkuka\b
Robotics & Autonomous Systems	ardupilot	\bardupilot\b
Robotics & Autonomous Systems	ros	\bros\b
Robotics & Autonomous Systems	robocorp	\brobocorp\b
Robotics & Autonomous Systems	motion planning	\bmotion planning\b
Robotics & Autonomous Systems	trajectory planning	\btrajectory planning\b
Robotics & Autonomous Systems	path planning	\bpath planning\b
Robotics & Autonomous Systems	obstacle avoidance	\bobstacle avoidance\b
Robotics & Autonomous Systems	da vinci	\bda vinci\b
Robotics & Autonomous Systems	da vinci surgical system	\bda vinci surgical system\b
Robotics & Autonomous Systems	davinci surgical system	\bdavinci surgical system\b
Robotics & Autonomous Systems	davincir surgical system	\bdavincir surgical system\b
Robotics & Autonomous Systems	digital surgery	\bdigital surgery\b
Robotics & Autonomous Systems	cyberknife	\bcyberknife\b
Robotics & Autonomous Systems	robo-advising	\brobo-advising\b
Autonomous Vehicles & ADAS	adas	\badas\b
Autonomous Vehicles & ADAS	adas/ads	\badas/ads\b
Autonomous Vehicles & ADAS	ads	\bads\b
Autonomous Vehicles & ADAS	self-driving	\bself-driving\b
Autonomous Vehicles & ADAS	driverless cars	\bdriverless cars\b
Autonomous Vehicles & ADAS	driverless fork trucks	\bdriverless fork trucks\b
Autonomous Vehicles & ADAS	av planning and navigation	\bav planning and navigation\b

Autonomous Vehicles & ADAS	av sensor and localization	\bav sensor and localization\b
Autonomous Vehicles & ADAS	av simulators	\bav simulators\b
Autonomous Vehicles & ADAS	av software	\bav software\b
Autonomous Vehicles & ADAS	active parking assist	\bactive parking assist\b
Autonomous Vehicles & ADAS	active safety	\bactive safety\b
Autonomous Vehicles & ADAS	driver assistance	\bdriver assistance\b
Autonomous Vehicles & ADAS	traffic jam assist	\btraffic jam assist\b
Autonomous Vehicles & ADAS	lane detection	\blane detection\b
Autonomous Vehicles & ADAS	lidar	\blidar\b
Autonomous Vehicles & ADAS	camera sensors	\bcamera sensors\b
Autonomous Vehicles & ADAS	sensor fusion	\bsensor fusion\b
Autonomous Vehicles & ADAS	ekf	\bekf\b
Autonomous Vehicles & ADAS	extended kalman filters	\bextended kalman filters\b
Autonomous Vehicles & ADAS	kalman filters	\bkalman filters\b
Autonomous Vehicles & ADAS	simultaneous localization	\bsimultaneous localization\b
Autonomous Vehicles & ADAS	slam	\bslam\b
Autonomous Vehicles & ADAS	environment model	\benvironment model\b
Autonomous Vehicles & ADAS	simulation-based validation	\bsimulation-based validation\b
Autonomous Vehicles & ADAS	simulated environments	\bsimulated environments\b
Autonomous Vehicles & ADAS	intelligent transport system	\bintelligent transport system\b
Autonomous Vehicles & ADAS	smart mobility	\bsmart mobility\b
Autonomous Vehicles & ADAS	intelligent mobility	\bintelligent mobility\b
Autonomous Vehicles & ADAS	guided vehicles	\bguided vehicles\b
Autonomous Vehicles & ADAS	laser guided vehicles	\blaser guided vehicles\b
Autonomous Vehicles & ADAS	asrs systems	\basrs systems\b
Autonomous Vehicles & ADAS	automine	\bautomine\b
UAVs & Drones	drone	\bdrone\b
UAVs & Drones	drones	\bdrones\b
UAVs & Drones	uav	\buav\b
UAVs & Drones	uavs	\buavs\b
UAVs & Drones	uav engineer	\buav engineer\b
UAVs & Drones	uav/uas	\buav/uas\b
UAVs & Drones	tuav	\btuav\b
UAVs & Drones	multi-rotor uav	\bmulti-rotor uav\b
UAVs & Drones	rov	\brov\b
UAVs & Drones	remotely operated vehicles	\bremotely operated vehicles\b
UAVs & Drones	remote operation	\bremote operation\b
UAVs & Drones	unmanned	\bunmanned\b
Automation & RPA	automation	\bautomation\b
Automation & RPA	automate	\bautomate\b
Automation & RPA	automatic	\bautomatic\b
Automation & RPA	automating	\bautomating\b
Automation & RPA	rpa	\brpa\b
Automation & RPA	digital workforce	\bdigital workforce\b
Automation & RPA	uipath	\buipath\b

Automation & RPA	ui path	\bui path\b
Automation & RPA	blue prism	\bblue prism\b
Automation & RPA	blueprism	\bblueprim\b
Automation & RPA	ansible	\bansible\b
Automation & RPA	selenium	\bselenium\b
Automation & RPA	camunda	\bcamunda\b
Automation & RPA	jbpm	\bjbpm\b
Automation & RPA	cofax	\bcofax\b
Automation & RPA	kofax	\bkofax\b
Automation & RPA	rule engine	\brule engine\b
Automation & RPA	decision table	\bdecision table\b
Automation & RPA	operational decision management	\boperational decision management\b
Automation & RPA	just walk out technology	\bjust walk out technology\b
Industrial Automation & Control Systems	plc control systems	\bplc control systems\b
Industrial Automation & Control Systems	plc controlled i/o	\bplc controlled i/o\b
Industrial Automation & Control Systems	plc logic	\bplc logic\b
Industrial Automation & Control Systems	plc programming	\bplc programming\b
Industrial Automation & Control Systems	plcs	\bplcs\b
Industrial Automation & Control Systems	programmable controllers	\bprogrammable controllers\b
Industrial Automation & Control Systems	programmable logic controller	\bprogrammable logic controller\b
Industrial Automation & Control Systems	programmable logic controllers	\bprogrammable logic controllers\b
Industrial Automation & Control Systems	rslogix	\brslogix\b
Industrial Automation & Control Systems	ladder logic	\bladder logic\b
Industrial Automation & Control Systems	scada	\bscada\b
Industrial Automation & Control Systems	dcs	\bdcs\b
Industrial Automation & Control Systems	hmi	\bhmi\b
Industrial Automation & Control Systems	hmi's	\bhmi's\b
Industrial Automation & Control Systems	hil	\bhil\b
Industrial Automation & Control Systems	factorytalk	\bfactorytalk\b
Industrial Automation & Control Systems	control logic	\bcontrol logic\b
Industrial Automation & Control Systems	control system upgrade	\bcontrol system upgrade\b
Industrial Automation & Control Systems	control technology	\bcontrol technology\b
Industrial Automation & Control Systems	supervisory control	\bsupervisory control\b

Industrial Automation & Control Systems	intelligent control strategies	\bintelligent control strategies\b
Industrial Automation & Control Systems	controls engineer	\bcontrols engineer\b
Industrial Automation & Control Systems	instrumentation & controls	\binstrumentation & controls\b
Industrial Automation & Control Systems	pid tuning	\bpid tuning\b
Industrial Automation & Control Systems	vfd	\bvfd\b
Industrial Automation & Control Systems	vfds	\bvfds\b
Industrial Automation & Control Systems	mechatronics	\bmechatronics\b
Industrial Automation & Control Systems	mechatronic systems	\bmechatronic systems\b
Industrial Automation & Control Systems	mechatronics concepts	\bmechatronics concepts\b
Industrial Automation & Control Systems	energy management systems	\benergy management systems\b
Industrial Automation & Control Systems	predictive maintenance	\bpredictive maintenance\b
Industrial Automation & Control Systems	industry 3.0	\bindustry 3\.0\b
Industrial Automation & Control Systems	industry 4.0	\bindustry 4\.0\b
Industrial Automation & Control Systems	iiot	\biiot\b
Industrial Automation & Control Systems	iot	\biot\b
Industrial Automation & Control Systems	aiot	\baiot\b
Knowledge Representation & Ontologies	knowledge graph	\bknowledge graph\b
Knowledge Representation & Ontologies	knowledge graphs	\bknowledge graphs\b
Knowledge Representation & Ontologies	knowledge modeling	\bknowledge modeling\b
Knowledge Representation & Ontologies	knowledge representation	\bknowledge representation\b
Knowledge Representation & Ontologies	ontologies	\bontologies\b
Knowledge Representation & Ontologies	ontology construction	\bontology construction\b
Knowledge Representation & Ontologies	taxonomies	\btaxonomies\b
Knowledge Representation & Ontologies	controlled vocabularies	\bcontrolled vocabularies\b
Knowledge Representation & Ontologies	skos	\bskos\b
Knowledge Representation & Ontologies	rdfs	\brdfs\b
Knowledge Representation & Ontologies	krr	\bkrr\b
Biometrics & Security AI	biometric services	\bbiometric services\b

Biometrics & Security AI	biometric system	\bbiometric system\b
Biometrics & Security AI	ecommerce fraud detection	\becommerce fraud detection\b
Biometrics & Security AI	self-harm detection	\bself-harm detection\b
Biometrics & Security AI	machine-based moderation	\bmachine-based moderation\b
Copilots & AI Productivity Tools	copilot	\bcopilot\b
Copilots & AI Productivity Tools	co pilot	\bco pilot\b
Copilots & AI Productivity Tools	co-pilot	\bco-pilot\b
Copilots & AI Productivity Tools	cursor	\bcursor\b
Copilots & AI Productivity Tools	coderabbit	\bcoderabbit\b
Copilots & AI Productivity Tools	codium	\bcodium\b
Copilots & AI Productivity Tools	code generation	\bcode generation\b
Augmented & Mixed Reality	augmented reality	\baugmented reality\b
Augmented & Mixed Reality	human-machine interaction	\bhuman-machine interaction\b
Adaptive & Personalised Learning	adaptive learning solutions	\badaptive learning solutions\b
Adaptive & Personalised Learning	adaptive learning systems	\badaptive learning systems\b
Adaptive & Personalised Learning	teacherai	\bteacherai\b
Adaptive & Personalised Learning	findfill.ai	\bfindfill\.ai\b
Adaptive & Personalised Learning	nanobana	\bnanobana\b
Research & Academic	aaai	\baaai\b
Research & Academic	icml	\bicml\b
Research & Academic	ijcai	\bijcai\b
Research & Academic	emnlp	\bemnlp\b
Research & Academic	machine reasoning	\bmachine reasoning\b
Research & Academic	machine perception	\bmachine perception\b
Research & Academic	gpr	\bgpr\b
Miscellaneous / Cross-Domain	agri-tech	\bagri-tech\b
Miscellaneous / Cross-Domain	intelligent customer service technologies	\bintelligent customer service technologies\b
Miscellaneous / Cross-Domain	predictive intelligence	\bpredictive intelligence\b
Miscellaneous / Cross-Domain	networked systems	\bnetworked systems\b
Miscellaneous / Cross-Domain	technology assisted review	\btechnology assisted review\b
Miscellaneous / Cross-Domain	mtpe	\bmtpe\b
Miscellaneous / Cross-Domain	ipa	\bipa\b
Miscellaneous / Cross-Domain	aoi	\baoi\b
Miscellaneous / Cross-Domain	solution engineer	\bsolution engineer\b
Miscellaneous / Cross-Domain	mle	\bmle\b

Table C2 presents the keyword list derived from conference call transcripts. Keywords were identified through a two-step process: GPT-5 was first prompted to read all 2025 conference call files and flag sentences containing AI-related discourse; the resulting candidate terms were then manually reviewed and consolidated into a final list. Compared to the Glassdoor keyword set in Table A1, the conference call keywords tend to reflect higher-level strategic and financial language alongside technical terminology, consistent with the investor-facing nature of earnings call communications.

Table C2: AI-Related Keywords for Conference Call AI Content

Category	Keyword	Regex Pattern
Part-of-Word Fragments	ai+	\bai\+
Part-of-Word Fragments	ai-	\bai\-
Part-of-Word Fragments	ai/	\bai/
Part-of-Word Fragments	gpt-	\bgpt\-
Part-of-Word Fragments	gpt4	\bgpt4\b
Part-of-Word Fragments	gpt40	\bgpt40\b
Part-of-Word Fragments	gpt5	\bgpt5\b
Part-of-Word Fragments	gpts	\bgpts\b
Part-of-Word Fragments	llm-	\bllm\-
Part-of-Word Fragments	llm/	\bllm/
Part-of-Word Fragments	ml-	\bml\-
Part-of-Word Fragments	ml/	\bml/
Part-of-Word Fragments	gen-	\bgen\-
Part-of-Word Fragments	gemini-	\bgemini\-
Part-of-Word Fragments	llama-	\bllama\-
Part-of-Word Fragments	nlp-	\bnlp\-
Part-of-Word Fragments	large-language	\blarge\-language\b
Core AI	ai	\bai\b
Core AI	a i	\ba\ i\b
Core AI	.ai	\.ai\b
Core AI	ai2	\bai2\b
Core AI	ai360	\bai360\b
Core AI	aigc	\baigc\b
Core AI	aiml	\baiml\b
Core AI	aiops	\baiops\b
Core AI	aiot	\baiot\b
Core AI	ai's	\bai's\b
Core AI	aiwx	\baiwx\b
Core AI	artificial intelligence	\bartificial\ intelligence\b
Core AI	artificial general intelligence	\bartificial\ general\ intelligence\b
Core AI	artificial neural	\bartificial\ neural\b
Core AI	artificial vision	\bartificial\ vision\b
Core AI	augmented intelligence	\baugmented\ intelligence\b
Core AI	machine intelligence	\bmachine\ intelligence\b

Core AI	agi	\bagi\b
Core AI	asi	\basi\b
Core AI	xai	\bxai\b
Core AI	digital-ai	\bdigital\-ai\b
Core AI	edge-ai	\bedge\-ai\b
Core AI	pre-ai	\bpre\-ai\b
Core AI	hpc/ai	\bhpc/ai\b
Generative AI	genai	\bgenai\b
Generative AI	genai's	\bgenai's\b
Generative AI	generative-ai	\bgenerative\-ai\b
Generative AI	generative search	\bgenerative\ search\b
Generative AI	generative adversarial networks	\bgenerative\ adversarial\ networks\b
Generative AI	diffusion model	\bdiffusion\ model\b
Generative AI	diffusion models	\bdiffusion\ models\b
Generative AI	stable diffusion	\bstable\ diffusion\b
Generative AI	image generation	\bimage\ generation\b
Generative AI	text generation	\btext\ generation\b
Generative AI	text to image	\btext\ to\ image\b
Generative AI	text to speech	\btext\ to\ speech\b
Generative AI	text to video	\btext\ to\ video\b
Generative AI	text-to-speech	\btext\-to\-speech\b
Generative AI	tts	\btts\b
Generative AI	dall-e	\bdall\-e\b
Generative AI	midjourney	\bmidjourney\b
Generative AI	mid journey	\bmid\ journey\b
Generative AI	firefly	\bfirefly\b
Large Language Models	llm	\bllm\b
Large Language Models	llms	\bllms\b
Large Language Models	llm's	\bllm's\b
Large Language Models	llmops	\bllmops\b
Large Language Models	ll ops	\bll\ ops\b
Large Language Models	large language model	\blarge\ language\ model\b
Large Language Models	large foundation models	\blarge\ foundation\ models\b
Large Language Models	large model	\blarge\ model\b
Large Language Models	large models	\blarge\ models\b
Large Language Models	long language models	\blong\ language\ models\b
Large Language Models	open language models	\bopen\ language\ models\b
Large Language Models	small language model	\bsmall\ language\ model\b
Large Language Models	small language models	\bsmall\ language\ models\b
Large Language Models	big model	\bbig\ model\b
Large Language Models	foundation model	\bfoundation\ model\b
Large Language Models	foundation models	\bfoundation\ models\b
Large Language Models	foundational model	\bfoundational\ model\b
Large Language Models	foundational models	\bfoundational\ models\b
Large Language Models	language model	\blanguage\ model\b

Large Language Models	language modeling	\blanguage\ modeling\b
Large Language Models	language models	\blanguage\ models\b
Specific LLM Models	gpt	\bgpt\b
Specific LLM Models	chatgpt	\bchatgpt\b
Specific LLM Models	chatgpts	\bchatgpts\b
Specific LLM Models	chat gpt	\bchat\ gpt\b
Specific LLM Models	chat gpt4	\bchat\ gpt4\b
Specific LLM Models	claude	\bclaude\b
Specific LLM Models	gemini	\bgemini\b
Specific LLM Models	geminis	\bgeminis\b
Specific LLM Models	google gemini	\bgoogle\ gemini\b
Specific LLM Models	bard	\bbard\b
Specific LLM Models	llama	\bllama\b
Specific LLM Models	mistral	\bmistral\b
Specific LLM Models	deepseek	\bdeepseek\b
Specific LLM Models	deepseek-r1	\bdeepseek\-r1\b
Specific LLM Models	deepseek-v2	\bdeepseek\-v2\b
Specific LLM Models	deepseeks	\bdeepseeks\b
Specific LLM Models	deep seek	\bdeep\ seek\b
Specific LLM Models	qwen	\bqwen\b
Specific LLM Models	qwen3	\bqwen3\b
Specific LLM Models	alibaba qwen	\balibaba\ qwen\b
Specific LLM Models	tongyi qwen	\btongyi\ qwen\b
Specific LLM Models	grok	\bgrok\b
Specific LLM Models	autogpt	\bautogpt\b
Specific LLM Models	searchgpt	\bsearchgpt\b
Specific LLM Models	perplexity	\bperplexity\b
Specific LLM Models	perplexity pro	\bperplexity\ pro\b
Specific LLM Models	h2o	\bh2o\b
AI Companies & Products	openai	\bopenai\b
AI Companies & Products	openais	\bopenais\b
AI Companies & Products	openai's	\bopenai's\b
AI Companies & Products	anthropic	\banthropic\b
AI Companies & Products	hugging face	\bhugging\ face\b
AI Companies & Products	huggingface	\bhuggingface\b
AI Companies & Products	cohere	\bcohere\b
AI Companies & Products	ibm watson	\bibm\ watson\b
AI Companies & Products	alexa	\balexa\b
AI Companies & Products	apple intelligence	\bapple\ intelligence\b
AI Companies & Products	jasper	\bjasper\b
AI Companies & Products	sagemaker	\bsagemaker\b
AI Companies & Products	bedrock	\bbedrock\b
AI Companies & Products	neuralink	\bneuralink\b
AI Companies & Products	altitudeai	\baltitudeai\b
AI Companies & Products	fastwebai	\bfastwebai\b

AI Companies & Products	sahabat-ai	\bsahabat\-ai\b
AI Companies & Products	solist-ai	\bsolist\-ai\b
AI Companies & Products	tulsa-ai	\btulsa\-ai\b
AI Companies & Products	epic-ai	\bepic\-ai\b
AI Companies & Products	winstonai	\bwinstonai\b
AI Companies & Products	totalai	\btotalai\b
AI Companies & Products	pathai	\bpathai\b
AI Companies & Products	cashai	\bcashai\b
AI Companies & Products	cemai	\bcemai\b
AI Companies & Products	captionai	\bcaptionai\b
AI Companies & Products	opsgpt	\bopsgpt\b
AI Companies & Products	bondgpt	\bbondgpt\b
AI Companies & Products	liongpt	\bliongpt\b
AI Companies & Products	maxigpt	\bmaxigpt\b
AI Companies & Products	plangpt	\bplangpt\b
AI Companies & Products	treasurygpt	\btreasurygpt\b
AI Companies & Products	audiencegpt	\baudiencegpt\b
AI Companies & Products	zoom virtual agent	\bzoom\ virtual\ agent\b
Copilot & Assistant Products	copilot	\bcopilot\b
Copilot & Assistant Products	co-pilot	\bco\-pilot\b
Copilot & Assistant Products	co pilot	\bco\ pilot\b
Copilot & Assistant Products	copilot+	\bcopilot\+
Copilot & Assistant Products	copilots	\bcopilots\b
Copilot & Assistant Products	github copilot	\bgithub\ copilot\b
Copilot & Assistant Products	github's copilot	\bgithub's\ copilot\b
Copilot & Assistant Products	microsoft copilot	\bmicrosoft\ copilot\b
Copilot & Assistant Products	microsoft copilots	\bmicrosoft\ copilots\b
Copilot & Assistant Products	microsoft 365 copilot	\bmicrosoft\ 365\ copilot\b
Copilot & Assistant Products	office copilot	\boffice\ copilot\b
Copilot & Assistant Products	teams copilot	\bteams\ copilot\b
Copilot & Assistant Products	intelligent assistant interface	\bintelligent\ assistant\ interface\b
Copilot & Assistant Products	intelligent virtual agent	\bintelligent\ virtual\ agent\b
Copilot & Assistant Products	virtual agents	\bvirtual\ agents\b
Machine Learning	ml	\bml\b
Machine Learning	machine learning	\bmachine\ learning\b
Machine Learning	machine learning-	\bmachine\ learning\-

Machine Learning	machine learnings	\bmachine\ learnings\b
Machine Learning	machine-learning	\bmachine\-learning\b
Machine Learning	mlops	\bmlops\b
Machine Learning	mlsecops	\bmlsecops\b
Machine Learning	jfrog ml	\bjfrog\ ml\b
Machine Learning	deep learning	\bdeep\ learning\b
Machine Learning	deep learning-based	\bdeep\ learning\-based\b
Machine Learning	deep-learning	\bdeep\-learning\b
Machine Learning	supervised learning	\bsupervised\ learning\b
Machine Learning	supervised machine learning	\bsupervised\ machine\ learning\b
Machine Learning	unsupervised learning	\bunsupervised\ learning\b
Machine Learning	reinforcement learning	\breinforcement\ learning\b
Machine Learning	reinforcement learning with human feedback	\breinforcement\ learning\ with\ human\ feedback\b
Machine Learning	rlhf	\brlhf\b
Machine Learning	federated learning	\bfederated\ learning\b
Machine Learning	transfer learning (implicit in fine-tuning)	\btransfer\ learning\ \(implicit\ in\ fine\-tuning\)\b
Machine Learning	automl	\bautoml\b
Machine Learning	model training	\bmodel\ training\b
Machine Learning	training models	\btraining\ models\b
Machine Learning	model deployment	\bmodel\ deployment\b
Machine Learning	fine-tuning	\bfine\-tuning\b
Machine Learning	fine tuning	\bfine\ tuning\b
Machine Learning	fine-tune	\bfine\-tune\b
Machine Learning	lora	\blora\b
Machine Learning	quantization	\bquantization\b
Machine Learning	tokenization	\btokenization\b
Machine Learning	inference	\binference\b
Machine Learning	inferences	\binferences\b
Machine Learning	inferencing	\binferencing\b
Machine Learning	embedding	\bembedding\b
Machine Learning	embeddings	\bembeddings\b
Machine Learning	embedding technology	\bembedding\ technology\b
Machine Learning	behavioral cloning	\bbehavioral\ cloning\b
Machine Learning	data science	\bdata\ science\b
Machine Learning	data scientist	\bdata\ scientist\b
Machine Learning	data scientists	\bdata\ scientists\b
Neural Networks & Transformers	neural	\bneural\b
Neural Networks & Transformers	neuronal networks	\bneuronal\ networks\b
Neural Networks & Transformers	multi-layer neural networks	\bmulti\-layer\ neural\ networks\b
Neural Networks & Transformers	deep neural	\bdeep\ neural\b

Neural Networks & Transformers	convolutional neural	\bconvolutional\ neural\b
Neural Networks & Transformers	recurrent neural networks	\brecurrent\ neural\ networks\b
Neural Networks & Transformers	cnn	\bcnn\b
Neural Networks & Transformers	autoencoder	\bautoencoder\b
Neural Networks & Transformers	transformer	\btransformer\b
Neural Networks & Transformers	transformers	\btransformers\b
Neural Networks & Transformers	transformer model	\btransformer\ model\b
Neural Networks & Transformers	transformer models	\btransformer\ models\b
Neural Networks & Transformers	transformer architecture	\btransformer\ architecture\b
Neural Networks & Transformers	transformer architectures	\btransformer\ architectures\b
Neural Networks & Transformers	transformer network	\btransformer\ network\b
Neural Networks & Transformers	transformer technology	\btransformer\ technology\b
Neural Networks & Transformers	transformer-based	\btransformer\-based\b
Neural Networks & Transformers	mixture of experts	\bmixture\ of\ experts\b
Neural Networks & Transformers	mixture-of-experts	\bmixture\-of\-experts\b
Neural Networks & Transformers	moe	\bmoe\b
NLP	nlp	\bnlp\b
NLP	natural language	\bnatural\ language\b
NLP	natural-language	\bnatural\-language\b
NLP	sentiment analysis	\bsentiment\ analysis\b
NLP	customer sentiment analysis	\bcustomer\ sentiment\ analysis\b
NLP	semantic analysis	\bsemantic\ analysis\b
NLP	semantic search	\bsemantic\ search\b
NLP	machine translation	\bmachine\ translation\b
NLP	machine translations	\bmachine\ translations\b
NLP	question answering	\bquestion\ answering\b
NLP	speech recognition	\bspeech\ recognition\b
NLP	speech to text	\bspeech\ to\ text\b
NLP	speech-to-text	\bspeech\-to\-text\b
NLP	voice recognition	\bvoice\ recognition\b
NLP	voice synthesis	\bvoice\ synthesis\b
NLP	optical character recognition	\boptical\ character\ recognition\b
NLP	intelligent character recognition	\bintelligent\ character\ recognition\b
NLP	conversational search	\bconversational\ search\b
NLP	ambient scribing	\bambient\ scribing\b

Computer Vision	computer vision	\bcomputer\ vision\b
Computer Vision	computer vision-enabled	\bcomputer\ vision\-enabled\b
Computer Vision	image recognition	\bimage\ recognition\b
Computer Vision	picture recognition	\bpicture\ recognition\b
Computer Vision	face recognition	\bface\ recognition\b
Computer Vision	facial recognition	\bfacial\ recognition\b
Computer Vision	facial recognition technology	\bfacial\ recognition\ technology\b
Computer Vision	object detection	\bobject\ detection\b
Computer Vision	object recognition	\bobject\ recognition\b
Computer Vision	target identification	\btarget\ identification\b
Computer Vision	multimodal	\bmultimodal\b
Computer Vision	multi-modal	\bmulti\-modal\b
Computer Vision	vision language	\bvision\ language\b
Computer Vision	vision systems	\bvision\ systems\b
Computer Vision	vision transformers	\bvision\ transformers\b
Computer Vision	vision-language model	\bvision\-language\ model\b
Computer Vision	visual language model	\bvisual\ language\ model\b
Computer Vision	vlm	\bvlm\b
Computer Vision	clip	\bclip\b
Computer Vision	machine vision	\bmachine\ vision\b
Computer Vision	radar and vision fusion system	\bradar\ and\ vision\ fusion\ system\b
ML Algorithms	decision tree	\bdecision\ tree\b
ML Algorithms	gradient boosting	\bgradient\ boosting\b
ML Algorithms	random forest (implicit)	\brandom\ forest\ \(implicit\)\b
ML Algorithms	anomaly detection	\banomaly\ detection\b
ML Algorithms	detection systems	\bdetection\ systems\b
ML Algorithms	predictive model	\bpredictive\ model\b
ML Algorithms	predictive models	\bpredictive\ models\b
ML Algorithms	predictive modeling	\bpredictive\ modeling\b
ML Algorithms	predictive analysis	\bpredictive\ analysis\b
ML Algorithms	predictive analytics	\bpredictive\ analytics\b
ML Algorithms	predictive demand	\bpredictive\ demand\b
ML Algorithms	predictive maintenance analytics	\bpredictive\ maintenance\ analytics\b
ML Algorithms	probabilistic modeling	\bprobabilistic\ modeling\b
ML Algorithms	process mining	\bprocess\ mining\b
ML Algorithms	recommendation engine	\brecommendation\ engine\b
ML Algorithms	recommendation engines	\brecommendation\ engines\b
ML Algorithms	recommendation system	\brecommendation\ system\b
ML Algorithms	recommendation systems	\brecommendation\ systems\b
ML Algorithms	recommender system	\brecommender\ system\b
ML Algorithms	recommender systems	\brecommender\ systems\b
Frameworks & Tools	pytorch	\bpytorch\b
Frameworks & Tools	tensorflow	\btensorflow\b
Frameworks & Tools	jax	\bjax\b
Frameworks & Tools	vllm	\bvllm\b

Frameworks & Tools	langchain	\blangchain\b
Frameworks & Tools	lang chain	\blang\ chain\b
Frameworks & Tools	llamaindex	\bllamaindex\b
Frameworks & Tools	lama index	\blama\ index\b
Frameworks & Tools	pinecone	\bpinecone\b
Frameworks & Tools	triton	\btriton\b
Frameworks & Tools	algorithm	\balgorithm\b
RAG & Vector	rag	\brag\b
RAG & Vector	rags	\brags\b
RAG & Vector	graph rag	\bgraph\ rag\b
RAG & Vector	retrieval augmented generation	\bretrieval\ augmented\ generation\b
RAG & Vector	retrievable augmented generation	\bretrievable\ augmented\ generation\b
RAG & Vector	vector database	\bvector\ database\b
RAG & Vector	vector databases	\bvector\ databases\b
RAG & Vector	vector embeddings	\bvector\ embeddings\b
RAG & Vector	vector knowledge base	\bvector\ knowledge\ base\b
RAG & Vector	vector search	\bvector\ search\b
RAG & Vector	vector store	\bvector\ store\b
RAG & Vector	context window	\bcontext\ window\b
RAG & Vector	context windows	\bcontext\ windows\b
RAG & Vector	context length	\bcontext\ length\b
Prompting & In-Context Learning	prompt	\bprompt\b
Prompting & In-Context Learning	prompting	\bprompting\b
Prompting & In-Context Learning	prompts	\bprompts\b
Prompting & In-Context Learning	chain of thought	\bchain\ of\ thought\b
Prompting & In-Context Learning	chain-of-thought	\bchain\-of\-thought\b
Prompting & In-Context Learning	model context protocol	\bmodel\ context\ protocol\b
Chatbots & Voice Bots	bot	\bbot\b
Chatbots & Voice Bots	bots	\bbots\b
Chatbots & Voice Bots	chatbot	\bchatbot\b
Chatbots & Voice Bots	chatbots	\bchatbots\b
Chatbots & Voice Bots	chat bot	\bchat\ bot\b
Chatbots & Voice Bots	chat bots	\bchat\ bots\b
Chatbots & Voice Bots	voice bot	\bvoice\ bot\b
Chatbots & Voice Bots	voice bots	\bvoice\ bots\b
Chatbots & Voice Bots	voicebot	\bvoicebot\b
Chatbots & Voice Bots	voicebots	\bvoicebots\b
Chatbots & Voice Bots	robo advisors	\brobo\ advisors\b
Automation & Robotics	automate	\bautomate\b
Automation & Robotics	automating	\bautomating\b
Automation & Robotics	automation	\bautomation\b

Automation & Robotics	automated	\bautomated\b
Automation & Robotics	automatic	\bautomatic\b
Automation & Robotics	automization	\bautomization\b
Automation & Robotics	autonomous	\bautonomous\b
Automation & Robotics	autonomy	\bautonomy\b
Automation & Robotics	rpa	\brpa\b
Automation & Robotics	robot	\brobot\b
Automation & Robotics	robot-	\brobot\-
Automation & Robotics	robots	\brobots\b
Automation & Robotics	robotic	\brobotic\b
Automation & Robotics	robotization	\brobotization\b
Automation & Robotics	robotaxi	\brobotaxi\b
Automation & Robotics	robotaxis	\brobotaxis\b
Automation & Robotics	cobots	\bcobots\b
Automation & Robotics	rov	\brov\b
Automation & Robotics	drones	\bdrones\b
Automation & Robotics	combat drone	\bcombat\ drone\b
Automation & Robotics	self-driving	\bself\-driving\b
Automation & Robotics	self-driving technology	\bself\-driving\ technology\b
Automation & Robotics	self-driving vehicles	\bself\-driving\ vehicles\b
Automation & Robotics	intelligent driving	\bintelligent\ driving\b
Automation & Robotics	smart driving	\bsmart\ driving\b
Agentic AI	agentic systems	\bagentic\ systems\b
Agentic AI	agenticai	\bagenticai\b
Agentic AI	agentic-ai	\bagentic\-ai\b
Other / Miscellaneous	a i technologies	\ba\ i\ technologies\b
Other / Miscellaneous	aav technologies	\baav\ technologies\b
Other / Miscellaneous	aav technology	\baav\ technology\b
Other / Miscellaneous	aavs	\baavs\b
Other / Miscellaneous	uam	\buam\b
Other / Miscellaneous	uam's	\buam's\b
Other / Miscellaneous	iot	\biot\b
Other / Miscellaneous	digital twins	\bdigital\ twins\b
Other / Miscellaneous	token	\btoken\b
Other / Miscellaneous	tokens	\btokens\b
Other / Miscellaneous	deep fakes	\bdeep\ fakes\b
Other / Miscellaneous	deepfake	\bdeepfake\b
Other / Miscellaneous	deepfakes	\bdeepfakes\b

Appendix D LLM Validation and Classification

To identify AI-related job postings, we employ GPT-5 to analyze job titles and job descriptions from Revelio Labs. Using a structured prompt, the model performs two independent classification tasks. First, it evaluates whether a job title is AI-related based on its semantic meaning and extracts relevant AI-related keywords that appear directly in the title. Second, it determines whether the job description contains a substantive discussion of AI-related technologies, guided by both a predefined AI keyword list from Babina et al. (2024), "Artificial intelligence, firm growth, and product innovation," and contextual semantic understanding. For descriptions identified as AI-related, the model extracts meaningful AI-related keywords or phrases exactly as they appear in the text. We then use the GPT-generated keyword list from a sample of 200,000 job postings to manually refine and compile a comprehensive AI keyword list disclosed in Appendix Table C1. Full prompt used for the analysis is presented in Figure 1.

```
You are an accounting expert research analyst.

The dataset has two fields:
    - jobtitle_translated
    - description

TASK 1 – JOB TITLE AI DETECTION
    - Determine whether 'jobtitle_translated' is AI-related
        - Ignore metaphorical mentions
        - AI KEYWORDS REFERENCE: {keyword_text}
    - If yes, extract all English keywords or phrases exactly
    - If not AI-related, keyword must be empty
    - Fill "jobtitle_is_ai_related" and "jobtitle_ai_keyword"

TASK 2 – DESCRIPTION AI DETECTION
    - Determine whether 'description' is AI-related
    - If yes, extract all English keywords or phrases exactly
    - If not AI-related, keyword must be empty
    - Fill "description_is_ai_related" and "description_ai_keyword"

OUTPUT FORMAT (JSON ONLY)
{
    "jobtitle_is_ai_related": 1 or 0,
    "jobtitle_ai_keyword": "",
    "description_is_ai_related": 1 or 0,
    "description_ai_keyword": ""
}
```

Figure 1: Prompt used for job postings

We apply the GPT-5 to process textual data from Glassdoor employee reviews. Figure 2 presents the prompt used to analyze the Glassdoor employee reviews. Each review contains four fields (review summary, advice, pros, and cons), which together provide structured but open-ended employee feedback. As shown in Figure 2, we implement a structured multi-task prompt that decomposes the analysis into sequential subtasks and leverages in-context learning through both explicit task instructions and embedded examples. As a first step, we preprocess the data by removing comments that appear to be AI-generated (e.g., those beginning with "ChatGPT/" or "ChatGPT\") to avoid contamination, since these reviews include the keyword "ChatGPT" but their content is not actually related to AI. The model then determines whether a review discusses AI using both keyword matching and contextual understanding, while excluding cases where AI is mentioned only as part of a job title or used metaphorically. Here, reviews that are not AI-related are excluded from further analysis. For AI-related reviews, the model extracts the relevant sentences supporting AI discussion, evaluates sentiment toward AI (positive, negative, or neutral) using contextual cues across fields (with "pros" generally indicating positive sentiment and "cons" indicating negative sentiment), and identifies sentiment-bearing keywords. It then assigns each review to one predefined category (including AI Job Insecurity, AI UpSkill, AI Operation & Productivity, and AI Leadership) guided by in-context examples. Finally, it classifies the overall sentiment of the full review to summarize the employee's overall opinion in the review (not only the AI-related content).

```
You are an accounting expert research analyst.

The dataset contains four fields:
    - review_summary
    - review_advice
    - review_pros
    - review_cons

Pre-processing rule: If any comment starts with "ChatGPT/" or "ChatGPT\\", consider it
                     AI-generated and exclude it from AI-related analysis.

Step 1 – AI RELEVANCE
    - Determine if the review is AI-related using keywords + semantic understanding
       - AI-related topics include AI, automation, machine learning, algorithms, etc.
       - Ignore metaphorical mentions of AI.
       - If AI keyword is only mentioned in summary and is only related to job title or
         firm, such as "AI intern", "ML engineer", "AI developer", "Nice LLM startup",
         "Lean Medical AI startup", this is not considered as AI-related comment, return 0
    - Output "ai-related-gpt": 1 (yes) or 0 (no).
    - If ai-related-gpt = 1, extract:
       - "ai_comment": all declarative sentence(s) that support the AI relevance decision
    - If ai-related-gpt = 0, return all other fields as "" and STOP.
    - AI KEYWORDS REFERENCE: {keyword_text}

Step 2 - AI SENTIMENT (if ai-related-gpt = 1)
    - Classify sentiment toward AI: positive / negative / neutral
    - Guidance for field-based sentiment:
       - Content in review_pros → generally positive.
       - Content in review_cons → generally negative.
       - Content in review_summary or review_advice → determine sentiment based on context
    - Extract:
```

```
       - "ai_sentiment": positive | negative | neutral
       - "sentiment_keywords": keywords that directly reflect the AI sentiment

Step 3 - AI CATEGORY
    - Assign exactly one category if applicable (or "" if none):
       1. AI Job Insecurity: If the review talks about layoffs or employee displacement
          by AI or robots.
       2. AI UpSkill: If the review talks about employees' experiences with and
          opportunities to use AI tools, as well as broader efforts toward AI-related skill
          development and training.
       3. AI Operation and Productivity: If the review talks about how AI or automation
          affects firm operation, production, services, or efficiency.
       4. AI Leadership: If the review talks about artificial intelligence and managers or
          Leadership.
    - Use the following in-context examples to guide your classification of the review:
      {examples_text}

Step 4 – OVERALL SENTIMENT
    - Classify the overall sentiment of the full review (not just AI part): positive /
      negative / neutral

OUTPUT FORMAT (JSON ONLY)
{
    "ai-related-gpt": 0 or 1,
    "ai_comment": "",
    "ai_sentiment": "",
    "sentiment_keywords": "",
    "ai_category": "",
    "overall_sentiment": ""
}
```
Figure 2: Prompt used for Glassdoor employee reviews

To automatically analyze the AI-related content, we employ GPT-5 to process textual data from conference call transcripts. Specifically, we design a structured multi-task prompting framework that enables the model to perform a sequence of coordinated analytical tasks within a unified prompt. The prompt first instructs GPT-5 to determine whether a transcript segment is related to artificial intelligence (AI), using both a predefined AI keyword list disclosed in Appendix Table C1 and contextual semantic understanding of AI-related concepts, such as machine learning, automation, and neural networks. For segments identified as AI-related, GPT-5 subsequently performs several information extraction tasks, including identifying the specific AI keywords mentioned, counting the frequency of those keywords, classifying the sentiment toward AI (positive, negative, or neutral), and detecting any explicitly disclosed monetary investments associated with AI initiatives. This structured prompting approach guides the model through a clear analytical workflow, improving consistency and transparency in the classification and extraction process. By organizing multiple analytical tasks into a unified prompt with explicit instructions and output constraints, this framework reduces ambiguity in model responses and facilitates systematic large-scale analysis of AI-related disclosures in conference call transcripts. Figure 3 presents the prompt used to analyze the conference call transcripts.

```
You are an accounting expert research analyst.

STEP 1 – AI RELEVANCE
    - Determine whether the input text is AI-related
        - Ignore metaphorical mentions
        - AI KEYWORDS REFERENCE: {keyword_text}
    - "mention_AI_gpt": 1 for AI, 0 for non-AI
    - If 0, set all other fields to empty

STEP 2 – KEYWORDS & COUNTS
    - Identify the specific AI keyword(s) in the input text
    - Count total mentions of AI keywords
    - Fill "mentioned_keyword_gpt" and "num_keyword_mentioned_gpt"

STEP 3 – SENTIMENT
    - Classify sentiment toward AI: positive / negative / neutral
    - Extract declarative sentences supporting sentiment
    - Fill "sentiment_gpt" and "supporting_sentiment_gpt"

STEP 4 – AI INVESTMENT
    - Identify the monetary amount.
    - Extract the declarative sentence(s) that clearly support this amount.
    - Fill "investment_amount_gpt" and "supporting_investment_amount_gpt"

OUTPUT FORMAT (JSON ONLY)
{
    "mention_AI_gpt": "",
    "mentioned_keyword_gpt": "",
    "num_keyword_mentioned_gpt": "",
    "sentiment_gpt": "",
    "supporting_sentiment_gpt": "",
    "investment_amount_gpt": "",
    "supporting_investment_amount_gpt": ""
}
```